LUMEN: (Lane-Use Monitoring for Event Notification): An Integrated Sensor and Computer Vision-Based System for Rural Blind Curves Lane Safety Enhancement and Accident Detection

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Lastly, to students, educators, educational institutions, and possible beneficiaries. It is our hope that LUMEN contributes to improving road safety through accessible and practical technological solutions and that it serves as a meaningful step toward safer and more responsive environments within and beyond the academic community.

ABSTRACT

Title: LUMEN: (Lane-Use Monitoring for Event Notification): An Integrated Sensor and Computer Vision-Based System for Rural Blind Curves Lane Safety Enhancement and Accident Detection

Keywords: Incident Aware System, Real-Time Event Notification Systems, YOLO Algorithm, Intelligent Transportation System, Road Safety, Wireless Sensor Network

Members: Carl Justin S. De Leon, Brad Gervin D. Eslabra, Ric Ardeil Sean C. Estrada, Richell Ann C. Gabis, and Mitch T. Torio.

Road traffic accidents remain a critical safety concern, particularly in rural blind-curve areas where limited visibility and the absence of advanced traffic infrastructure significantly increase collision risks. This study presents LUMEN (Lane-Use Monitoring for Event Notification), an incident-aware traffic warning system designed to enhance lane safety and driver awareness in accident-prone rural settings. The system integrates computer vision, radar detection, and real-time event notification to actively monitor vehicle presence and potential accidents at blind curves.

LUMEN employs a Raspberry Pi 5 as the central processing unit, serving as the main node for sensor data aggregation, computer vision inference, and MQTT broker hosting. Vehicle proximity and approach direction are detected using HLK-LD2451 24GHz millimeter-wave radar sensors deployed through ESP32 microcontroller nodes, which transmit data wirelessly to the Raspberry Pi via HC12 RF modules operating at 433MHz. A day/night wide-angle OV5647 camera module streams live footage through MediaMTX as an RTSP feed, while a custom-trained YOLOv8 NCNN accident detection model runs inference directly on the Raspberry Pi 5. Device communication is managed via an MQTT publish-subscribe architecture hosted locally on the Raspberry Pi, enabling low-latency data exchange between sensor nodes, the processing unit, and a remote

monitoring dashboard. The dashboard, built using C# and the Avalonia framework, receives real-time radar and vision data over a Tailscale VPN tunnel, allowing remote monitoring regardless of network topology. Warning signals are delivered through NeoPixel LED matrix mounted at a blinker light enclosure, triggered by the Raspberry Pi upon vehicle detection. A solar-powered setup ensures energy efficiency and suitability for off-grid rural environments.

Prototype testing focused on detection accuracy, response time, RF communication reliability, and system stability under varying conditions. Results demonstrate that the system can reliably detect approaching vehicles through radar, classify accident events through computer vision, trigger timely LED warning signals, and transmit real-time alerts to a remote dashboard, thereby improving situational awareness at blind curves. The findings confirm the technical feasibility of LUMEN as a low-cost, scalable, and energy-efficient safety solution with strong potential to reduce road accidents and support safer transportation in rural communities with minimal existing infrastructure.

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CHAPTER I
INTRODUCTION

Background of the Study

Safety in roads has long been a global challenge, with traffic-related accidents continuing to affect human lives whether for short-term or long-term, and worst possibly claim their lives. According to the World Health Organization (2023), traffic-related accidents as a whole, claims about 1.19 million lives annually and injures 20-50 million each year. Among the most common accident-prone areas are blind curves specifically in Rural Areas, where drivers have limited visibility of incoming traffic. While urban areas often rely on advanced traffic management systems, rural roads remain neglected, relying mostly on basic reflectors, signages, and timely human labor that provide only minimal safety development.

Within the Philippines, road safety concerns are deepened by the existing road/terrain problems such as narrow provincial roads, either unsafe or limited infrastructure, and also the lack of human efforts in rural areas. As reported by WHO (2023), despite the continuous promotion of Road safety measures by the Department of Transportation and Land Transportation Office, the data statistics that the Philippine Statistics Authority (2023) states that road traffic deaths increased from *39%* from *7,938 deaths in 2011* to *11,096 deaths in 2021.* During the 2020 *Pandemic*, it shows how severe traffic accidents occur as it happens that *8,746 deaths* were calculated despite the Pandemic Lockdown.

Locally, Barangay Tebeng, Dagupan City, Pangasinan serves as a representative example of the road safety challenges prevalent in rural Philippine communities. The barangay's road network features multiple blind and sharp curves that severely restrict driver visibility, preventing motorists from detecting oncoming vehicles until they are already in dangerously close proximity. The tight bends and narrow lane widths typical of this area leave little to no room for evasive maneuvers, significantly increasing the likelihood of head-on collisions. While road corner reflectors are present along several of these curved sections, their effectiveness remains limited. They rely on adequate lighting conditions to function, offer no real-time indication of an actively approaching vehicle, and are rendered ineffective during nighttime, heavy rain, or when obscured by dirt and weathering. These passive measures, though present, are insufficient in providing the level of situational awareness needed to meaningfully reduce collision risks at blind curves. Consequently, the curved stretches of Tebeng Road remain a recognized accident-prone zone within the locality, embodying the very infrastructure limitations that continue to claim lives on rural Philippine roads.

To mitigate this persistent traffic-related global crisis, taking advantage of the rushing advancement and innovations of technology offers a lot of opportunities to solve this problem even just by targeting rural areas for a while would provide a significant impact.The development of an incident-aware monitoring and warning system for blind curves introduces an active approach to road safety by replacing passive visual aids such as corner reflective mirrors.This proposed system, designed for blind curves, actively

signals the presence of approaching vehicles to provide the drivers with efficient and real-time awareness, reducing the risk of accidents in rural settings.

Rural road environments continue to experience elevated traffic-related risks due to challenging roadway conditions such as blind curves, narrow lanes, and limited visibility, often compounded by the absence of active safety infrastructure. In many of these areas, drivers rely primarily on passive warning measures, which are frequently insufficient in preventing collisions, particularly during low-light or adverse weather conditions. Addressing these limitations requires the adoption of accessible and adaptive safety technologies that can operate effectively in resource-constrained settings. A low-cost, energy-efficient, and automated lane-use monitoring and warning system offers a practical approach to enhancing driver awareness by providing timely alerts, thereby contributing to improved road safety and reducing the likelihood of accidents in rural environments.

Thus, this study aims to provide a formula to solve the timely related issue. Introducing LUMEN, an innovative, collision avoidance system capable of detecting accidents with live monitoring designed for safer lane management in rural blind curves. By integrating priority lane-use monitoring with active signaling, the project delivers its purpose of a practical and cost-effective solution aiming to reduce accidents and improve driver awareness tailored to the needs of rural communities and even aligning with broader road safety goals for both locally and worldwide.

Objectives of the Study

1. To design and construct a functional prototype that integrates lane-use monitoring and visual warning mechanisms suitable for rural blind-curve conditions.
2. To develop and integrate detection and monitoring capabilities that enable systems to identify accidents in real time.
3. To incorporate a real-time event notification feature that provides immediate warning to drivers and communicates incident alerts to relevant authorities.
4. To evaluate LUMEN's performance in terms of detection , response speed, power efficiency, and reliability under varying traffic and environmental conditions.
5. To assess the feasibility and sustainability of the system for long term rural deployment.

Scope and Delimitation

Scope

This study involves the conceptualization, design, development, and evaluation of an incident-aware monitoring and warning system for accident-prone rural blind curves, with Barangay Tebeng, Dagupan City, Pangasinan as the primary deployment and testing site. The system consists of one primary monitoring unit and two auxiliary detection units. The primary unit is responsible for vehicle monitoring and accident detection using a day/night wide-angle OV5647 camera module integrated with a custom-trained YOLOv8n NCNN computer vision model, enabling continuous analysis under varying

lighting conditions. The auxiliary units utilize HLK-LD2451 24 GHz millimeter-wave(mmWave) radar sensors deployed through ESP32 microcontroller nodes to detect approaching vehicles, their speed, and distance. All three units are interconnected through HC-12 RF modules operating at 433 MHz, ensuring low-latency wireless communication between components. A Raspberry Pi 5 serves as the central processing unit, managing sensor data aggregation, computer vision inference, and MQTT-based communication. Warning signals are delivered through NeoPixel LED matrix blinker lights triggered upon vehicle detection, while a remote monitoring dashboard built using C# and the Avalonia framework receives real-time data over a Tailscale VPN tunnel. To support off-grid rural deployment, the system is powered by a solar energy setup consisting of solar panels, a solar charge controller, and a LiFePO4 battery.

The scope further covers prototype construction, circuit design, flowchart creation, programming, and pilot testing under different ambient conditions. Key parameters measured include detection accuracy, response time, energy efficiency, and hardware reliability. The objective of this study is to determine the technical feasibility of implementing a localized, low-cost safety solution to reduce collisions in blind curve areas where visibility is limited and permanent traffic infrastructure is not feasible. The materials and methods employed emphasize accessibility, cost-effectiveness, and adaptability to rural conditions, making the system relevant to drivers, pedestrians, and local communities in high-risk zones. Furthermore, the study seeks to contribute to safer transportation in rural accident-prone areas by demonstrating the effectiveness of

integrating computer vision, radar sensing, wireless communication, and renewable energy into a cohesive and reliable warning system.

Delimitation

The conduct of this study is delimited to accident-prone rural blind curves, specifically within the road network of Barangay Tebeng, Dagupan City, Pangasinan, where limited visibility significantly increases the risk of collisions. The system is confined to detecting vehicles in real time and triggering immediate LED blinker light warnings to alert drivers. It does not extend to urban intersections, multi-lane highways, or other complex roadway environments that demand broader infrastructure and more advanced traffic management systems. Furthermore, the evaluation is delimited to prototype development and short-term localized testing, focusing on functional parameters such as detection accuracy, response time, power efficiency, and hardware reliability under varying environmental conditions. Predictive analytics, large-scale congestion modeling, and advanced vehicle-to-infrastructure communications are excluded from the current scope of investigation.

The study is further delimited by not addressing the long-term impacts of extreme environmental conditions, such as typhoons or heavy rainfall, which may affect sensor performance and hardware durability over extended periods. Instead, the focus remains solely on establishing technical feasibility within blind curve applications, ensuring that the findings are directly relevant to improving safety in this specific roadway context.

The LUMEN system, in its current form, does not incorporate cellular-based SMS alerting or integration with existing government traffic management platforms, as these fall outside the bounds of the present prototype. While the system shows potential for broader applications, its present design and testing are strictly limited to the defined deployment site in rural blind curve conditions. Most importantly, the project is delimited to functioning as a proof-of-concept prototype for feasibility testing only and is not intended for immediate large-scale or permanent field deployment. By setting these delimitations, the study provides a focused and realistic assessment of the proposed system's operational viability within its intended rural context.

Significance of the Study

In a global context, road safety remains one of the leading causes of injury and death worldwide, with rural areas being particularly vulnerable due to underdeveloped infrastructure and limited access to advanced traffic management systems. Blind curves in particular are recurring accident-prone zones where conventional safety measures as signages are often insufficient. The pursuit of accessible, adaptable, and cost-conscious solutions tailored for blind curves, this study contributes to international efforts to reduce traffic-related accidents and fatalities.

In an economic context, accidents impose substantial costs, particularly in rural communities where resources are already scarce. Families face expenses from medical treatment, insufficient income, while local governments bear costs associated with

emergency and accident response. Developing cost-effective and scalable safety solutions alleviates these pressures by reducing the frequency and severity of accidents.

In the environmental context, traditional solutions to improve road safety, such as road widening or extensive reconstruction, often damage natural landscapes, consume large amounts of resources, and might cause long-term ecological disruptions. Sustainable environmental alternatives minimize these constraints by focusing on small-scale, energy efficient implementations. By integrating these technologies into rural areas, it demonstrates how road safety improvements can align with environmental preservation.

In the societal context, mitigating the risks posed by blind curves contribute to the overall stability and resilience of rural communities. Its significance can be seen in several ways:

- Motorists and pedestrians benefit from improved awareness and safer travel conditions, reducing their exposure to accidents and strengthening their confidence in everyday mobility.
- Local government units and transportation authorities gain access to a scalable cost-effective means of enhancing road safety, reinforcing effective governance and responsible public service delivery.
- Academe and research institutions acquire a valuable case study in applying embedded systems, engineering innovations, thereby enriching both theoretical study and practical innovation in aspects of transportation safety.

Conceptual Framework

INPUT	PROCESS	OUTPUT
USERS ● Vehicle Drivers ● Pedestrians ● Administrators **HARDWARE** ● Raspberry Pi 5 ● ESP32 Microcontroller Nodes ● HLK-LD2451 24 GHz Millimeter-Wave Radar Sensor ● ArduCam OV5647 Day/Night Wide-Angle Camera Module ● HC-12 RF Module ● NeoPixel LED Matrix Blinker Lights ● 12V 30AH/36AH LiFePO4 Battery ● 30W / 100W Solar Panels	**RESEARCH AND DEVELOPMENT** ● Create circuit diagrams, system architecture, and flowcharts ● Identify deployment site and assess road conditions at Barangay Tebeng **PROTOTYPING** ● Label dataset and train YOLOv8n NCNN model ● Program Raspberry Pi, ESP32, and monitoring dashboard **TESTING AND DEBUGGING** ● Conduct unit,	**LUMEN: (Lane-Use Monitoring for Event Notification): An Integrated Sensor and Computer Vision-Based System for Rural Blind Curves Lane Safety Enhancement and Accident Detection**

• Solar Charge Controller **SOFTWARE** • Thonny IDE • Microsoft Visual Studio • Avalonia Framework • YOLOv8n NCNN Computer Vision Model • MediaMTX (RTSP Stream Server) • MQTT Broker (Hosted on Raspberry Pi 5) • Tailscale VPN • Roboflow (Dataset Labeling) • Google Colab (Model Training)	integration, and pilot field testing • Measure detection accuracy, response time, and power efficiency	

Figure 1: Research Paradigm of the Study

Utilizing a Raspberry Pi 5 as the central processing unit and ESP32 microcontroller nodes as auxiliary detection units, LUMEN integrates a range of

hardware components including an ArduCam OV5647 day/night wide-angle camera for real-time visual data, HLK-LD2451 24 GHz millimeter-wave radar sensors for vehicle proximity and approach direction detection, NeoPixel LED matrix blinker lights for active driver warnings, and a solar-powered off-grid energy setup ensuring continuous operation in rural environments. Powered by a custom-trained YOLOv8n NCNN computer vision model and supported by an MQTT-based communication architecture, LUMEN effectively detects approaching vehicles and identifies accident events, triggering timely warning signals and transmitting real-time alerts to a remote monitoring dashboard accessible by authorities over a Tailscale VPN tunnel.

The development process involves careful research, prototyping, programming, and rigorous testing to ensure accurate vehicle and accident detection, reliable wireless communication, and optimal warning signal response time. By delivering an active, incident-aware lane safety system, LUMEN aims to reduce the risk of collisions and improve driver awareness at accident-prone rural blind curves where conventional passive safety measures remain insufficient.

Definition of Terms

Avalonia Framework: A cross-platform user interface framework used to develop the LUMEN remote monitoring dashboard, enabling graphical display of real-time detection events and system status.

Computer Vision: A field of artificial intelligence that enables systems to interpret and analyze visual data from cameras or images, used in LUMEN for real-time vehicle and accident detection.

Embedded System: A specialized computer designed to perform dedicated functions within larger mechanical or electronic systems, such as the Raspberry Pi 5 and ESP32 used in LUMEN.

ESP32: A microcontroller with wireless communication capabilities used in LUMEN's auxiliary detection units to process radar sensor data and transmit detection signals to the primary monitoring unit via HC-12 RF modules.

HC-12 RF Module: A wireless serial communication transceiver operating at 433 MHz, used to enable long-range data transmission between the ESP32 auxiliary units and the Raspberry Pi 5 central processing unit.

HLK-LD2451: A 24 GHz millimeter-wave Doppler radar sensor used in LUMEN to detect the presence, speed, and approach direction of vehicles at blind curve road sections.

Incident-Aware System: A system designed to actively monitor and respond to unexpected road events, such as detecting approaching vehicles or identifying accidents in real time.

MediaMTX: A real-time media streaming server used in LUMEN to broadcast live camera footage as an RTSP feed accessible through the remote monitoring dashboard.

MQTT (Message Queuing Telemetry Transport): A lightweight publish-subscribe communication protocol hosted on the Raspberry Pi 5, enabling low-latency data exchange between LUMEN's sensor nodes, processing unit, and monitoring dashboard.

NeoPixel LED Matrix: An addressable LED lighting component mounted within the blinker light enclosure, triggered by the Raspberry Pi 5 upon vehicle detection to deliver visible warning signals to approaching drivers.

Prototype: An initial working version of a system used to validate design, functionality, and feasibility before full-scale development or deployment.

Raspberry Pi 5: A single-board computer serving as the central processing unit of LUMEN, responsible for running the YOLOv8n computer vision model, managing MQTT communication, aggregating sensor data, and controlling warning signal outputs.

Real-Time Processing: The capability of a system to instantly analyze input data, such as radar signals or camera footage, and produce immediate output, enabling timely actions and responses in critical applications.

Roboflow: A cloud-based platform used for collecting, organizing, and labeling the image dataset used to train LUMEN's custom YOLOv8n accident detection model.

RTSP (Real-Time Streaming Protocol): A network protocol used to stream live video footage from the LUMEN camera module to the remote monitoring dashboard.

Sensor: A device that detects physical or environmental changes and converts them into electrical signals, such as the millimeter-wave radar sensors used in LUMEN for vehicle detection.

Solar Charge Controller: An electronic device that regulates the flow of electricity from the solar panels to the LiFePO4 battery, ensuring safe and efficient charging for LUMEN's off-grid power setup.

Tailscale VPN: A virtual private network service used to establish a secure remote connection between the LUMEN system and the monitoring dashboard, enabling real-time data access regardless of network topology.

YOLOv8n NCNN: A lightweight, optimized variant of the You Only Look Once object detection algorithm, custom-trained and deployed on the Raspberry Pi 5 to perform real-time accident detection from live camera footage.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter includes the pertinent studies and literature that the researchers took into account to support the significance of the current investigation. In order to better grasp the study, it also provides the synthesis of the art to fully comprehend the research.

Incident Aware System

Accidents significantly affect road safety and the efficiency of transportation networks, making rapid detection and response essential. Gannina et al. (2024) propose a road accident and incident detection system that leverages traffic API monitoring and continuous analysis of real-time data from sources such as GPS-enabled devices. By tracking live vehicle movements, the system can identify abnormal traffic patterns—such as sudden road closures or irregular slowdowns—that may indicate an accident or incident. The proposed approach uses numerical traffic pattern analysis to detect anomalies and notifies users immediately, enabling timely responses to mitigate impacts. Beyond individual drivers, the system provides valuable insights for authorities by identifying accident-prone areas where preventive safety measures can be implemented. Furthermore, it assists law enforcement in recognizing drivers at higher risk of causing accidents, thereby informing targeted educational and enforcement programs.According to Gannina et al. (2024), this method enhances preparedness and reduces the negative consequences of traffic incidents, ultimately contributing to improved road safety and more efficient traffic management.

Research conducted by Toan Dinh Trinh (2025), presents an Uncertainty-Aware Decision Support System (UA-DSS) for incident-responsive ramp metering in urban expressways. The system integrates a multi-stage fuzzy logic controller (MS-FLC) with a fuzzy-defined risk evaluation mechanism to address real-time decision-making under data uncertainty. A linguistic Risk Factor is formulated based on incident severity, time-of-day, and control effectiveness, enabling adaptive control decisions through fuzzy inference. Simulation experiments using a representative network and diverse incident scenarios demonstrate that UA-DSS significantly outperforms conventional strategies such as ALINEA and non-risk-adaptive MS-FLC. The proposed approach shows strong robustness, improving mainline traffic flow and minimizing ramp queue spillback, even under delayed or imprecise input conditions. This work contributes to the field of intelligent transportation systems by providing a practical and interpretable AI-driven traffic control architecture, aligned with EAAI's emphasis on engineering applications of fuzzy logic and real-time decision support.

According to the study of Muhammed Ali Basli et al. (2025),Vehicle Ad Hoc Networks emerge as a new research area. Vehicle ad hoc networks (VANETs) are widely used in intelligent transportation systems to reduce traffic congestion as well as to ensure safety and security by using vehicle-to-vehicle(V2 V) and vehicle-to-roadside(V2R) unit communications. Many people are seriously injured or even die in traffic accidents due to human errors, including driver errors (*e.g.* driver inattention and distraction, careless driving and poor driving skills) and errors of other road users (*e.g.* traffic violations). According to TÜİK data, the fact that the ratio of driver fault to total fault has never

fallen below 88% for decades shows that this problem has become a chronic problem and needs to be thought about and solution(s) produced. The main factor that increases the possibility of drivers making mistakes is the roads they navigate with the help of navigation that they have not experienced before. At this point, a driver traveling on a road he has not experienced before should be warned as he approaches the point where many accidents have occurred on the relevant route before. By warning the driver who is approaching accident black spots, where many accidents have occurred in the past, more attention can be paid to points with high accident risk. Radius information of accident black spots can be estimated using a fuzzy-based model. As a result, the aim is to minimize accidents caused by driver error. This is done with the Accident Information System (KABİS) application.

Additionally, in the paper proposed by Leila Bouchrit et al. (2023),Vehicular ad hoc networks (VANETs) safety applications aim to mitigate the growing frequency of daily road accidents. These applications require the prompt dissemination of warning messages among vehicles, regardless of traffic conditions or environment. The limited terrestrial coverage in sparsely populated rural regions challenges the timely receipt of warnings during accidents. To address these issues, integrating unmanned aerial vehicles (UAVs) into VANETs offers a promising solution. UAVs can function as flying relays, meeting necessary delay limitations and overcoming communication problems. In prior work, we introduced a unified UAVs-VANET architecture, using UAVs to relay safety messages among vehicles in rural settings. This paper extends that work by integrating UAVs in rural and urban environments. Our approach includes determining the minimal

residual UAV energy needed to connect to a charging station, ensuring continuous UAV operation. This extended framework ensures all vehicles on a 25 km urban highway receive alerts within 1.95 seconds.

The emergence of vehicular networks has enabled early road safety warnings, as vehicles can access real-time safety information through networked communication. However, Wang et al. (2023) note that the topology of vehicular networks is often fragmented due to high vehicle mobility, which can disrupt continuous communication. To address this, vehicular data is typically uploaded to infrastructures, allowing vehicles to download information when needed. Yet, in areas such as suburban regions with sparse infrastructure, data delivery is often delayed, and reception may fail. To overcome these limitations, the authors propose leveraging parking vehicles as virtual infrastructures to support data delivery to moving vehicles. Their approach, called Vehicular Data Delivery based on Virtual Infrastructures (DDVI), elects parking vehicles at the periphery of parking lots to act as gateways, enabling surrounding vehicles to access safety data via name-centric reverse paths in a single data delivery process. Experimental evaluation of DDVI demonstrates that this method effectively reduces data delivery delays and minimizes communication failures, highlighting the potential of using underutilized parking vehicles to enhance the reliability and efficiency of vehicular network-based road safety systems.

Road traffic accidents result in over 1.19 million deaths annually (WHO, 2023), emphasizing the urgent need for more effective predictive safety systems. El-Gezawi (2025) proposes an integrated framework for intelligent traffic risk prediction and

accident prevention, combining real-time object detection (via YOLOv4), adaptive lighting, federated learning, and vehicle-to-infrastructure (V2I) communication.A key contribution of the study is a mathematically grounded accident probability model, based on neural network classifiers and trained on real-world traffic datasets from urban Saudi roads. The model incorporates variables such as traffic density, vehicle speed, weather conditions, and driver behavior. Experimental results show that the system achieves 88.7% predictive accuracy with a validated F1-score of 0.87, providing hazard alerts up to 600 meters in advance. These results outperform conventional AI-based traffic prediction systems in terms of both accuracy and latency. According to El-Gezawi (2025), the framework's scalability and low-latency edge deployment make it suitable for real-time risk mitigation in diverse urban environments, highlighting its potential as a robust tool for accident prevention and intelligent traffic management.

The proportion of rural road fatalities is gradually increasing, with traffic safety problems being particularly pronounced in town-rural and town-center areas. Based on accident data from Hunan Province, Li (2022) conducted a chi-square test to examine the correlation between various risk factors and accident severity, stratified by gender. A binary logistic regression model was then used to identify significant factors influencing accident severity in each area. The results indicate that in town-rural areas, severe accidents are significantly associated with single-vehicle crashes, motorcycle involvement, and intersections. In contrast, in town-center areas, severe accidents are primarily linked to elderly drivers, single-vehicle crashes, and nighttime conditions. According to Li (2022), identifying these risk factors provides a foundation for targeted

safety improvement measures in key areas, contributing to enhanced traffic safety and the sustainability of rural transportation networks.

Road Traffic Accidents (RTAs) are a leading cause of fatalities and injuries worldwide, particularly in developing countries where infrastructure and traffic management are often inadequate. According to Ashraf (2025), the spatial patterns and determinants of RTAs along the Chiniot-Sargodha Road, Pakistan, can be examined by integrating multiple spatial analysis methods to identify accident hotspots and their correlation with demographic, infrastructural, and environmental factors, addressing gaps in the existing literature. The study uses spatial and statistical techniques, including Kernel Density Estimation (KDE), Standard Deviational Ellipse (SDE), and Average Nearest Neighbor (ANN), to analyze RTA clustering patterns from 2012 to 2021. Data from traffic records and geographical databases were integrated with GIS tools to visualize hotspots and assess their association with demographic, infrastructural, and environmental factors. The findings show that RTA hotspots concentrated in central urban areas of Chiniot-Sargodha Road mainly affect younger males (21–30). The year 2012 stands out due to infrastructure changes, traffic volume, and poor weather. Poor road conditions, inadequate management, and human errors are vital contributors, with a multi-method approach offering a comprehensive understanding of RTA dynamics. The study's novelty lies in its multi-method approach, combining spatial and statistical analyses to comprehensively understand RTA dynamics in semi-urban areas. It highlights the need for a multifaceted approach to road safety, combining engineering, traffic management, and education, while also providing a framework with practical insights for

policymakers and planners. Ashraf (2025) further suggests that future research should use real-time data and machine learning for improved accident prediction and prevention strategies.

Road traffic accidents (RTAs) pose serious safety challenges that jeopardize socio-economic growth and sustainable development of the road transport sector in particular, and Nigeria's overall economic development in general. According to Ibe et al. (2025), this necessitates knowledge-based approaches for proactively implementing controls aimed at significantly reducing RTAs and their associated injury, disability, and death burdens. Their study used secondary RTA data in Nigeria to map the geospatial probability of road accidents, with the goal of prioritizing road safety awareness and mitigating the consequences of accidents in specific geospatial locations. The objectives included determining the occurrence probability of RTAs, along with the probabilities of death, injury, and disability across Nigeria's 36 states, the FCT, and major road corridors, while also providing knowledge-based information for efficient and effective road safety resource planning and administration (RSRPA). The results, as highlighted by Ibe et al. (2025), are crucial for prioritizing concentrations of road safety patrols and enforcement of traffic regulations. The disproportionate accident rates across Nigeria's geospatial zones suggest that varying the concentrations and approaches to road safety enforcement—giving higher priority to regions with greater occurrence probabilities—will yield more effective results. Specifically, the study shows that the North West, North Central, and South West regions have the highest occurrence probability coefficients of RTAs, followed by the North East region and the FCT Abuja.

In contrast, the South East and South South regions have the lowest occurrence probability coefficients, along with reduced injury and death consequences. Furthermore, Ibe et al. (2025) emphasize that mapping these findings onto Nigeria's outline map and embedding them within a geographic information system (GIS) could provide authorities and road users with actionable insights for enforcing compliance with road safety measures and policies across different geospatial locations. If effectively implemented, this framework has the potential to achieve significant reductions in RTA incidents and their consequences across Nigeria and West Africa in the long run.

Real-Time Event Notification Systems

In a study, Bozomitu et al. (2021) emphasized the crucial role of real-time event notification systems in enhancing drivers' cognitive processing by supplying accurate situational data on roadway conditions. Their findings highlighted the effectiveness of timely alerts in improving situational awareness and decision-making when faced with unforeseen hazards. When drivers receive immediate notifications about weather changes, obstacles, or roadwork, their ability to process critical information improves significantly, thereby reducing reaction times that often lead to accidents. This fosters a safer driving environment by helping drivers shift from reactive responses to proactive anticipation of risks. Moreover, the integration of such systems cultivates a learning environment where drivers refine their awareness and develop stronger anticipatory skills over time. The study underscored that real-time alerts are not merely supplementary

measures but fundamental tools in advancing traffic safety and efficiency, forming the groundwork for future developments in intelligent transportation systems.

Odeleye and Umar (2021) investigated the effectiveness of real-time notifications in unpredictable driving environments, where uncertainty often compromises road safety. Their research demonstrated that immediate event alerts reduce confusion and hesitation, which improves traffic dynamics by enabling safer and more confident driving behavior. Enhanced situational awareness through these systems helps drivers make informed decisions under pressure, thereby minimizing risks associated with panic or indecision. Their study also noted that such technologies alleviate the psychological burden on drivers, especially in complex traffic situations, promoting smoother flow and reducing accidents. By confirming the importance of real-time notifications, their work validates current technologies while encouraging further innovation to maximize clarity, reliability, and inclusivity. These findings also carry implications for traffic regulation and infrastructure planning, emphasizing the role of advanced notification systems in shaping safer transportation networks.

Focusing on predictive safety, Xu et al. (2024) explored the integration of deep learning into Internet of Vehicles (IoV) systems to anticipate and prevent accidents. Their study demonstrated that advanced algorithms can process vast data streams in real time, analyzing driver behavior, environmental conditions, and traffic flow to generate predictive alerts. Unlike conventional systems that react after an incident occurs, predictive models allow for proactive responses, offering drivers enough time to adjust and avoid hazards. This shift marks a major advancement in road safety, introducing

long-term prevention strategies rather than short-term reactions. Xu et al. also discussed the ethical and technical considerations of deploying AI in transportation, particularly regarding data privacy and accountability. Their findings illustrate both the technical feasibility and the broader implications of predictive alerts, emphasizing their potential to reshape roadway safety practices.

Zhang and Kianfar (2022) examined advanced detection algorithms designed to identify abnormal traffic conditions in real time. Their research found that these algorithms enhance both the speed and accuracy of detecting congestion, sudden stops, or irregular driver behavior, enabling earlier interventions by drivers and traffic authorities. Machine learning plays a key role in refining these systems, as algorithms continuously improve with exposure to historical traffic data, enhancing predictive accuracy. This iterative learning process ensures that alerts remain precise and reliable, strengthening overall traffic monitoring. Beyond individual safety, their findings suggest that such tools can inform long-term planning and data-driven policies for safer urban infrastructure. The study underscores the importance of computational intelligence in building proactive monitoring systems that address risks before they escalate into serious incidents.

Mokhtaru et al. (2022) highlighted the value of mobile networks in anomaly detection and real-time traffic response. Their research revealed that integrating mobile communication technologies facilitates faster reporting of accidents, congestion, and sudden hazards, ensuring quicker responses from both drivers and authorities. By leveraging mobile platforms, systems can provide additional functionalities such as rerouting guidance, emergency coordination, and driver-to-driver communication,

improving overall efficiency. This accessibility makes mobile-based notification platforms highly scalable and adaptable across different regions, even those with limited infrastructure. Their study emphasized that mobile connectivity strengthens roadway resilience and broadens safety access, making it a critical component of intelligent transport systems. Such innovations demonstrate how mobile technology continues to play a pivotal role in enhancing safety and efficiency in modern traffic management.

Rocha et al. (2023) explored the impact of sensor-based vehicular communication on roadway safety through immediate hazard alerts. Their findings showed that IoT-enabled vehicles can exchange real-time data regarding road conditions, enabling drivers to anticipate and respond to dangers more effectively. This collaborative exchange fosters collective awareness within traffic networks, improving safety at both individual and systemic levels. The study also emphasized the role of IoT in emergency response, where vehicles transmit detailed information about incidents, locations, and driver conditions to first responders. Such communication significantly reduces intervention times and improves rescue outcomes. Rocha et al. underscored IoT's role as a cornerstone of scalable and interconnected real-time notification systems, highlighting its potential to shape future transportation infrastructure.

In another significant advancement, Patil (2024) applied machine learning techniques to traffic sign detection within real-time event notification systems. His study demonstrated that automated recognition reduces reliance on human interpretation, minimizing errors caused by distraction or misinterpretation. These systems ensure drivers receive timely reminders about regulations, promoting compliance and confidence

while reducing accidents linked to overlooked signs. Beyond enforcement, Patil emphasized the educational value of these systems, which can provide real-time feedback on new or updated signage, enhancing long-term driver awareness. By reducing errors and adapting to evolving infrastructure, machine learning-supported traffic sign detection enhances safety and reinforces intelligent transportation systems.

Moreover, Almassar and Girsang (2022) examined cloud-based machine learning for processing real-time traffic data streams. Their findings indicated that cloud infrastructure enables fast and large-scale data analysis, allowing accident-prone situations to be identified early without compromising performance. Cloud computing's scalability makes it possible to share traffic information across multiple stakeholders, from government agencies to emergency responders, fostering coordinated responses. Their research highlighted that such systems improve responsiveness during peak traffic while ensuring continuous coverage across urban environments. Moreover, they stressed the importance of public-private partnerships to maximize the potential of cloud resources for safety applications. Their work demonstrates how cloud adoption provides a strong foundation for reliable, future-ready traffic management systems.

Expanding on cloud applications, Alhilal et al. (2022) investigated distributed machine learning models designed to improve roadway alert dissemination. Their study revealed that cloud-based systems can efficiently deliver notifications across wide geographical areas, extending safety services to both urban and rural drivers. By ensuring inclusivity, these technologies address disparities in access to traffic safety resources. Their findings also pointed to the efficiency of cloud platforms in reducing

communication delays, ensuring alerts are timely and actionable. Importantly, their study emphasized the sociopolitical implications of equitable safety access, promoting road safety as a universal resource rather than one limited to developed regions. Their analysis affirms the scalability of cloud systems as vital infrastructure for nationwide transport safety strategies.

Lastly, Zißner et al. (2023) stressed the importance of heterogeneous data fusion in improving the reliability of real-time notification systems. Their study demonstrated that combining data from multiple sources—such as sensors, vehicular communication, and traffic reports, minimizes false alarms and produces alerts that are more accurate and actionable. This integration creates a more comprehensive view of roadway dynamics, allowing drivers to make safer and more informed decisions. The researchers also emphasized that data fusion supports continuous system refinement, as feedback from diverse sources improves adaptability to evolving conditions. By promoting greater trust and usability, heterogeneous data integration strengthens proactive safety measures and ensures the sustainability of intelligent transportation systems. Their findings align with the broader move toward data-driven models in transportation, reinforcing that multi-source integration is essential to advancing roadway safety.

YOLO Algorithm

The YOLO (You Only Look Once) algorithm has emerged as a transformative tool in real-time object detection, particularly within transportation safety systems. Gao (2022) demonstrated that YOLO significantly improves the accuracy of vehicle detection at critical intersections, where rapid recognition is crucial for preventing accidents. By detecting multiple objects in a single neural network pass, the algorithm ensures both speed and reliability, which are indispensable for safety applications. Gao's research indicated that YOLO reduced false detection rates while enhancing the ability of traffic monitoring systems to manage complex environments. This is especially important in busy intersections where pedestrian movement and vehicle interactions create high-risk conditions. The study highlighted YOLO as a practical technology for accident prevention strategies and reinforced its role in building safer road networks.

Zhang et al. (2024) expanded upon these findings by integrating YOLO within broader smart transportation management frameworks. Their study highlighted that the algorithm enhances system responsiveness by providing immediate recognition of vehicles, obstacles, and hazards. This real-time capability allows for adaptive traffic regulation, improving both flow efficiency and safety in high-density environments. Zhang and colleagues also emphasized YOLO's potential to reduce congestion by supporting dynamic traffic light control and vehicle prioritization systems. Their research positioned YOLO not only as a detection tool but also as a fundamental component of intelligent traffic ecosystems. These results underscore the growing importance of

incorporating YOLO into modern urban mobility solutions designed to balance safety, efficiency, and sustainability.

The adaptability of YOLO across diverse environments was examined by Niu et al. (2023), who demonstrated its effectiveness in both urban and rural traffic management. In rural areas, where infrastructure is limited, the ability of YOLO to function without relying heavily on external systems ensures that traffic monitoring remains feasible. Conversely, in urban centers with heavy traffic congestion, its precision supports timely regulation and the prevention of roadway conflicts. The study further highlighted YOLO's flexibility in detecting vehicles, pedestrians, and obstacles across varied conditions, making it an adaptable solution for different contexts. Niu et al. argued that this versatility makes YOLO suitable for developing countries where infrastructure constraints limit the implementation of traditional monitoring systems. The findings confirm YOLO's relevance as a practical and scalable tool for diverse transportation settings.

Tao (2024) examined the development of specialized versions of the algorithm, such as NRT-YOLO, designed to optimize detection performance while reducing computational complexity. This version enables YOLO to function efficiently even on devices with limited processing power, making it suitable for embedded systems used in traffic applications. The research demonstrated that NRT-YOLO maintained strong accuracy while requiring fewer computational resources, ensuring feasibility in real-time traffic monitoring. Tao emphasized that reducing computational demands is critical for expanding YOLO's applications to portable systems such as mobile traffic cameras and

roadside units. These improvements allow for broader adoption, particularly in areas where high-performance computing infrastructure is unavailable. The study reflects how algorithmic refinements contribute to extending YOLO's applicability in real-world transportation environments.

Yu (2024) introduced MR-YOLO, another modified version designed to balance high detection accuracy with efficiency in resource-constrained conditions. The study revealed that MR-YOLO performed effectively in monitoring road traffic while maintaining lower energy and processing requirements. This advancement ensures its applicability in autonomous vehicles and portable monitoring stations that require lightweight yet powerful detection systems. Yu's findings underscored the importance of refining algorithms to meet practical deployment challenges rather than focusing solely on theoretical performance. The research also emphasized that efficient models enable scalability across different types of transportation systems, from urban smart networks to rural traffic monitoring units. By prioritizing balance between speed and resource use, MR-YOLO extends YOLO's relevance to broader applications within intelligent transportation.

Further performance improvements were highlighted in studies focusing on YOLOv8. Darmadi et al. (2024) found that YOLOv8 achieved significantly faster processing speeds while maintaining high accuracy in detecting vehicles and pedestrians. These improvements are particularly valuable in dynamic traffic scenarios where even slight delays in detection could compromise safety outcomes. The study revealed that YOLOv8 is well-suited for real-time decision-making processes at intelligent

intersections and in autonomous driving systems. Faster recognition also contributes to more efficient traffic regulation, reducing congestion and ensuring smoother mobility. By demonstrating how upgraded versions of YOLO can address prior limitations, the research underscores the continued evolution of detection systems for advanced traffic management.

Complementing these findings, Li et al. (2023) evaluated YOLOv8's performance under varied conditions, including crowded roadways and mixed traffic flows. Their study demonstrated that the algorithm maintained high detection accuracy for multiple object categories, including vehicles, pedestrians, and traffic signs. This reliability is especially important for autonomous vehicles, which rely on precise recognition to ensure passenger and pedestrian safety. Li and colleagues also noted that YOLOv8's adaptability extends to adverse conditions such as poor lighting, further broadening its applicability. The research concluded that YOLOv8 sets a benchmark for future detection algorithms, combining accuracy, speed, and robustness. Such qualities reinforce its critical role in advancing intelligent transportation technologies.

The application of YOLO to enforcement contexts has also been studied in depth. Jiao and Wang (2022) applied the algorithm to real-time detection and tracking of speeding vehicles. Their research demonstrated that YOLO could provide continuous monitoring by simultaneously identifying vehicle type and speed, enhancing traffic enforcement capabilities. Unlike traditional radar systems, which are often limited in coverage, YOLO offers a more comprehensive and adaptive solution. The study highlighted how vision-based monitoring can strengthen regulatory frameworks by

ensuring compliance and improving safety. These findings illustrate how YOLO contributes not only to accident prevention but also to broader transportation governance.

YOLO's contribution to pedestrian safety was further explored by Gündüz and Işık (2023), who applied the algorithm to urban traffic monitoring systems. Their study revealed that YOLO improved the detection of pedestrians and vehicles in crowded areas, thereby reducing collision risks in high-density traffic environments. By integrating the algorithm into surveillance systems, municipalities were able to enhance pedestrian protection without compromising traffic flow. The research emphasized that such applications are vital in cities experiencing rapid population growth and increasing vehicle numbers. Gündüz and Işık concluded that YOLO provides a practical solution for protecting vulnerable road users in modern transportation systems. Their findings position the algorithm as a valuable tool for urban safety management.

Pawar et al. (2024) explored an innovative integration of YOLO with image captioning and audio analysis, aiming to improve situational awareness in traffic management. Their research showed that this multimodal system allowed for real-time alerts by transforming visual detections into descriptive outputs. This capability enabled traffic managers to respond more effectively to emergencies and unusual events. The study also emphasized that combining YOLO with other artificial intelligence technologies expanded its utility beyond detection, positioning it as part of a larger traffic intelligence framework. Pawar and colleagues argued that such integrations pave the way for more comprehensive safety solutions by enhancing both monitoring and

decision-making. Their work highlights the potential of YOLO to evolve into a key component of advanced intelligent transportation systems.

Intelligent Transportation System (ITS)

One study stated that the Intelligent Transportation System (ITS) expands rapidly aligning with the rising demand of safer, more efficient, and sustainable transportation solutions (Elassy et al., 2024). With this system, traffic management and control are among the cover of its various applications, it aims to enhance mobility experiences. ITSs illustrates its significance fostering sustainable smart cities which most likely focuses on urbanization. Hence, the Intelligent Transportation System within rural areas is a gap, extending ITS implementation to rural context holds significant potential, as these regions often face distinct safety challenges and limited infrastructure, making intelligent solutions crucial for improving connectivity, reducing accidents, and ensuring equitable access to advanced transportation technologies for future generations.

According to Diderot et al. (2023), the implementation of Intelligent Transportation Systems is embarked by Developed countries, improving transport's safety, efficiency, and its environment. This paper outlines the relevance of ITSs in developing countries, differentiating the transportation scenarios between developing and developed countries. Vehicles per Capita (VpC) was indicated, where between 2004 and 2014, VpC strongly increased in various areas just like the 95% increase in East Asia, 73% increase in Brazil, etc. Tailored with this increase is also an increase in terms of concerns with safety. According to the World Health Organization (WHO), road traffic

incidents are the leading cause of deaths worldwide, where most victims live in developing countries. Thus, applying ITS improves mobility, efficiency, road safety and security but doing such means taking considerations concerning its affordability since developing countries often experience financial constraints.

In a study. Khalil et al. (2024) stated that Intelligent Transportation System (ITS) transpires indispensable components within modern transportation, offering its highly complex spatiotemporal environments suitable for research and development, influenced by external variables such as weather, social events, and holidays, making modeling and adaptation particularly challenging. This emphasizes how advanced learning technologies, especially Deep Learning (DL) and emerging large language models can be harnessed within ITS to tackle tasks such as traffic flow forecasting, vehicle detection and classification, road condition monitoring, traffic sign recognition, and autonomous driving. Architectural choices and problem-specific design decisions are examined to shape Deep Learning methods in Intelligent Transportation System, and key hurdles are flagged to be its explainability, transfer learning, hybrid modelling, privacy and security, and ultra-reliable low-latency communications. Overall, the importance of integrating advanced data-driven learning methods with real-world operational requirements to address the methodological and practical barriers in the advancement of intelligent transportation systems is emphasized.

An intelligent transportation system (ITS) using roadside infrastructure can tame the problems in transportation such as accidents, slow traffic flow, and pollution (Creß et al., 2023). A knowledge gap is stated, where “no current systematic review of existing

ITS solutions exists." Current ITS systems are able to provide real-time information to people in traffic scenarios. However, further ITS research should focus on more reliable traffic perception through the use of contemporary sensors, plug-and-play systems, and secured decentralized real-time distribution of digital twins. ITS in general, could prevent approximately 400,000 to 600,000 accidents per year, in-parallel, reducing travel time, fuel consumption, and greenhouse gases around 15%.

In (2021), Rudskoy et al. stated that introducing Intelligent Transport System (ITS) solves the existing transport problems, where efficient management of transport flow problems arises due to increasing vehicle-to-population ratio and intensified use of individual transport. As transport problems multiply, taking advantage of the rapid advancement and modernization should be a priority area of development. Digital twin is an ITS key element that uses mathematical modeling methods to analyze transport networks and develop solutions for enhancement of traffic and pedestrian flows, traffic management, optimization of traffic lights, etc. With such elements, combining strategic plans to existing infrastructure and systems creates rational direction for development in the future.

According to Du et al. (2021), the advancement of an Intelligent Transport System heavily relies on precise vehicle positioning, with Precise Point Positioning (PPP) standing out as a potential enabler of accuracy. While PPP offers the advantage of operating globally with a single receiver, its performance significantly degrades in urban settings due to multipath interference, non-line-of-sight reception, and atmospheric errors. These vulnerabilities threaten the integrity and safety of ITS applications,

especially in traffic management and autonomous driving. This emphasizes that integrating PPP with inertial navigation systems and regional augmentation methods can improve positioning availability, moreover, there is still a pressing need to address integrity monitoring in dense urban environments. With this, insight regarding how vehicle localization emerges, though often taken for granted, is truly a foundation of Intelligent Transportation System effectiveness and remains an open research challenge.

With Yuan et al. (2021) study, machine learning can be noticeable for becoming a transformative driver for Intelligent Transportation System development, as it supports advanced sensing, prediction, and traffic management capabilities. Machine Learning (ML) techniques make ITS smarter by enabling vehicles and infrastructure to perceive their environment, forecast traffic flows, and optimize route or signal control. However, applying ML in ITS introduces new concerns, including the handling of heterogeneous and large-scale data, ensuring robustness in safety-critical conditions, and maintaining privacy of collected mobility information. This highlights that while ML enhances the adaptability and efficiency of ITS, the field must also balance innovation with reliability, interpretability, and scalability. Thus, the Intelligent Transportation System powered by Machine Learning holds great promise, but it requires carefully designed models that can operate seamlessly within complex, real-world transportation networks.

Based on Chowdhury et al. (2022), Intelligent Transport System (ITS) does not only lie in the ability to collect vast amounts of transport data but more importantly in turning this data into meaningful, actionable decisions. This supports that functional requirements such as data quality, explainability, and timeliness are critical in ensuring

ITS analytics are not just theoretically accurate but also operationally useful. Integration of cloud and edge computing, combined with privacy-preserving techniques, forms an essential backbone for making ITS analytics both trustworthy and practical in real-world deployments. Gathering all presented information, ITS must focus on building reliable data pipelines that consider uncertainty, feedback loops, and decision support so that predictive models genuinely improve transport safety, traffic efficiency, and user experience.

Laña et al. (2021) stated that although predictive modeling is a strong component of an Intelligent Transportation System, many studies tend to overlook the operational realities of implementing these models in live transportation settings. It is notable that most ITS studies concentrate on accuracy and model performance without adequately addressing essential issues such as latency, interpretability, and the role of human decision-makers. In order for ITS to achieve practical value, models must be evaluated not only on technical precision but also on how actionable, understandable, and responsive their outputs are in real time. The future of ITS designs should integrate human-in-the-loop considerations, transparent decision frameworks, and deployment-focused evaluation metrics to bridge the gap between laboratory research and real-world adoption.

Lastly, as per Garg & Kaur (2023), applications of an Intelligent Transport System are wide-ranging, covering traffic management, traveler information services, public transport optimization, and road safety improvements. However, while ITS innovations are often driven by technology such as sensing devices, communication platforms, and

V2X networks. Non-technical challenges like institutional collaboration, funding, policy-making, and social equity also play a decisive role in determining the success of ITS deployment. This discussion led to a conclusion that ITS cannot thrive in isolation; instead, it requires holistic planning that integrates technical advances with governance, standards, and public engagement. ITS is not merely a technological system but a socio-technical ecosystem that balances innovation with long-term sustainability and accessibility.

Road Safety

According to the study of Rosen et al., (2025), road traffic injuries (RTIs) claim 1.35 million lives and affect 50 million humans with non-fatal injuries on the world's roads annually. Individuals aged 5-29 years old are said to be the leading fatalities caused by road traffic crashes and these accidents burden mostly low- and middle-income countries (LMICs) where over 90% of RTIs occur. Between 2011 and 2017, the enforcement of speed laws and legislation of child restraint increased. However, a correlation between death rate and enforcement of preventative measures for speeding, drunk driving, seatbelts, helmets, and child restraints became somewhat irrelevant. There is also slow progress in adopting and implementing policies and practices such as road safety measurements, safer roads and mobility, safer vehicles, safer road users, and post-crash response. This calls for accelerating the global commitment for implementing proven interventions , meeting the target of the ongoing road safety agenda.

Based on Imran et al. (2023), road safety have been a concern on Intersections where nodes of traffic networks and collision-prone area exist, due to intricate environment of unpredictable events. Within transport related research, the assessment of Road Safety has been a consistent concern in intersections. Existing methods are used mostly for high-risk intersections only, neglecting minor instances. Many existing research have already integrated many specifications such as video-based systems, trajectory-based analyses, conflict analysis techniques, and many more. These findings underlined the benefits of real-time monitoring for more accurate risk profiling, integration of significant progress like computer vision, crash data analytics and such strengthens safety management for intersections.

The study of Musselwhite et al. (2025) stated that rural areas have a higher number of road safety concerns, rating road traffic collisions in Wales at their highest - 291.3 and 285.9 deaths and injuring 100,000 individuals respectively. Rural areas drivers tend to spend longer driving time within those roads, also extending their exposure to danger of less complex networks with fewer roads, cross sections, and intersections etc. and many have less traffic control, rules, and priorities identified (Coffin et al., 2021, Cox et al., 2017, Musselwhite et al., 2010). Road safety can be relatively urban centric, which makes rural areas less as the beneficiary, where no system or efforts are being implemented on sharp bends, narrow roads, and more factors. Hence, extending the implementations of road safety strategies decreases the risk and impact of traffic incidents.

Road safety is a complex balance between the different components of the road system: user, vehicle, environment, and infrastructure (Cantisani et al., 2023). Potential issues present in a road system imposes safety concerns and disrupts the balance. Europe's Directive 2008/96/EC or also known as "road infrastructure safety management" introduces procedures which later on used for the improvement of European Transport Network (TEN-T) safety, identifying the vital role of safety inspection but only focused on freeways and highways. With this, proposals were embarked on that current procedures are also to be used in secondary and local rural roads, like for example, the "Manual for safety Inspections of Secondary and Local Rural Roads", aiming to implement strategic inspections and reading of situations for traffic. These inspections could be the countermeasure for rural roads critical potential factors for accidents such as single carriageway and one lane, insufficient passing sight distance, repeated small radius curves, poor road markings, etc.

A study in 2022, Babić et al. stated due to the dynamic nature and complexity of road traffic, road safety is still one of the major issues worldwide, given that approximately 1.3 million people die annually in road crashes, additionally, 20-50 million people get injured with short- or long-term consequences. Road safety really depends on the interactions of the three main factors which are the human, vehicle, and road. Other than human error, vehicle-related issues, and roadway characteristics, the entire road infrastructure and surrounding environment raises concerns too. Preventive measures such as influencing road users and human behaviors, law enforcement, implementation of vehicle safety systems, and improvement on roads and their environment. Road

markings, as an example, represent basic means of communication between road users and authorities, but due to several reasons like visibility conditions, road markings become somewhat inefficient. Perhaps, fulfilling the gaps with innovative systems providing the same function in a consistent manner would make an impact with road safety.

Bamidele & Baah (2022) stated with their study that road safety in Nigeria can be substantially improved through the strategic adoption of Internet of Things (IoT) technologies, which provide new ways of monitoring traffic violations, identifying unsafe driving behavior, and responding quickly to emergencies. The study discussed IoT applications such as RFID-based vehicle registration, automatic number plate recognition (ANPR), and telematics, which collectively help in detecting speed violations, tracing stolen vehicles, and reducing accident response times. The integration of these systems not only strengthens enforcement but also fosters a culture of compliance among road users. What stood out was the evidence that even partial deployment of IoT solutions can already lead to noticeable safety improvements, although, long-term success still depends on supportive policies, inter-agency collaboration, and extending coverage beyond major cities. This highlighted that road safety is not just a matter of enforcement or driver education, but also about leveraging technology in a coordinated and inclusive way.

According to Masello et al. (2022), the potential of Advanced Driver Assistance Systems (ADAS) to improve road safety is significant, but their effectiveness varies greatly depending on driving conditions and geographic context. The quantitative assessment revealed that the widespread deployment of six key ADAS features such as

lane keeping, automatic braking, blind-spot monitoring and such, could reduce accident frequency by about 23.8% in the United Kingdom. What I understood from their findings is that these technologies are particularly effective under urban-clear and rural-clear daylight conditions, where reductions can reach nearly 29%. However, performance gains diminish in adverse conditions such as poor visibility, wet roads, or high-speed rural highways, this means that while ADAS adoption promises substantial safety benefits, it is not a universal solution, and implementation strategies should account for environmental and contextual limitations. In the end, enhancing road safety requires both technological innovation and careful tailoring of these innovations to diverse traffic environments.

As stated by McCarty & Kim (2024), demographic variables exert a strong indirect influence on road safety outcomes, something that is often overlooked in transport policy. By applying regression-based machine learning to accident data across England, the study revealed that census-level demographic features such as age, gender, and population density accounted for over 28% of the variance in accident rates per capita. Road safety is not only a matter of vehicle technology or infrastructure design but also deeply connected to the social composition of communities. For instance, areas with younger populations or higher traffic exposure may inherently carry greater accident risks. This underscores the importance of tailoring safety interventions to specific socio-demographic profiles, whether through targeted education campaigns, enforcement strategies, or infrastructure investments. Road safety strategies become more effective when demographic realities are considered alongside traditional engineering and enforcement measures.

According to Tavakkoli et al. (2022), body of evidence on road safety interventions in low- and middle-income countries (LMICs) remains limited, with significant variation in both the types of interventions implemented and the methods used to evaluate them. The review covering a decade of interventions showed that most strategies are law-based or enforcement-driven such as speeding laws, helmet mandates, and drink-driving restrictions, but often lack rigorous evaluation frameworks to assess real-world impact. These interventions are theoretically sound and have shown effectiveness in high-income countries, their outcomes in LMICs are often inconsistent due to weak enforcement, inadequate infrastructure, and resource constraints. The importance of not only adopting proven interventions but also investing in proper monitoring, evaluation, and local adaptation are highlighted, ensuring that these measures achieve the intended safety improvements. Road safety in LMICs is not solely about copying global best practices, but about contextualizing them to local social, cultural, and institutional realities.

Moreover, Singh et al. (2021), stated that advancing road safety requires the digitalization of highways through the integration of Internet of Things (IoT), renewable energy, and artificial intelligence. It highlights innovative strategies such as smart highway lighting, real-time traffic and emergency management systems, and the use of renewable sources like solar, wind, and piezoelectric technologies to power safety applications. Inclusion of smart display boards and AI-driven models for vulnerable road users (VRUs) can provide proactive alerts and adaptive responses that enhance protection for pedestrians and cyclists. The study reinforces that future road safety does not solely

depend on infrastructure upgrades but also on embedding intelligence, sustainability, and connectivity into highway systems to create safer, greener, and more adaptive transport environments.

Wireless Sensor Network

Wireless Sensor Networks (WSNs) have increasingly been recognized as a foundational technology for road safety monitoring, enabling distributed, low-power sensing and real-time data transmission across multiple nodes without reliance on fixed physical infrastructure. A study reviewed the potential of low-power wireless networks to improve road safety by characterizing available WSN technologies and their applications in road transport, presenting an innovative concept of area-based vehicle speed monitoring, and conducting simulations to assess the radio coverage of proposed deployment models. The findings highlight the practical viability of deploying distributed wireless sensor nodes along road corridors to continuously monitor vehicle movement, a principle that directly supports the multi-node architecture employed in the LUMEN system. According to this study of Rychlicki et al. (2024), low-power wireless networks represent one of the most promising technologies for expanding road safety applications beyond urban centers into areas with limited infrastructure.

The integration of WSNs with IoT frameworks has proven effective in enabling real-time accident detection and automated emergency notification. A comprehensive review of IoT-based accident detection systems examined 101 peer-reviewed articles and categorized approaches based on sensor type, incorporation frameworks, and recognition

techniques, finding that multi-sensor utilization significantly enhances recognition precision and dependability, while also identifying persistent challenges such as implementation costs, adverse weather conditions, and data accuracy issues. (Eduzonejournal) The study further emphasized the need for stronger communication systems to improve emergency responsiveness and advocated for the integration of sensor fusion approaches and artificial intelligence into future IoT-enabled road safety architectures. According to this review (2022), while substantial progress has been made in IoT-based accident detection, standardization gaps and communication reliability remain critical areas for continued development.

Building on IoT-based detection frameworks, studies have also demonstrated the effectiveness of ESP32 microcontrollers as low-cost, low-power nodes within distributed wireless sensor networks. An ESP32-based low-power and low-cost wireless sensor network was developed and presented, demonstrating its suitability for deployment in environments where energy efficiency and cost constraints are primary considerations. The ESP32's built-in wireless communication capabilities, combined with its compatibility with a wide range of sensors and its low power consumption profile, make it a practical choice for roadside sensor node deployment in off-grid rural environments. According to Perdomo-Campos et al. (2023), ESP32-based WSN architectures offer a viable and accessible foundation for implementing distributed sensing systems in resource-limited settings.

The use of MQTT-based communication protocols within WSN architectures has also been widely validated as an effective approach for low-latency data exchange

between distributed sensor nodes and central processing units. A study established a wireless sensor mesh network using ESP32 microcontrollers and demonstrated that the ESP-MESH networking protocol offers a low-cost, easy-to-implement, medium-range, and low-power option for interconnecting numerous sensor nodes spread over a large physical area, enabling continuous real-time monitoring of environmental parameters across distributed nodes. The study confirmed that ESP32-based multi-node networks can reliably transmit sensor data to a central base station in real time, validating the communication architecture used in LUMEN where ESP32 beacon nodes relay radar detection data to the Raspberry Pi 5 central processing unit. According to this study of Hillol et al. (2022), ESP32-based mesh and hub-and-spoke WSN configurations are well-suited for scalable, cost-effective multi-node deployments in environments where wired infrastructure is unavailable.

The application of WSNs in vehicle detection and traffic event reporting has further demonstrated the relevance of sensor-driven, networked architectures for road safety. An IoT-based system was developed to automatically detect vehicle accidents and report them to rescue operators without human involvement, using on-vehicle sensors to transmit accident data to a remote server for visual representation of the accident scene, while also addressing false reporting issues and providing alerts to the victim's family. The study underscored the importance of reliable wireless data transmission and automated alert mechanisms in reducing emergency response times, reinforcing the value of real-time event notification as a core function of road safety systems. According to Aslam et al. (2024), IoT-based sensor networks capable of autonomous detection and

immediate remote reporting represent a critical advancement in minimizing the consequences of road accidents, particularly in areas where human monitoring is not continuously available.

Synthesis

The reviewed literature on intelligent transportation systems reveals their proven effectiveness in enhancing road safety, functionality, and usability across diverse traffic environments. A consistent finding across local and international studies is that ITS solutions leveraging artificial intelligence (AI) and Internet of Things (IoT) technologies excel at real-time vehicle detection, incident awareness, and automated event notifications that significantly improve response times and safety outcomes. Researchers consistently emphasize automation as a critical advantage that reduces human operator workload while maintaining practical usability for drivers and administrators. Multiple studies highlight the particular value of adaptive systems in rural contexts where limited infrastructure prevents conventional safety measures from succeeding. These findings collectively demonstrate that ITS technologies achieve optimal performance when integrating automation, real-time responsiveness, and contextual adaptability into unified frameworks, providing a strong conceptual foundation for LUMEN's development.

Although ITS benefits are widely recognized, the literature also identifies persistent gaps limiting their full potential. A recurring challenge is implementation in resource-limited rural areas where budget constraints, poor connectivity, and inadequate infrastructure hinder scalability. Many deployed systems suffer from poor integration

across detection, monitoring, and notification components, resulting in fragmented operations. Existing solutions often address isolated traffic issues—vehicle detection OR incident response OR notifications—rather than comprehensive frameworks. Studies also note insufficient adaptation to local road conditions, driving behaviors, and demographic factors, particularly in high-risk rural blind curves. These limitations underscore the ongoing need for integrated, context-sensitive systems that can effectively serve developing regions and infrastructure-scarce environments.

LUMEN (Lane-Use Monitoring for Event Notification) directly responds to these identified challenges and gaps. By seamlessly integrating lane-use monitoring, computer vision-based incident detection, and real-time event notification into a single framework, LUMEN overcomes the fragmentation prevalent in existing studies. Its emphasis on usability ensures accessibility for diverse stakeholders, while its rural-optimized design addresses infrastructure limitations through solar-powered, cost-effective deployment. Unlike narrow-focused systems, LUMEN combines detection accuracy, automated alerting, and environmental adaptability into one cohesive solution specifically targeting blind curve safety challenges. By building on literature strengths while directly addressing documented weaknesses, LUMEN validates its technical approach and establishes its meaningful contribution to advancing rural road safety through intelligent transportation practices.

CHAPTER III

METHODOLOGY

This chapter presented the designs, materials, procedures and statistical procedures and analysis used in this study. This will also provide the outlines of the project's methodology, the research and development method, project design, operation flow, material descriptions, project development, testing and evaluation methods.

RESEARCH METHOD USED

The method of investigation used for this study is Research & Development (R&D). It refers to a series of activities that organizations carry out to create new ideas and improve existing products. R&D is typically the first step in the development process, involving market research, product design, and testing (K. Will, 2024).

DESIGN PHASE

The design of LUMEN (Lane-Use Monitoring for Event Notification) focuses on combining key hardware and software components to create an incident-aware, minimal traffic management system.The system is composed of a main traffic pole responsible for accident detection and two roadside beacons capable of detecting incoming vehicles. The main traffic pole will use a Raspberry Pi 5 as the main compute module, while two

ESP32 will be used in the beacons. The device will feature a sturdy frame ensuring stability, with appropriate heat dissipation and protection against rainy weather to maintain safe operation under various environmental conditions. Portability will be addressed by including wheels for easier deployment, and the main post will be adjustable and bendable, allowing height and angle modifications for optimal positioning.

OPERATION FLOW

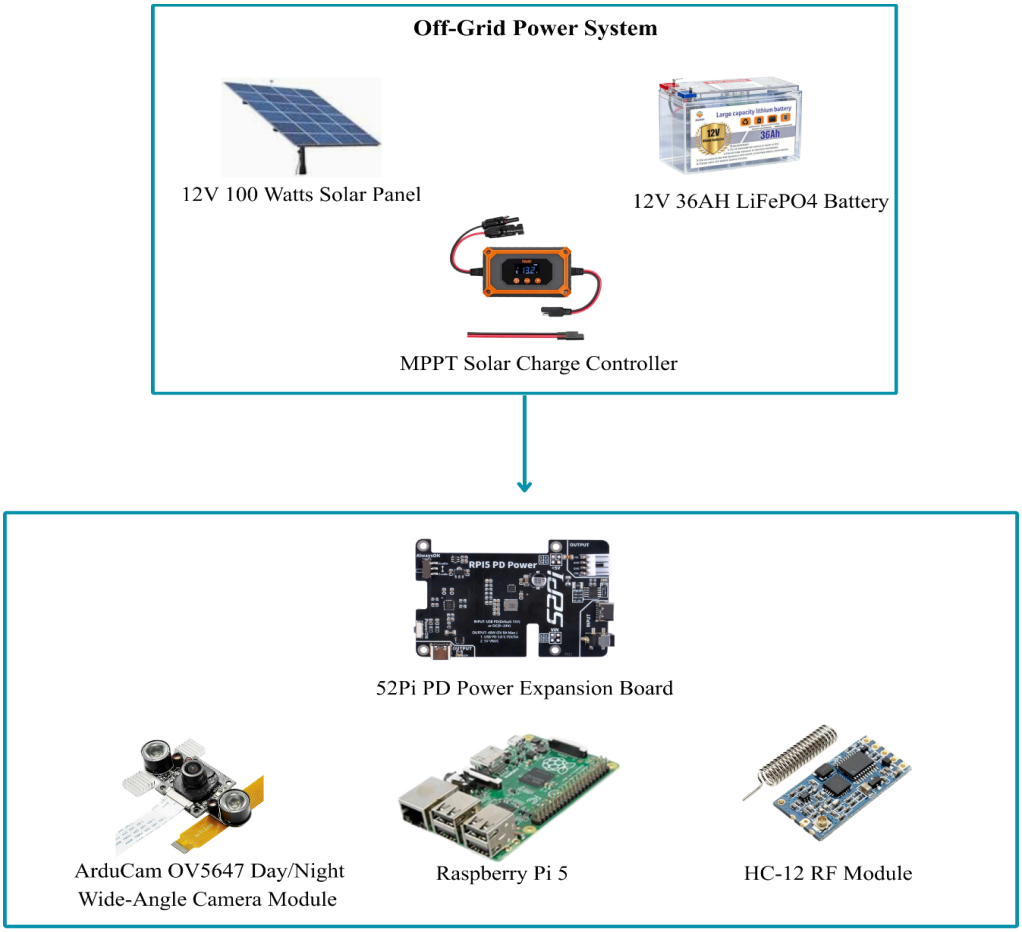


Figure 3.1: Picture Diagram of Components used in Lumen's main Traffic Pole

The system operation begins with the 100W solar panel charging the 12V 36AH LiFePO4 battery through the solar charge controller, which regulates and ensures safe battery charging and power management. The solar charge controller is connected to a 52Pi RPi 5 PD module to step down voltage and provide power delivery capabilities for the Raspberry Pi 5, which acts as the central controller, connected to various peripherals. A Day/Night Camera continuously captures real-time video for monitoring. The Pi processes this data to make decisions about accident detection.

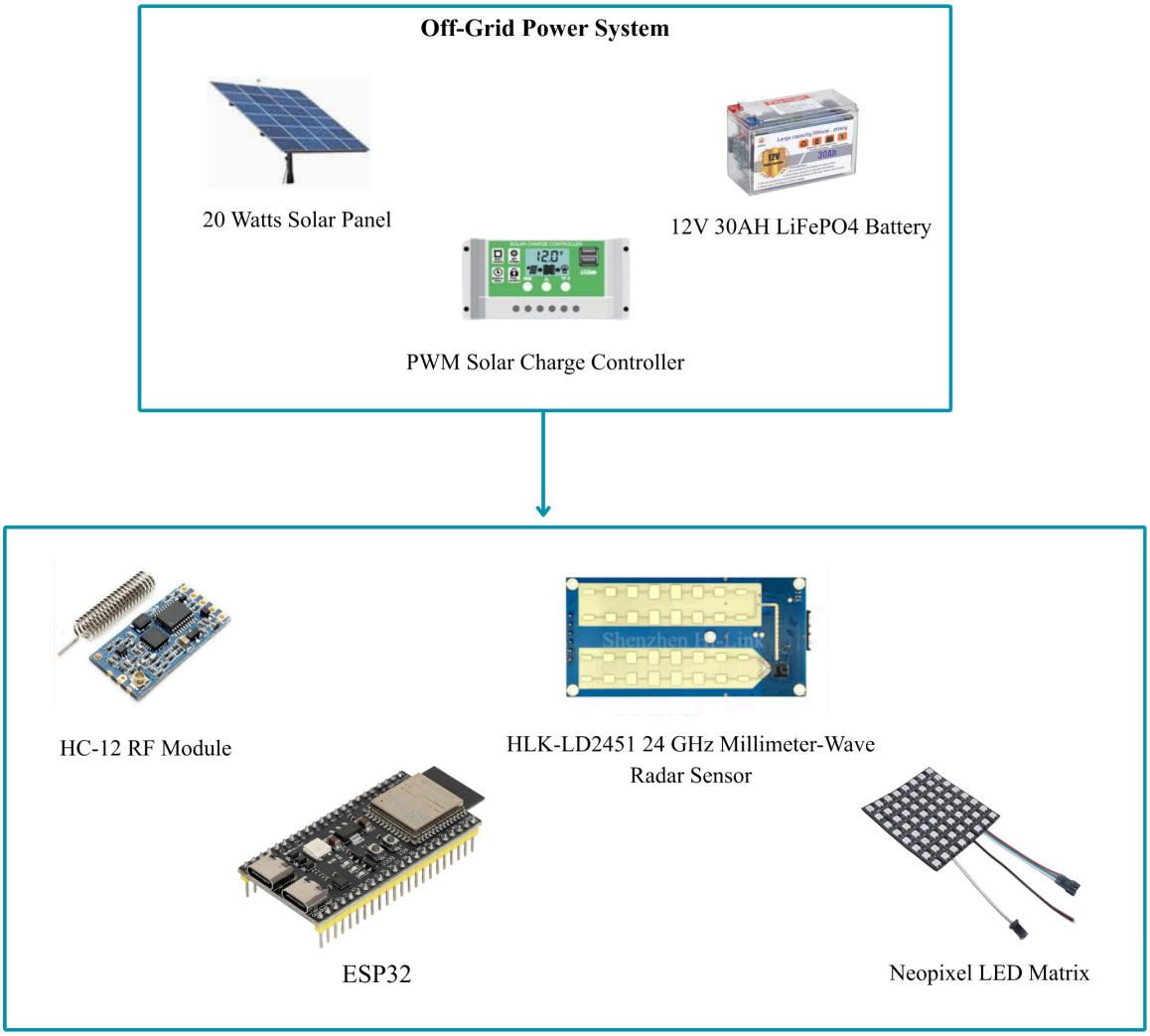


Figure 3.2: Picture Diagram of Components used in Lumen's Road Beacons

Each road beacon operates autonomously powered by a solar energy system comprising a 20W solar panel, a solar charge controller, and a 12V 30AH LiFePO4 battery, ensuring continuous power supply without reliance on external grids. The ESP32 microcontroller serves as the brain of the beacon, processing inputs from the HLK-LD2451 24GHz Radar Module that detects nearby vehicles, their speed, and the direction they are going . Upon detecting a vehicle, the ESP32 activates traffic blinker lights and to alert drivers or pedestrians of approaching traffic or hazards.

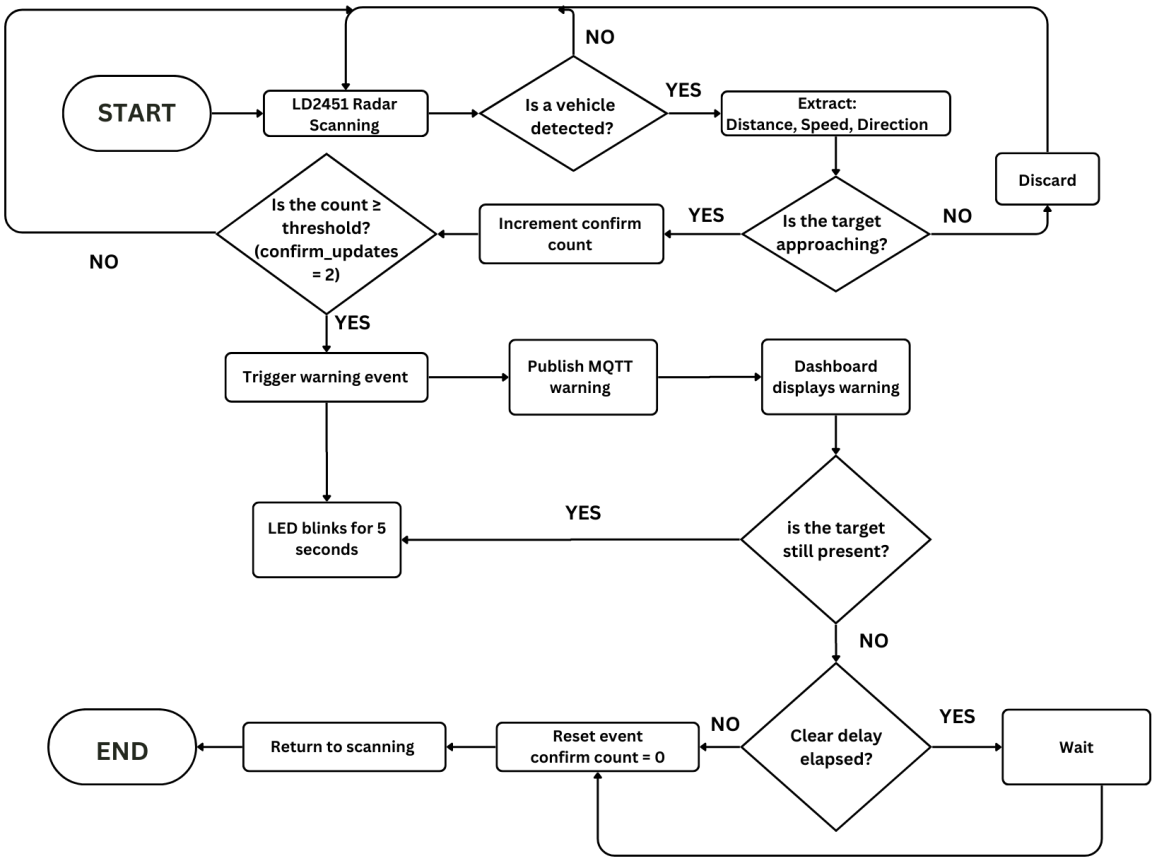


Figure 3.3: Flow Chart of Vehicle Detection and Visual Warning

The LUMEN radar system constantly scans the road using the HLK-LD2451 radar. When a vehicle is spotted, it checks if the vehicle is approaching and if not, it ignores it and keeps scanning. If the vehicle is approaching, it waits for 2 consecutive confirmations to make sure it is a real detection and not a false reading. Once confirmed, it triggers a warning and the dashboard shows an alert and the LED on the opposite node blinks amber for 5 seconds to warn oncoming drivers. The system then watches to see if the vehicle is still there. Once the vehicle leaves and a short delay passes, everything resets and the radar goes back to scanning normally.

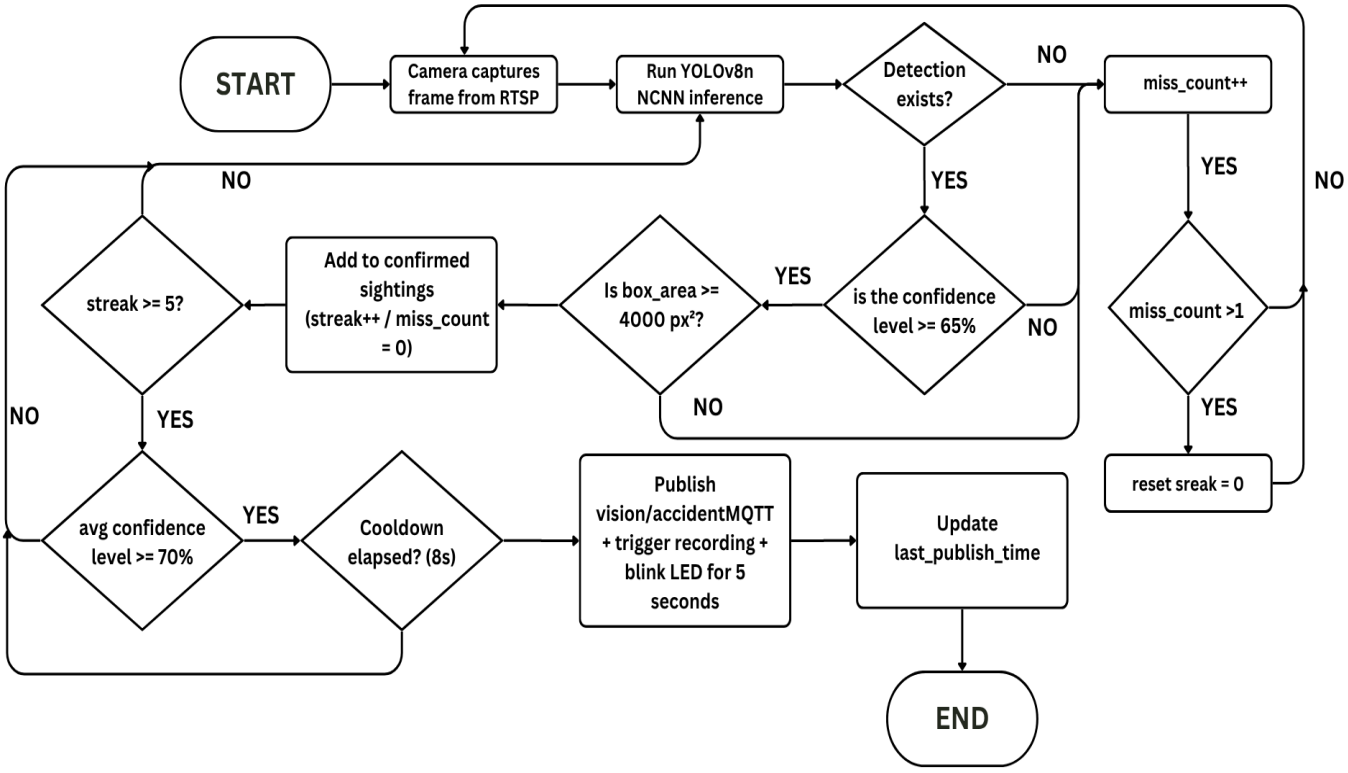
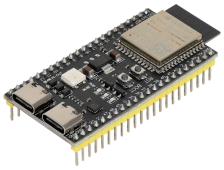
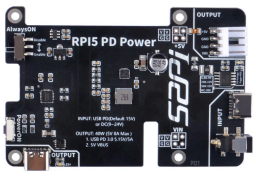
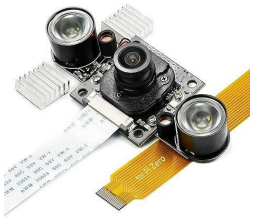


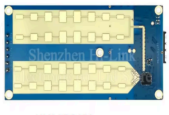
Figure 3.4: Flow Chart of Accident Detection

The LUMEN accident detection system operates in a continuous loop, beginning when the camera captures a frame from the RTSP stream and immediately passes it through the YOLOv8n NCNN inference engine. If no object is detected in the frame, a miss counter is incremented, and the system checks whether too many consecutive misses have occurred — if so, the streak is fully reset to zero before looping back to capture the next frame, preventing a false build-up of detections across unrelated frames. If an object is detected, the system applies two qualification gates before accepting it: first, the bounding box area must be at least 4,000 pixels squared to filter out tiny, distant, or irrelevant objects in the background, and second, the confidence level reported by the model must be at least 65% to ensure only reasonably certain detections are counted. A frame that passes both gates is added to the confirmed sightings, incrementing the streak counter while resetting the miss counter to zero. Once the streak reaches 5 consecutive qualifying frames, the system evaluates the average confidence across the entire streak and requires it to be at least 70% before proceeding — this final gate rejects weak streaks where detections barely passed the individual threshold. If the average confidence is sufficient and 8 seconds have elapsed since the last alert was sent, the system publishes an accident event to the MQTT broker, send LED blink command on two esp32 sub nodes and triggers the video recording module, after which the last publish time is updated and the loop continues to the next frame.

Table 3.1: List of Electronic Component Used for the Prototype

Component	Image	Description	Application
Raspberry Pi 5		A microcomputer that acts as the compute module for the project.	Running vehicle detection algorithms, data processing, and traffic light control logic.
ESP32		A microcontroller with wifi capabilities.	Used in traffic beacons for vehicle detection.
52Pi RPi 5 PD Power Expansion Board		Provides power delivery capabilities for Raspberry Pi	Used to step down voltage from 12V of battery to 5V5A for Raspberry Pi 5.

ArduCam Day/Night Wide Angle		Camera module used for accident detection.	Monitoring lane usage and accidents.
12V (30AH and 36AH) LiFePO4 Battery		Main source of power for LUMEN.	Primary power source for LUMEN, providing consistent supply.
LM2596s Adjustable Buck Converter		Used to step down voltage.	Steps down 12V voltage from battery to 5V for the system(two sub-nodes).
20 & 100 Watts Solar Panel		Alternative source of power for LUMEN.	Backup and supplementary power source to charge battery and reduce grid dependency.

Solar Charge Controller		Regulates the flow of electricity from solar panels to the battery	Ensures that the LiFePO4 battery receives safe and efficient charging from the solar panels
HLK-LD2451 24GHz Millimeter-Wave Radar Module		24 GHz Millimeter-Wave radar sensor.	Provides precise vehicle presence detection, including its distance and speed with proper calibration.
Traffic Blinker Lights		Roadside warning lights	Alerts pedestrians or drivers of incoming vehicles from the other side of the curve.

HC-12 Transceiver Module		Radio Frequency Module	Use for communication between the two ESP32 and Raspberry Pi

Table 3.1 outlines the electronic components used in the LUMEN system. It includes the component images, product descriptions, and their specific functions or roles within the system.

Table 3.2: Bill of Materials

Bill of Materials			
Product Name	Estimated Unit Price (₱)	Quantity	Total Price
Raspberry Pi 5 4GB RAM Variant Set(includes cooling fan and SD Card)	5,300.00	1	5,300.00

ESP32	250.00	2	500.00
ArduCam Day/Night Wide Angle	850.00	1	850.00
52Pi RPi 5 PD Expansion Board	1350	1	1,350.00
HLK-LD2451 24GHz mmWave Radar	650	2	1,300.00
12V 30AH LiFePO4 Battery	2,000.00	1	2,000.00
12V 36AH LiFePO4 Battery	2,200.00	1	2,200.00
12V 50W Solar Panel	1,100	1	1,100.00
12V 20W Solar Panel	670	2	1,340
PWM Solar Charge Controller	130	2	260

MPPT Solar Charge Controller	1,100	1	1,100
Neopixel 8x8 LED Matrix	150	2	300
Traffic Signal Blinker Lights	490	2	900.00
HC-12 RF Module	400	3	1,200.00

Estimated Total Cost: ₱19,700.00.

Table 3.2 presents the Bill of Materials for the LUMEN system, which has an estimated cost of approximately **₱19,700.00**, depending on component quality, supplier pricing, and quantity purchased. This estimate covers essential hardware needed for a reliable and portable lane-use monitoring traffic light solution.

FABRICATION PHASE

The researchers have procured the hardware and materials required to create the project. A better solution will be needed after the procurement process is done by the researchers to ensure product quality. After completion, the programmers, designers, and researchers of the team will collaborate in the testing and validating phase using a

prototype version of the project with breadboards and Arduino to ensure the system works.

Table 3.3 : Gantt Chart

Activities	AUGUST				SEPTEMBER				OCTOBER				NOVEMBER				DECEMBER				JANUARY				FEBRUARY				MARCH			
	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4
Conducting of Research																																
Title Presentation																																
Title Defense																																
Preparation, Organization, and Documentation																																
Architecture and Design Planning																																
Prototype Developipment																																
Testing and Validation																																
Finalization of Documents																																
Product and Document Presentation																																

The research team conducted the research project from the 1st week of August to the lastweek of September. The researchers presented their titles on the 4th day of the 2nd week of August and defended it on the 3rd week of October. After presenting how the project will move forward, it was approved by the research adviser. The project's prototyping was worked on for over 3 months and submitted on time before the due date. The development of the project's prototype was completed in the 3rd week of March. The

preparation and submission of paper and prototype for the final defense was on the 3rd week of March.

TESTING METHODS

The system was subjected to several tests to determine its overall performance and reliability. Functional testing was conducted to verify that each component, such as the Raspberry Pi, ArduCam, sensors, LED display, and solar unit, worked properly. After this, integration testing was carried out to ensure that the hardware and software modules communicated effectively.

To simulate real-world use, simulation testing and field testing were conducted with the participation of licensed drivers, pedestrians, and authorities. These tests evaluated detection accuracy, signal responsiveness, and user perception. In addition, stress testing was done under challenging conditions such as low light and multiple vehicles, while failure handling tests were used to validate that the system could log incidents and notify authorities in case of accidents.

The researchers evaluated the performance of LUMEN system to ensure that each criteria that concerns the system satisfies all expectations. The process highlighted both opportunities for improvement and potential challenges. Prior to the assessment, respondents were introduced to the

project and its key features and functions. Structured feedback on the system's usability and effectiveness was collected using an assessment sheet.

A. Licensed Drivers (8)

B. Traffic and Safety Personnels (0)

C. Authorities (2)

D. Pedestrians (9)

Evaluation Criteria

The criteria evaluated by the respondents during the system assessment are shown below:

Table 3.4: Criteria for System Evaluation

Criteria	Indicator
Functionality	Correct detection of road accidents,vehicles, speed estimation, and activation of warning lights
Reliability	Stable and consistent operation across different scenarios
Usability	Visibility, clarity, and comprehensibility of signals for drivers and pedestrians
Visibility and Clarity	Effectiveness of the warning lights and visual indicators in terms of brightness, visibility distance, and clarity.
Safety Impact	Ability of the system to improve driver awareness and reduce the risk of collisions.
Efficiency	Speed and responsiveness of the system in detecting vehicles and activating warning signals with minimal delay to ensure timely alerts for road users.
Sustainability	Efficiency of solar-powered operation and durability of materials under long-term use
Economic Feasibility	Cost-effectiveness for local government units and rural deployment
Extensibility	Potential to expand the system for urban intersections, multi-lane highways, or smart road use

INSTRUMENT AND TECHNIQUE USED

A. Unit Test

Unit Testing was conducted to evaluate the functionality of each individual component of the LUMEN system before full integration. This method ensured that every module, such as vehicle detection, warning signals, emergency handling, power supply, and cloud logging, performed according to its intended purpose. By isolating and testing each unit, errors were identified and corrected early, thereby improving the reliability and overall performance of

Table 3.5 Unit Test for Wireless Communication

Test Case Name:	HC-12 Test		**Test ID:**	HC-01
Description:	This covers key functional and reliability aspects of HC-12 Transceiver Module			
No.	Test Scenario	Expected Output	Result Pass/Fail	Comments
1	Point-to-Point Data Transparency	Data sent from Module A (TX) appears identically on Module B (RX) via Serial Monitor.		
2	AT Command Entry (Command Mode)	Module returns "OK" when SET pin is pulled LOW and "AT" is sent.		
3	Baud Rate Configuration	Changing baud rate (e.g., AT+B115200) updates the module; communication resumes at new speed.		
4	Channel/Frequency	Changing channel (e.g.,		

	Switching	AT+C001 to AT+C100) successfully avoids local 433MHz interference.		
5	Line-of-Sight Range Test	Maintain stable data link at a distance of 100m (or rated distance for specific FU mode).		
6	Power Mode Transition (FU1-FU4)	The module successfully enters low-power modes (e.g., AT+FU2) and reduces current consumption.		

Table 3.6: Unit Test for Accident Detection and Emergency Protocol

Test Case Name:	Accident & Emergency Protocol Test		**Test ID:**	ACC-02
Description:	This test covers detection accuracy, alert handling, and system reliability aspects essential for testing YOLOv8n-based accident detection systems.			
No.	Test Scenario	Expected Output	Result Pass/Fail	Comments
1	Accident occurs in camera view	YOLOv8n detects accident; alert triggered		
2	Multiple accident vehicles	All involved vehicles detected; accident confirmed		
3	False positives (e.g., stopped vehicles)	No accident alert triggered; system filters false alarms		
5	Alert delivery failure	System retries or logs failure while ensuring alerts sent later		
6	Real-time performance	Detection and alert generation occur within seconds		

Table 3.7: Unit Test for Radar Node Detection and Filtering Test

Test Case Name:	Radar Node Detection and Filtering Test		**Test ID:**	HLK-03
Description:	This test evaluates the ability of the radar node to detect approaching vehicles, measure their distance and speed, filter irrelevant targets, and transmit valid detection data to the Raspberry Pi base node.			
No.	Test Scenario	Expected Output	Result Pass/Fail	Comments
1	A vehicle enters the radar detection range	The radar detects the vehicle and generates valid detection data		
2	A vehicle approaches the radar sensor	The system identifies the target as approaching and updates distance and speed values		
3	A vehicle moves away from the radar	The system does not trigger an approaching detection event		
4	Non-vehicle movement (e.g., pedestrian or noise) is introduced	The system filters out irrelevant targets and avoids false detection		
5	Continuous valid frames are received from one vehicle	Detection is confirmed only after the required number of valid frames		

Table 3.8: Unit Test for Base Node Data Processing and Alert Control

Test Case Name:	Base Node Data Processing and Alert Control Test		**Test ID:**	RPI-04
Description:	This test evaluates the capability of the Raspberry Pi to receive radar data, process detection events, manage system logic, and send appropriate warning commands to the corresponding radar node.			
No.	Test Scenario	Expected Output	Result Pass/Fail	Comments
1	Valid data is received from a radar node	The Raspberry Pi correctly parses and displays/logs the received data		
2	Radar Node A detects an approaching vehicle	The Raspberry Pi sends a BLINK_YELLOW command to Radar Node B		
3	Radar Node B detects an approaching vehicle	The Raspberry Pi sends a BLINK_YELLOW command to Radar Node A		
4	No updates are received after detection	The system clears the detection event after the configured delay		
5	Continuous detections from the same node	The system maintains or refreshes the warning state correctly		
6	Invalid or corrupted data is received	The system ignores the data without crashing or triggering false alerts		

Table 3.9: Unit Test for Off-Grid Power System Reliability

Test Case Name:	Off-Grid Power System Reliability Test		**Test ID:**	SOL-05
Description:	This test evaluates the reliability and adequacy of the off-grid power system in supplying stable and continuous power to all system components under normal and extended operation.			
No.	Test Scenario	Expected Output	Result Pass/Fail	Comments
1	Power is supplied from the battery to the system	All components power on successfully		
2	All modules (ESP32, radar, HC-12, NeoMatrix) operate simultaneously	The system runs without interruption or instability		
3	System runs continuously for an extended period	The system remains operational within expected runtime		
4	NeoMatrix alert is triggered during operation	No voltage drop or system reset occurs		
5	Battery level decreases over time	System continues functioning until near depletion		
6	Solar/off-grid charging is applied	Battery charges properly and supports system operation		

B. Integration Test

Integration testing will verify that all components—whether it is the Raspberry Pi with computer vision, the ESP32 with doppler radar and blinker lights—work together as a complete system according to the overall requirements. This approach will evaluate how these modules interact in an actual road setting, demonstrated in a pilot test, to ensure seamless performance before considering larger-scale deployment

Table 3.10: Integration Test

Parameter	**Description**
Duration	Total time taken for the test session (e.g., in minutes or hours).
Average Detection & Processing Time	The average time taken for vehicle detection and warning signal activation, calculated from total duration divided by detections.
Warning Signals Activated	Number of successful warning signals displayed on the blinker and buzzer during detections.
Emergency Event Handling	The number of simulated accident or obstruction events logged and reported to dashboard/ authorities.
Power Supply Status	Operational status of the solar panel and battery during the test (e.g., day/night, low-power conditions).
Network Conditions	Description of the network environment during the test (e.g., connection type, speed, and stability).
User Experience Feedback	Qualitative feedback from drivers, pedestrians, and test observers regarding clarity of signals, responsiveness, and

	overall system performance

PHINMA University of Pangasinan

Arellano St,. Dagupan City, Pangasinan

LUMEN (Lane-Use Monitoring for Event Notification): An Integrated Sensor and Computer Vision-Based System for Rural Blind Curve Roads Lane Safety and Accident Detection

Evaluation Sheet

Name of Respondent: _____

Type of Respondent: ☐ Driver ☐ Traffic/Safety Personnel

☐ Authority ☐ Pedestrians

Instructions: Evaluate the system by placing a check (✓) in the box that corresponds to your assessments of each criterion.

Scale:

- **5 - Outstanding**
- **4 - Above Satisfactory**
- **3 - Satisfactory**
- **2 - Below Satisfactory**
- **1 - Needs Improvements**

Criteria	Description	5	4	3	2	1
Functionality	The system performs its intended purpose (lane monitoring, incident awareness, and traffic signaling) accurately.					

Reliability	The system operates consistently with minimal downtime and errors.					
Usability	Signals and interfaces are easy to understand and use for drivers and administrators.					
Visibility and Clarity	Traffic lights/signals are clear and visible in day and night conditions.					
Safety Impact	The system effectively reduces risks of accidents, particularly at blind curves.					
Efficiency	The system responds quickly to detected events while minimizing resource use.					
Sustainability	The system is environmentally sustainable, cost-efficient in power usage, and built for long-term service.					
Economic Feasibility	The cost of implementing and maintaining the system is practical and affordable.					
Extensibility	The system can be scaled or adapted to other road conditions or smart city systems.					

Recommendations:

Analysis of Data Gathering Methodology

To evaluate the performance and acceptability of the LUMEN (Lane-Use Monitoring for Event Notification) system, the responses gathered from the survey instrument were subjected to statistical analysis. Since the study employed a five-point Likert scale ranging from 1 (Needs Improvement) to 5 (Outstanding), appropriate tools were selected to summarize, interpret, and classify the data. These methods ensured that the results were processed systematically, yielding both quantitative precision and qualitative insights into respondent perceptions.

Weighted Mean

The **weighted mean** was employed to evaluate the survey responses and determine the overall assessment of the LUMEN (Lane-Use Monitoring for Event Notification) system. This statistical measure considers both the value assigned to each response and the frequency with which it occurs, ensuring that the results provide an accurate reflection of collective feedback. In this study, the weighted mean was particularly useful in analyzing criteria such as functionality, reliability, usability, safety, and efficiency.

Formula:

$$\overline{X} = \frac{\Sigma (w_i \bullet x_i)}{\Sigma x_i}$$

Where:

$\overline{X}$ = Weighted Mean

w_i = Weight assigned to each response option

x_i = Frequency of responses

Σx_i = Total number of responses

Percentage

The **percentage method** was used to show how responses were distributed across the five-point Likert scale, providing a clearer understanding of user feedback and system performance. This measure was particularly useful in evaluating user satisfaction, signal visibility, and the overall acceptance of the LUMEN system. A higher percentage of favorable responses indicated that users considered the system effective and reliable, while lower percentages pointed to areas requiring further improvement. This approach made it possible to identify trends in perception and to determine the system's impact on lane monitoring and traffic safety. To convert responses from the Likert scale (Outstanding, Above Satisfactory, Satisfactory, Below Satisfactory, Needs Improvement) into percentages, the following formula was used:

$$P = \frac{x}{n} \times 100$$

Where:

P = Percentage

x = Frequency of responses in a given category

n = Total number of responses

Interval Calculation

The **interval calculation** was applied to interpret the weighted mean results by dividing the range of the Likert scale into equal categories. This method allowed the numerical values to be translated into descriptive ratings, ensuring consistency in analysis and interpretation. In this study, the interval calculation provided the framework for classifying responses regarding the effectiveness, usability, and safety of the LUMEN system. By establishing fixed ranges, the researchers were able to determine whether the system's performance was rated as outstanding, satisfactory, or in need of improvement. The formula used to compute the interval width was:

$$I = \frac{Highest\ Score - Lowest\ Score}{Number\ of\ Categories}$$

Where:

Highest Score = 5 (Outstanding)

Lowest Score = 1 (Needs Improvement)

Number of Categories = 5

Substituting the values:

$$I = \frac{5-1}{5} = 0.80$$

Based on this computation, the descriptive categories were defined as follows

Table 3.11 : Likert Scale

Description	Scale	Range
Outstanding	5	5.00 – 4.21
Above Satisfactory	4	3.41 – 4.20
Satisfactory	3	2.61 – 3.40
Below Satisfactory	2	1.81 – 2.60
Needs Improvement	1	1.00 – 1.80

CHAPTER IV

SYSTEM DESIGN, RESULTS, AND DISCUSSION

This chapter outlines the implementation of the LUMEN **(**Lane-Use Monitoring Event Notification**)** system in rural blind curves, specifically in Tebeng, Dagupan City, Pangasinan—presenting the findings and discussions derived from the conducted study. It elaborates on the system's functionalities, testing results, and respondent's feedback based on the study's objectives. The results are analyzed using descriptive statistics—in order to evaluate the system's acceptance, applicability, and overall performance.

4.1 System Design

4.1.1 Development Tools and Techniques

The LUMEN System was developed integrating various hardware and software technologies to function effectively.

Hardware components:

- **Raspberry Pi** - Main Processing Unit of the system that is responsible for running the system and processing data from sensors and the camera.
- **ESP32** - Manage sensor data collection and wireless communication between system components.
- **ArduCam** - Capture real-time video footage of the monitored road segment.

- **HLK-LD2451 24GHz Doppler Radar Module** - Detects moving objects such as vehicles within its detection range.
- **HC-12 Transceiver Module** - Enables long-range wireless communication between system modules.
- **Traffic Blinker Lights** - Visual warning mechanism for approaching drivers.
- **30-50 Watts Solar Panel** - Supplies renewable energy to power up the system.
- **Solar Charge Controller** - Regulates the power flow from the solar panel to the battery.
- **12V LiFePO4 Battery** - Stores energy to provide continuous power to the system.

Softwares:

- **Computer Vision Algorithms** *(YOLO)* - Identifies vehicles from camera footage in real time.
- **AI Model Training** *(Google Colab)* - Train the YOLO-based vehicle detection model using computing resources
- **IDE** *(Thonny & Arduino IDE)*- Developing and debugging the system software
- **Dataset** *(Roboflow)* **-** dataset preparation and labeling of vehicle images for AI model training.
- **Database** - Stores system data and detection records
- **Dashboard** *(Microsoft Visual Studio)*- Displays real-time monitoring and system notifications,
- **Frontend** *(Avalonia)* - Framework for user interface for system monitoring.

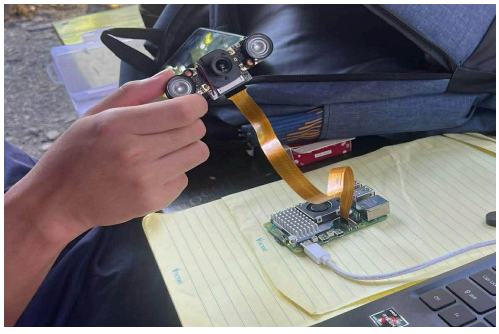
4.1.2 System Fabrication

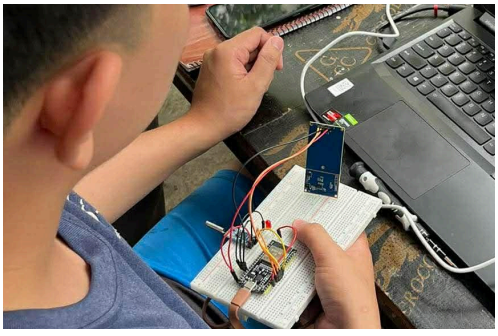
Component Setup and Initial Testing

The initial phase involved assembling the required hardware components and integrating them into a functional prototype. Sensors and Cameras were positioned to detect approaching vehicles from opposite directions of the blind curves. The Blinker Lights were installed to provide visible warnings to drivers. Initial testing was conducted to ensure proper wiring, connectivity, and data transmission between components

➢ **Components Placement and Assembly**

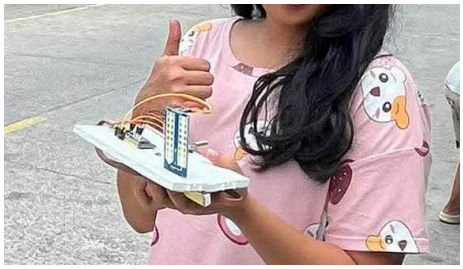
The components were arranged on a temporary prototyping setup to establish the basic system layout. During this stage, the camera and radar sensor were positioned to simulate their intended placement for monitoring a road segment, while the processing and communication modules were organized to allow efficient connections

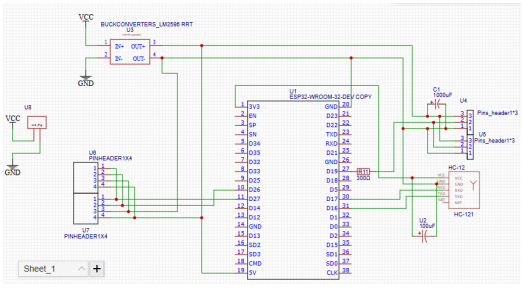




➢ **Wiring and Hardware Interconnections**

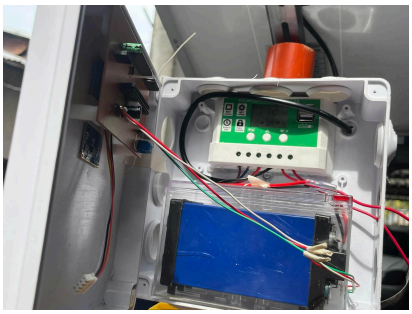
Electrical connections between the components were established using jumper wires and appropriate connectors. The ESP32 was connected to the radar sensor and communication modules, while the Raspberry Pi was configured to receive and process detection data. Proper wiring ensured stable communication between sensing, processing, and signal activation components.





➢ **Integration into the System Enclosure**

The hardware was mounted into a protective system enclosure. This enclosure organizes the internal components such as the camera and sensing modules, communication modules, and power system while protecting them from environmental conditions during roadside deployment.





➢ **Power System Setup**

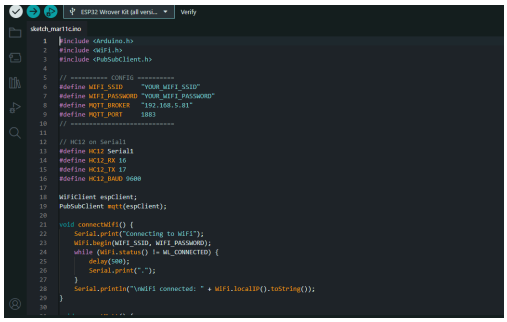
The solar-powered energy system was installed simultaneously with the integration into the enclosure, to provide an independent power source for the LUMEN system. The solar panel was connected to the charge controller to regulate voltage and safely charge the LiFePO4 battery, which supplies continuous power to the system components.

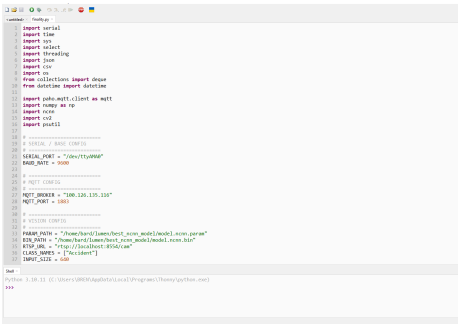




➢ **Integration with the Software Environment**

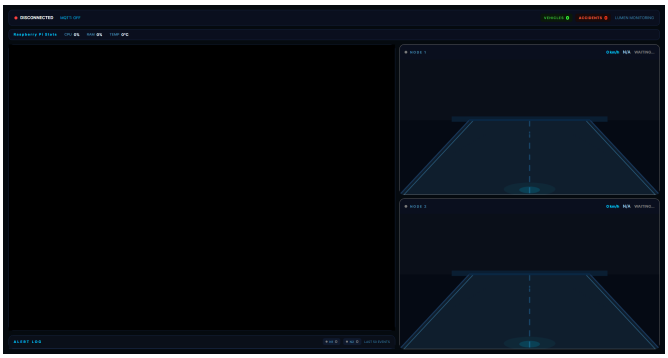
Once the hardware components were secured inside the enclosure, the system was integrated with the required software environment. The Raspberry Pi programs were developed and executed using Thonny, while the ESP32 firmware was uploaded through the Arduino IDE to enable communication with sensors and other modules.





➢ **Dashboard Development and System Monitoring**

After establishing hardware and software integration, a monitoring interface was developed to observe system operation. The dashboard, created using Microsoft Visual Studio with the Avalonia framework, allows monitoring of detection events, system status, and overall performance of the LUMEN system.



4.1.3 System Features and Functionalities

The LUMEN system is designed to provide several essential features:

1. **Vehicle Detection**
    - Detects approaching vehicles using radar sensing
    - Supports real-time traffic monitoring in blind-curve location

- Helps determine potential traffic conflicts between opposing lanes

2. **Real-Time Warning Signals**
    - Activates traffic blinker lights when a vehicle is detected
    - Provides clear visual warnings to approaching drivers
    - Ensures warnings are visible even in low-visibility conditions
3. **Event Notification**
    - Generates system alerts when vehicles approach simultaneously from opposite directions
    - Logs detection events for monitoring and analysis through the Dashboard
    - Enables real-time system monitoring by administrators/authorities
    - Sends real-time notification for notifications for administrators/authorities in case of any traffic conflicts.
    - Detects accidents using Camera with Computer Vision (YOLO).
4. **Solar-Powered Operation**
    - Uses solar panels and battery storage to provide sufficient continuous power
    - Allows the system to operate in remote/rural road location

4.1.4 System Architecture

The architecture of the LUMEN System is composed of three primary layers:

1. **Sensing Layer:** responsible for collecting real-time data from the road environment. It detects approaching vehicles and gathers information that will be processed by the system.
    - **ArduCam Camera Module:** Captures real-time video footage and provides visual input for the detection system.
    - **HLK-LD2451 24GHz Doppler Radar Module:** Detects moving vehicles within its sensing range.
    - **ESP32 Microcontroller:** Sends detection signals and data to the main processing unit.

2. **Processing Layer:** responsible for analyzing the collected data and making system decisions. It processes the sensor inputs and determines whether warning signals should be activated.
    - **Raspberry Pi:** the main processing unit of the system which receives data from the camera and sensors for analysis.
    - **Computer Vision Model (YOLO):** identifies vehicles entering the monitored area.
    - **Scripts (Thonny IDE):** Handles system logic and controls the interaction between sensors, AI model, and signaling devices.
    - **Model Training Environment (Google Colab):** Train the YOLO vehicle detection model during system development.

- **Dataset Management and Labelling (Roboflow):** Creates and labels image dataset required for training the detection model

3. **Notification Layer:** responsible for informing drivers and allowing system monitoring. It communicates detection results through warning signals and the monitoring dashboard.
    - **Traffic Blinker Lights:** Activates upon detection and provides visual warning signals to approaching drivers.
    - **HC-12 Transceiver Module:** Enables wireless communication between system modules and transmits detection data and signal commands.
    - **Dashboard:** Displays real-time monitoring data of vehicle detections and system status, allowing administrators/authorities to observe.
    - **Data Storage:** Stores system logs and detection records for monitoring and analysis.

4.1.5 System Testing and Debugging

To ensure reliability and functionality of the LUMEN system, each components underwent testing:

- **Unit Testing:** Evaluate individual components such as Camera, Sensors, Blinker Lights, and processing modules.

- **Integration Testing:** To confirm that all components were working together seamlessly, specifically Vehicle Detection process correctly and triggers the signaling mechanism.
- **User Acceptance Testing (UAT):** Evaluated the system's several criterias and was conducted among selected respondents.

A Likert Scale Survey was used to assess user satisfaction.

- **Functionality -** Measures whether the system performs its intended task correctly, such as detecting vehicles and activating warning signals
- **Reliability -** Evaluates the system's consistency and stability during operations, ensuring it works properly without frequent errors or failures.
- **Usability -** Assess how easy the system is to understand and interact with.
- **Visibility and Clarity -** Determines whether the warning signals are clearly visible and understandable.
- **Safety Impact -** Measures the system's effectiveness in improving road safety—helping drivers avoid potential collisions.
- **Efficiency -** Evaluates how quickly and effectively the system responds when detecting vehicles and activating alerts.
- **Sustainability -** Assess whether the system can operate continuously with minimal environmental impact.
- **Economic Feasibility -** Determines whether the system is cost-effective and practical to implement.

- **Extensibility -** Measures the system's ability to be expanded or upgraded.

4.1.6 Deployment Process

The system was deployed on various platforms to ensure the software is correctly configured and operational.

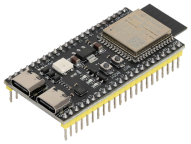
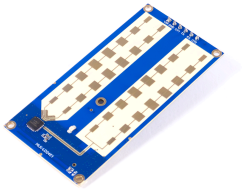
SOFTWARE

	Web Services	Description and Usage
Dashboard	 Microsoft Visual Studio	An integrated development environment (IDE) used for developing applications and software interfaces. In the LUMEN system, it was used to design and develop the **system dashboard**, which allows administrators to monitor vehicle detections, system status, and event notifications in real time.
Frontend Framework	 Avalonia	A cross-platform user interface framework used for building graphical desktop applications. It was used in the LUMEN system to develop the **frontend interface of the monitoring dashboard**, enabling a clear and interactive display of system data and alerts.
Raspberry Pi Logic	 Thonny	A Python IDE designed for beginners and commonly used for Raspberry Pi development. In this project, it was used to **write, test, and execute Python scripts responsible for the system's main logic**, including running the AI model and processing detection results.

ESP-32 Logic	 Arduino IDE	A software platform used for programming microcontrollers such as the ESP32. It was used to **develop and upload the firmware controlling the ESP32**, which manages sensor inputs and communication between hardware components of the system.
Dataset + Labelling	 Roboflow	A cloud-based platform used for computer vision dataset management. In this project, it was used to **collect, organize, and label images of vehicles**, preparing the dataset required for training the vehicle detection model used in the LUMEN system.
AI Model Training	 Google Colab	A cloud-based computing platform that provides access to powerful GPUs for machine learning tasks. It was used to **train the YOLO-based vehicle detection model using the prepared dataset**, enabling faster training and development of the AI component of the system.

HARDWARE

 Raspberry Pi 5	Serves as the **main processing unit of the LUMEN system**. During deployment, it is installed inside the system enclosure where it processes vehicle detection data, runs the AI model, and controls the activation of warning signals.

 ESP-32	Deployed to handle **sensor data acquisition and communication between hardware modules**. It assists in transmitting detection data and coordinating system responses.
 ArduCam	Mounted facing the **monitored road segment or blind curve** to capture real-time video footage. The captured images are used by the AI model to detect approaching vehicles.
 HLK-LD2451 24GHz MillimeterWave Radar Module	Module positioned to **detect moving vehicles within the detection range**. It provides motion detection data that complements the camera-based detection system.

 HC-12 Transceiver Module	Module deployed to enable **long-range wireless communication** between different system units, allowing signals and detection data to be transmitted across the system.

4.2 Results and Discussions

4.2.1 Data Presentation and Analysis

The study presents the findings derived from the survey responses of **19 participants**, including licensed drivers, traffic and safety personnel, pedestrians, and authorities. The data was analyzed using **descriptive statistics** and the **Likert Scale** to assess the numerous system's criterias.

Usability Testing and System Performance Evaluation

The LUMEN (Lane-Use Monitoring Event Notification) system underwent initial testing conducted by the researchers to assess its functionality, usability, and overall system performance. These preliminary tests aimed to verify that the system operates correctly under different conditions and that both the hardware and software components function as expected. The evaluation focused on key aspects such as (1) the reliability of wireless communication between system components and (2) the effectiveness of the system in triggering emergency response protocols. This also includes (3) the functionality of the radar node in accurately detecting approaching vehicles and filtering

irrelevant targets, (4) the performance of the Raspberry Pi base node in processing incoming data and transmitting appropriate warning signals, and (5) the reliability of the off-grid power setup in providing stable and continuous power to the system.

This process allowed the researchers to identify and resolve potential issues before proceeding to further system evaluation.

Table 4.1: Unit Test for Wireless Communication

Test Case Name:	HC-12 Test		**Test ID:**	HC-01
Description:	This covers key functional and reliability aspects of HC-12 Transceiver Module			
No.	Test Scenario	Expected Output	Result Pass/Fail	Comments
1	Point-to-Point Data Transparency	Data sent from Module A (TX) appears identically on Module B (RX) via Serial Monitor.	Pass	
2	AT Command Entry (Command Mode)	Module returns "OK" when SET pin is pulled LOW and "AT" is sent.	Pass	
3	Baud Rate Configuration	Changing baud rate (e.g., AT+B115200) updates the module; communication resumes at new speed.	Pass	
4	Channel/Frequency Switching	Changing channel (e.g., AT+C001 to AT+C100) successfully avoids local 433MHz interference.	Pass	
5	Line-of-Sight Range Test	Maintain stable data link at a distance of 100m (or rated	Pass	

		distance for specific FU mode).		
6	Power Mode Transition (FU1-FU4)	The module successfully enters low-power modes (e.g., AT+FU2) and reduces current consumption.	Pass	

Table 4.1 evaluates the functionality of the HC-12 Transceiver Module, ensuring that it correctly handles point-to-point data transparency and AT command entries . The test confirmed that data sent from the transmitter appears identically on the receiver and that baud rates and channels can be configured successfully to avoid local interference . Additionally, the line-of-sight range test and power mode transitions functioned as expected, maintaining stable data links and properly reducing current consumption in low-power modes . All test cases passed, confirming that the wireless warning signal communication operates as intended .

Table 4.2: Unit Test for Accident Detection and Emergency Protocol

Test Case Name:	Accident & Emergency Protocol Test		**Test ID:**	ACC-02
Description:	This test covers detection accuracy, alert handling, and system reliability aspects essential for testing YOLOv8n-based accident detection systems.			
No.	Test Scenario	Expected Output	Result Pass/Fail	Comments
1	Accident occurs in camera view	YOLOv8n detects accident; alert triggered	Pass	Model can be trained more to improve accuracy.

2	Multiple accident vehicles	All involved vehicles detected; accident confirmed	Pass	
3	False positives (e.g., stopped vehicles)	No accident alert triggered; system filters false alarms	Pass	
4	Alert delivery failure	System retries or logs failure while ensuring alerts sent later	Pass	
5	Real-time performance	Detection and alert generation occur within seconds	Pass	

Table 4.2 evaluates the functionality of the YOLOv8n-based accident detection system, ensuring that it correctly identifies accidents and multiple involved vehicles within the camera's view . The test confirmed that the system accurately triggers alerts for actual incidents while successfully filtering out false alarms, such as stopped vehicles . Additionally, the alert delivery and real-time performance features functioned as expected, ensuring that failure logs are generated when necessary and that visual detections prompt alerts within seconds. All test cases passed, confirming that the accident detection protocol operates as intended, though minor model training enhancements were noted for future accuracy improvements .

Table 4.3: Unit Test for Radar Node Detection and Filtering Test

Test Case Name:	Radar Node Detection and Filtering Test		**Test ID:**	HLK-03
Description:	This test evaluates the ability of the radar node to detect approaching vehicles, measure their distance and speed, filter irrelevant targets, and transmit valid detection data to the Raspberry Pi base node.			
No.	Test Scenario	Expected Output	Result	Comments

			Pass/Fail	
1	A vehicle enters the radar detection range	The radar detects the vehicle and generates valid detection data	Pass	Consecutive Vehicles that are near each other triggers only one detection
2	A vehicle approaches the radar sensor	The system identifies the target as approaching and updates distance and speed values	Pass	
3	A vehicle moves away from the radar	The system does not trigger an approaching detection event	Pass	
4	Non-vehicle and non-pedestrian movement (e.g., noise) is introduced	The system filters out irrelevant targets and avoids false detection	Pass	
5	Continuous valid frames are received from one vehicle	Detection is confirmed only after the required number of valid frames	Pass	

Table 4.3 evaluates the functionality of the HLK-LD2451 radar node, ensuring that it correctly detects approaching vehicles, updates distance and speed values, and appropriately ignores vehicles moving away . The test confirmed that the sensor successfully filters out non-vehicle and non-pedestrian noise, generating valid detection

data only for actual targets . Additionally, the continuous frame validation functioned as expected, confirming detections only after the required number of valid frames were received. All test cases passed, confirming that the radar node operates accurately as intended .

Table 4.4: Unit Test for Base Node Data Processing and Alert Control

Test Case Name:	Base Node Data Processing and Alert Control Test		**Test ID:**	RPI-04
Description:	This test evaluates the capability of the Raspberry Pi to receive radar data, process detection events, manage system logic, and send appropriate warning commands to the corresponding radar node.			
No.	Test Scenario	Expected Output	Result Pass/Fail	Comments
1	Valid data is received from a radar node	The Raspberry Pi correctly parses and displays/logs the received data	Pass	
2	Radar Node A detects an approaching vehicle	The Raspberry Pi sends a BLINK_YELLOW command to Radar Node B	Pass	
3	Radar Node B detects an approaching vehicle	The Raspberry Pi sends a BLINK_YELLOW command to Radar Node A	Pass	
4	No updates are received after detection	The system clears the detection event after the configured delay	Pass	

5	Continuous detections from the same node	The system maintains or refreshes the warning state correctly	Pass	
6	Invalid or corrupted data is received	The system ignores the data without crashing or triggering false alerts	Pass	

Table 4.4 evaluates the functionality of the Raspberry Pi base node, ensuring that it correctly parses received radar data and sends appropriate warning commands, such as "BLINK_YELLOW," to the corresponding opposing radar nodes . The test confirmed that the system effectively maintains warning states during continuous detections and clears detection events after a configured delay when no updates are received. Additionally, the data handling features functioned as expected, allowing the system to ignore invalid or corrupted data without crashing or triggering false alerts . All test cases passed, confirming that the central base node processing operates as intended .

Table 4.5:Unit Test for Off-Grid Power System Reliability

Test Case Name:	Off-Grid Power System Reliability Test		**Test ID:**	
Description:	This test evaluates the reliability and adequacy of the off-grid power system in supplying stable and continuous power to all system components under normal and extended operation.			
No.	Test Scenario	Expected Output	Result Pass/Fail	Comments
1	Power is supplied from the battery to the system	All components power on successfully	Pass	
2	All modules (ESP32,	The system runs without	Pass	

	radar, HC-12, NeoMatrix) operate simultaneously	interruption or instability		
3	System runs continuously for an extended period	The system remains operational within expected runtime	Pass	
4	NeoMatrix alert is triggered during operation	No voltage drop or system reset occurs	Pass	
5	Battery level decreases over time	System continues functioning until near depletion	Pass	
6	Solar/off-grid charging is applied	Battery charges properly and supports system operation	Pass	

The final unit test evaluates the reliability of the off-grid power system, ensuring that it correctly supplies stable and continuous power from the battery to all system components . The test confirmed that all modules, including the ESP32, radar, HC-12, and NeoMatrix, operate simultaneously without interruption or voltage drops, even when visual alerts are actively triggered. Additionally, the system continued to function properly as the battery gradually depleted over time and successfully recharged when solar power was applied. All test cases passed, confirming that the solar-powered setup operates as intended.

Overall, the unit testing and system performance evaluation demonstrate that the LUMEN system functions reliably under realistic operational conditions . While minor model training optimizations were identified to further improve camera-based detection, core functionalities—including radar precision, wireless data transmission, and continuous solar power delivery—remained intact, ensuring smooth and uninterrupted operation. The successful execution of test cases across all hardware and software modules confirms that the system meets its intended objectives, providing an efficient, incident-aware, and highly responsive safety mechanism for rural blind curves.

Integration Testing

The LUMEN (Lane-Use Monitoring Event Notification) system underwent integration testing to evaluate its overall performance through both unit testing and a pilot test. While the unit tests assessed individual components such as the HC-12 communication module and the accident and emergency response protocol, the pilot test examined how these components functioned together as a complete system. During testing, several parameters were observed, including the system's operational duration, average detection and processing time, activation of warning signals, emergency event handling, power supply stability from the solar-powered system, network conditions affecting wireless communication, and overall user experience feedback. These evaluations helped verify that the system components operated cohesively and responded effectively during detection and emergency scenarios.

Table 4.6: Comparison of Pilot Test and Unit Test Results

Parameter	**Pilot Test**	**Unit Test**
Duration	Operated for 6-7 hours continuously with no major issues	Each component ran 1-3 hours stably
Average Detection & Processing Time	Around 0.7–1.0 seconds detection time.	Around 1.0–1.5 seconds detection time.
Warning Signals Activated	Activated consistently, but slight delay in fast-moving cases	Lights responded immediately in testing
Emergency Event Handling	Alerts were successfully transmitted and logged	Protocol triggered correctly but tested separately
Power Supply Status	Stable, minor drops during low sunlight.	Stable power using controlled supply
Network Conditions	Reliable, slight delay at longer range.	Fast and stable at short range
User Experience Feedback	Generally positive, with noted limitations on fast vehicles.	Not applicable

Interview Results and Summary

User Evaluation Results (Likert Scale 1-5):

Table 4.7 - Functionality Evaluation Results

Functionality The system correctly performs its intended tasks such as detecting vehicles and activating warning signals.			
Likert scale	**Likert Scale Values**	**FREQUENCY**	**Mean**
Outstanding	5	6	30

Above Satisfactory	4	12	48
Satisfactory	3	1	3
Below Satisfactory	2	0	0
Needs Improvement	1	0	0
		Weighted Mean	4.26

The Functionality of the system received a weighted mean of 4.26, which is rated as Outstanding. This means users found the system highly accurate and effective in performing its intended tasks, such as detecting vehicles and activating warning signals.

Table 4.8 - Reliability Evaluation Results

Reliability The system is able to operate consistently and accurately under different conditions.			
Likert scale	**Likert Scale Values**	**FREQUENCY**	**Mean**
Outstanding	5	2	10
Above Satisfactory	4	11	44
Satisfactory	3	5	15
Below Satisfactory	2	1	2
Needs Improvement	1	0	0
		Weighted Mean	3.74

The Reliability of the system received a weighted mean of 3.74, which is rated as Above Satisfactory. This means users found the system stable and consistent in its operation across various conditions with minimal downtime.

Table 4.9 - Usability Evaluation Results

Usability The system is easy to be understood and interpreted by drivers, pedestrians, or operators.			
Likert scale	**Likert Scale Values**	**FREQUENCY**	**Mean**
Outstanding	5	7	35
Above Satisfactory	4	8	32
Satisfactory	3	3	9
Below Satisfactory	2	1	2
Needs Improvement	1	0	0
		Weighted Mean	4.11

The Usability of the system received a weighted mean of 4.11, which is rated as Above Satisfactory. This means users found the system easy to interact with, and its signals were clear and straightforward for both drivers and pedestrians.

Table 4.10 - Visibility and Clarity Evaluation Results

Visibility and Clarity The warning signals are clearly visible and easily recognizable by approaching road users.			
Likert scale	**Likert Scale Values**	**FREQUENCY**	**Mean**
Outstanding	5	13	65
Above Satisfactory	4	6	24
Satisfactory	3	0	0
Below Satisfactory	2	0	0

Needs Improvement	1	0	0
		Weighted Mean	4.68

The Visibility and Clarity of the system received a weighted mean of 4.68, which is rated as Outstanding. This means users found the system highly visible, providing bright and clear visual indicators for approaching road users during day and night.

Table 4.11 - Safety Impact Evaluation Results

Safety Impact The system contributes to reducing potential road accidents and improving road safety.			
Likert scale	**Likert Scale Values**	**FREQUENCY**	**Mean**
Outstanding	5	12	60
Above Satisfactory	4	7	28
Satisfactory	3	0	0
Below Satisfactory	2	0	0
Needs Improvement	1	0	0
		Weighted Mean	4.63

The Safety Impact of the system received a weighted mean of 4.63, which is rated as Outstanding. This means users found the system highly effective in improving driver awareness and reducing the risks of collisions at blind curves.

Table 4.12 - Efficiency Evaluation Results

Efficiency The system detects events and activates the appropriate response quickly and effectively.			
Likert scale	**Likert Scale Values**	**FREQUENCY**	**Mean**
Outstanding	5	4	20
Above Satisfactory	4	13	52
Satisfactory	3	2	6
Below Satisfactory	2	0	0
Needs Improvement	1	0	0
		Weighted Mean	4.11

The Efficiency of the system received a weighted mean of 4.11, which is rated as Above Satisfactory. This means users found the system fast and responsive, providing timely warnings to road users with minimal delay.

Table 4.13 - Sustainability Evaluation Results

Sustainability The effectiveness of the solar-powered design and the system's ability to operate continuously with minimal environmental impact.			
Likert scale	**Likert Scale Values**	**FREQUENCY**	**Mean**
Outstanding	5	11	55
Above Satisfactory	4	8	32
Satisfactory	3	0	0

Below Satisfactory	2	0	0
Needs Improvement	1	0	0
		Weighted Mean	4.58

The Sustainability of the system received a weighted mean of 4.58, which is rated as Outstanding. This means users found the system highly energy-efficient, praising its solar-powered design and capability for continuous, eco-friendly operation.

Table 4.14 - Economic Feasibility Evaluation Results

Economic Feasibility The system is affordable and practical for deployment by local government units or rural communities.			
Likert scale	**Likert Scale Values**	**FREQUENCY**	**Mean**
Outstanding	5	1	5
Above Satisfactory	4	7	28
Satisfactory	3	7	21
Below Satisfactory	2	2	4
Needs Improvement	1	2	2
		Weighted Mean	3.15

The Economic Feasibility of the system received a weighted mean of 3.16, which is rated as Satisfactory. This means users found the system reasonably cost-effective and practical enough for local government units and rural deployment, though there may be room for cost optimization.

Table 4.15 - Extensibility Evaluation Results

Extensibility The system's potential to be expanded or adapted for other road monitoring or smart traffic applications.			
Likert scale	**Likert Scale Values**	**FREQUENCY**	**Mean**
Outstanding	5	8	40
Above Satisfactory	4	6	24
Satisfactory	3	5	15
Below Satisfactory	2	0	0
Needs Improvement	1	0	0
		Weighted Mean	4.16

The Extensibility of the system received a weighted mean of 4.16, which is rated as Above Satisfactory. This means users found the system to have strong potential to be scaled, expanded, or adapted for other road applications and smart city infrastructures.

Table 4.16 Computer Survey Table

Factor	**Weighted Mean**	**Percentage**
Functionality	4.26	85.20%
Reliability	3.74	74.80%
Usability	4.11	82.20%
Visibility and Clarity	4.68	93.60%
Safety Impact	4.63	92.60%
Efficiency	4.11	82.20%

Sustainability	4.58	91.60%
Economic Feasibility	3.16	63.20%
Extensibility	4.16	83.20%
Grand Mean	**4.16**	**83.20%**

USER EVALUATION FINDINGS AND SUMMARY

Overall, LUMEN system achieved a Grand Mean of 4.16, which is rated as Above Satisfactory on the Likert Scale. This shows that beneficiaries specifically licensed drivers are highly satisfied with how the system works, successfully monitoring the specified blind curve and sending real-time signals for them to be notified.

The Likert Scale Reference provides a structured method for capturing user feedback, ensuring that evaluations are grounded in actual data. This approach not only demonstrates the system's effectiveness and reliability but also highlights opportunities for future improvements.

CHAPTER V

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

5.1 Summary of Findings

This study aimed to develop the LUMEN (Lane-Use Monitoring Event Notification) system that is an innovation from existing ideas—providing an efficient and reliable minimal traffic light that provides real-time signaling and accident detection, meant for Rural Roads. The system would maximize the integration of computer vision technology, sensors, and LED signaling mechanism, supported by a solar-powered system for sustainable operation.

The key findings of the study are as follows:

- The system successfully detects approaching vehicles using a combination of camera-based detection and radar sensing, enabling accurate monitoring of road activity.
- The integration of the Raspberry Pi, ESP32 microcontroller, and HC-12 transceiver modules ensures efficient processing and reliable wireless communication between system components.
- The system effectively activates traffic blinker lights as warning signals, improving visibility and alerting drivers of potential hazards in blind-curve or high-risk areas.

- The implemented accident and emergency response protocol allows the system to respond appropriately during critical events, enhancing road safety.
- The use of a solar-powered system enables continuous operation in remote or rural locations without dependence on conventional power sources.
- User evaluation results indicate that the system is functional, reliable, and effective in providing real-time road monitoring and warning signals.
- Various testing methodologies, including unit testing and integration testing, confirmed the system's functionality, communication reliability, and overall performance.
- The system was evaluated using key criteria such as functionality, reliability, usability, safety impact, efficiency, sustainability, and economic feasibility, all of which yielded positive results.

5.2 Conclusions

The findings demonstrate that the Lane-Use Monitoring Event Notification (LUMEN) provides an effective, reliable, and sustainable solution for real-time road monitoring and hazard warning, particularly in rural and high-risk road environments. By integrating computer vision, radar sensing, wireless communication, and automated signaling, the system

significantly enhances road safety by providing timely alerts to approaching drivers and enabling appropriate responses during emergency situations.

Through testing and evaluation, the system has been proven to be technically feasible, efficient, and adaptable for deployment in areas with limited infrastructure. The use of solar power further strengthens its practicality by ensuring continuous operation while minimizing energy costs and environmental impact. Overall, the LUMEN system shows strong potential as a scalable and cost-effective solution for improving traffic safety and reducing road accidents.

5.3 Recommendations

Based on the results of the study, the following recommendations are proposed for future development and research:

1. Future versions of the system may support deployment across multiple road segments or intersections, allowing centralized monitoring and control for wider coverage
2. The system can be enhanced by incorporating a GSM module to enable real-time alert notifications to authorities during detected accidents or emergency situations, improving response time and coordination.
3. The physical enclosure of the system should be improved to ensure better protection against environmental factors such as heat, rain, and dust, especially for outdoor roadside installation.
4. Collaboration with local government units (LGUs) and transportation agencies is

encouraged to support real-world implementation and scalability of the system.

5. The Use of more advanced and higher-quality sensors is recommended to improve detection accuracy and system reliability, as the current implementation utilized cost-effective components that may have limitations in precision.

6. Additional sensors may be integrated to improve detection reliability, especially in low-visibility environments such as fogs or heavy rains.

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APPENDICES

Appendix A

SURVEY

PHINMA University of Pangasinan Arellano St.
Dagupan City, Pangasinan

LUMEN (Lane-Use Monitoring for Event Notification): An Integrated Sensor and Computer Vision-Based System for Rural Blind Curve Roads Lane Safety and Accident Detection
Evaluation Sheet

Name of Respondent: _____(OPTIONAL)
Type of Respondent: ☐ Driver ☐ Traffic/Safety Personnel

☑ Authority ☐ Pedestrian

Instructions: Evaluate the system by placing a check (✓) in the box that corresponds to your assessments of each criterion.
Scale:

- **5 - Outstanding**
- **4 - Above Satisfactory**
- **3 - Satisfactory**
- **2 - Below Satisfactory**
- **1 - Needs Improvements**

Criteria	Description	5	4	3	2	1
Functionality	It watches the road, spots trouble, and changes the lights exactly when it needs to.			✓		
Reliability	It stays turned on and working almost all the time.			✓		
Usability	Drivers know exactly what the lights mean, and the workers running it don't get confused.			✓		
Visibility and Clarity	You can easily see the lights whether it is bright daylight or pitch black outside.		✓			
Safety Impact	It keeps people safe, especially on sharp turns where you can't see what's coming..		✓			
Efficiency	When events happen on the road, it reacts right away without wasting energy.		✓			
Sustainability	It saves power, doesn't hurt the environment, and will work for a very long time.	✓				
Economic Feasibility	It doesn't cost too much money to set up or to fix.				✓	
Extensibility	We can easily put this same setup on other streets or connect it to new city upgrades later on.			✓		

Recommendations:

PHINMA University of Pangasinan Arellano St,.
Dagupan City, Pangasinan

LUMEN (Lane-Use Monitoring for Event Notification): An Integrated Sensor and Computer Vision-Based System for Rural Blind Curve Roads Lane Safety and Accident Detection
Evaluation Sheet

Name of Respondent: _____(OPTIONAL)
Type of Respondent: ☐ Driver ☐ Traffic/Safety Personnel

☑ Authority ☐ Pedestrian

Instructions: Evaluate the system by placing a check (✓) in the box that corresponds to your assessments of each criterion.
Scale:

- **5 - Outstanding**
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Criteria	Description	5	4	3	2	1
Functionality	It watches the road, spots trouble, and changes the lights exactly when it needs to.		✓			
Reliability	It stays turned on and working almost all the time.		✓			
Usability	Drivers know exactly what the lights mean, and the workers running it don't get confused.		✓			
Visibility and Clarity	You can easily see the lights whether it is bright daylight or pitch black outside.	✓				
Safety Impact	It keeps people safe, especially on sharp turns where you can't see what's coming..	✓				
Efficiency	When events happen on the road, it reacts right away without wasting energy.		✓			
Sustainability	It saves power, doesn't hurt the environment, and will work for a very long time.	✓				
Economic Feasibility	It doesn't cost too much money to set up or to fix.			✓		
Extensibility	We can easily put this same setup on other streets or connect it to new city upgrades later on.		✓			

Recommendations:

SHOULD USE REFLECTORS FOR NIGHTIME INSTANCES.

PHINMA University of Pangasinan Arellano St.
Dagupan City, Pangasinan

LUMEN (Lane-Use Monitoring for Event Notification): An Integrated Sensor and Computer Vision-Based System for Rural Blind Curve Roads Lane Safety and Accident Detection
Evaluation Sheet

Name of Respondent: JAYSON PAULINO (OPTIONAL)
Type of Respondent: ☑ Driver ☐ Traffic/Safety Personnel

☐ Authority ☐ Pedestrian

Instructions: Evaluate the system by placing a check (✓) in the box that corresponds to your assessments of each criterion.
Scale:

- **5 - Outstanding**
- **4 - Above Satisfactory**
- **3 - Satisfactory**
- **2 - Below Satisfactory**
- **1 - Needs Improvements**

Criteria	Description	5	4	3	2	1
Functionality	It watches the road, spots trouble, and changes the lights exactly when it needs to.		✓			
Reliability	It stays turned on and working almost all the time.		✓			
Usability	Drivers know exactly what the lights mean, and the workers running it don't get confused.				✓	
Visibility and Clarity	You can easily see the lights whether it is bright daylight or pitch black outside.	✓				
Safety Impact	It keeps people safe, especially on sharp turns where you can't see what's coming..	✓				
Efficiency	When events happen on the road, it reacts right away without wasting energy.			✓		
Sustainability	It saves power, doesn't hurt the environment, and will work for a very long time.	✓				
Economic Feasibility	It doesn't cost too much money to set up or to fix.	✓				
Extensibility	We can easily put this same setup on other streets or connect it to new city upgrades later on.			✓		

Recommendations:

EDUCATE EVERYONE ABOUT THE SIGNIFICANCE AND PURPOSE OF THE SYSTEM, SO EVERYONE, ESPECIALLY DRIVERS, CAN MAXIMIZE ITS USAGE.

PHINMA University of Pangasinan Arellano St,.
Dagupan City, Pangasinan

LUMEN (Lane-Use Monitoring for Event Notification): An Integrated Sensor and Computer Vision-Based System for Rural Blind Curve Roads Lane Safety and Accident Detection
Evaluation Sheet

Name of Respondent: _____(OPTIONAL)
Type of Respondent: ☑ Driver ☐ Traffic/Safety Personnel

☐ Authority ☐ Pedestrian

Instructions: Evaluate the system by placing a check (✓) in the box that corresponds to your assessments of each criterion.
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- **5 - Outstanding**
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- **3 - Satisfactory**
- **2 - Below Satisfactory**
- **1 - Needs Improvements**

Criteria	Description	5	4	3	2	1
Functionality	It watches the road, spots trouble, and changes the lights exactly when it needs to.	✓				
Reliability	It stays turned on and working almost all the time.		✓			
Usability	Drivers know exactly what the lights mean, and the workers running it don't get confused.	✓				
Visibility and Clarity	You can easily see the lights whether it is bright daylight or pitch black outside.	✓				
Safety Impact	It keeps people safe, especially on sharp turns where you can't see what's coming..	✓				
Efficiency	When events happen on the road, it reacts right away without wasting energy.		✓			
Sustainability	It saves power, doesn't hurt the environment, and will work for a very long time.	✓				
Economic Feasibility	It doesn't cost too much money to set up or to fix.		✓			
Extensibility	We can easily put this same setup on other streets or connect it to new city upgrades later on.	✓				

Recommendations:

PHINMA University of Pangasinan Arellano St,.
Dagupan City, Pangasinan

LUMEN (Lane-Use Monitoring for Event Notification): An Integrated Sensor and Computer Vision-Based System for Rural Blind Curve Roads Lane Safety and Accident Detection
Evaluation Sheet

Name of Respondent: _____(OPTIONAL)
Type of Respondent: ☑ Driver ☐ Traffic/Safety Personnel

☐ Authority ☐ Pedestrian

Instructions: Evaluate the system by placing a check (✓) in the box that corresponds to your assessments of each criterion.
Scale:

- **5 - Outstanding**
- **4 - Above Satisfactory**
- **3 - Satisfactory**
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Extensibility	We can easily put this same setup on other streets or connect it to new city upgrades later on.			✓		

Recommendations:

PHINMA University of Pangasinan Arellano St,.
Dagupan City, Pangasinan

LUMEN (Lane-Use Monitoring for Event Notification): An Integrated Sensor and Computer Vision-Based System for Rural Blind Curve Roads Lane Safety and Accident Detection
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Extensibility	We can easily put this same setup on other streets or connect it to new city upgrades later on.			✓		

Recommendations:

PHINMA University of Pangasinan Arellano St,.
Dagupan City, Pangasinan

LUMEN (Lane-Use Monitoring for Event Notification): An Integrated Sensor and Computer Vision-Based System for Rural Blind Curve Roads Lane Safety and Accident Detection
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Extensibility	We can easily put this same setup on other streets or connect it to new city upgrades later on.	✓				

Recommendations:

PHINMA University of Pangasinan Arellano St,.
Dagupan City, Pangasinan

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Economic Feasibility	It doesn't cost too much money to set up or to fix.			✓		
Extensibility	We can easily put this same setup on other streets or connect it to new city upgrades later on.		✓			

Recommendations:

PHINMA University of Pangasinan Arellano St,.
Dagupan City, Pangasinan

LUMEN (Lane-Use Monitoring for Event Notification): An Integrated Sensor and Computer Vision-Based System for Rural Blind Curve Roads Lane Safety and Accident Detection
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Type of Respondent: ☑ Driver ☐ Traffic/Safety Personnel

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Sustainability	It saves power, doesn't hurt the environment, and will work for a very long time.		✓			
Economic Feasibility	It doesn't cost too much money to set up or to fix.			✓		
Extensibility	We can easily put this same setup on other streets or connect it to new city upgrades later on.		✓			

Recommendations:

PHINMA University of Pangasinan Arellano St,.
Dagupan City, Pangasinan

LUMEN (Lane-Use Monitoring for Event Notification): An Integrated Sensor and Computer Vision-Based System for Rural Blind Curve Roads Lane Safety and Accident Detection
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Name of Respondent: _____(OPTIONAL)
Type of Respondent: ☑ Driver ☐ Traffic/Safety Personnel

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Extensibility	We can easily put this same setup on other streets or connect it to new city upgrades later on.	✓				

Recommendations:

PHINMA University of Pangasinan Arellano St,.
Dagupan City, Pangasinan

LUMEN (Lane-Use Monitoring for Event Notification): An Integrated Sensor and Computer Vision-Based System for Rural Blind Curve Roads Lane Safety and Accident Detection
Evaluation Sheet

Name of Respondent: RICA MAE ABLUENCIA (OPTIONAL)
Type of Respondent: ☐ Driver ☐ Traffic/Safety Personnel

☐ Authority ☑ Pedestrian

Instructions: Evaluate the system by placing a check (✓) in the box that corresponds to your assessments of each criterion.
Scale:

- **5 - Outstanding**
- **4 - Above Satisfactory**
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Criteria	Description	5	4	3	2	1
Functionality	It watches the road, spots trouble, and changes the lights exactly when it needs to.		✓			
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Sustainability	It saves power, doesn't hurt the environment, and will work for a very long time.		✓			
Economic Feasibility	It doesn't cost too much money to set up or to fix.			✓		
Extensibility	We can easily put this same setup on other streets or connect it to new city upgrades later on.		✓			

Recommendations:

PUT IT ON A SPOT WHERE EVERYONE CAN NOTICE IT.

PHINMA University of Pangasinan Arellano St,.
Dagupan City, Pangasinan

LUMEN (Lane-Use Monitoring for Event Notification): An Integrated Sensor and Computer Vision-Based System for Rural Blind Curve Roads Lane Safety and Accident Detection
Evaluation Sheet

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Recommendations:

PHINMA University of Pangasinan Arellano St,.
Dagupan City, Pangasinan

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Recommendations:

PHINMA University of Pangasinan Arellano St,.
Dagupan City, Pangasinan

LUMEN (Lane-Use Monitoring for Event Notification): An Integrated Sensor and Computer Vision-Based System for Rural Blind Curve Roads Lane Safety and Accident Detection
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PHINMA University of Pangasinan Arellano St,.
Dagupan City, Pangasinan

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PHINMA University of Pangasinan Arellano St.
Dagupan City, Pangasinan

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Recommendations:

ted with CamScanner

PHINMA University of Pangasinan Arellano St,.
Dagupan City, Pangasinan

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Recommendations:

d with CamScanner

PHINMA University of Pangasinan Arellano St,.
Dagupan City, Pangasinan

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Recommendations:

ith CamScanner

PHINMA University of Pangasinan Arellano St,.
Dagupan City, Pangasinan

LUMEN (Lane-Use Monitoring for Event Notification): An Integrated Sensor and Computer Vision-Based System for Rural Blind Curve Roads Lane Safety and Accident Detection
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☐ Authority ☑ Pedestrian

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Scale:

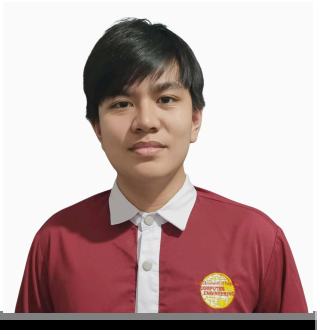
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Extensibility	We can easily put this same setup on other streets or connect it to new city upgrades later on.		✓			

Recommendations:

Appendix B

CURRICULUM VITAE



Carl Justin S. De Leon

Email: casa.deleon.up@phinmaed.com
Mobile: +63 963 472 4591

Education

PHINMA - University of Pangasinan | B.S. Computer Engineering 2021 - Present
Electrical Installation and Maintenance (EIM) NC II 2024

Experience

On-the-Job-Training
DOST Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA)

1st SIRIB Research Colloquium

Projects

Smart Home
Developed a miniature smart home with basic automation and IoT control.

Sound Sensor System (Microphone + LEDs)
Built a sound-reactive LED system using a microphone sensor.

IoT Water Level Monitoring System
Created a water level monitoring system with real-time mobile tracking via Blynk

Skills

Programming Languages: C++, Basic Python, Ladder Diagram (LD)
Electronics & Hardware: Circuit Design and Prototyping
Development Tools: Arduino IDE, Thonny IDE
Networking: Configuration, IP Addressing and subnetting
Other Skills: Troubleshooting and debugging, Basic System Integration

CURRICULUM VITAE



Brad Gervin D. Eslabra

Email: brde.eslabra@phinmaed.com
Mobile: +63 967 210 6913

Education

PHINMA - University of Pangasinan | B.S. Computer Engineering 2021 - Present

Experience

On-the-job Training (OJT)
Layad Circuits
-Hands-on experience in electronics design and PCB development
- Experience working on real-world engineering projects
- Improved problem-solving and troubleshooting skills
- Familiarity with industry tools and workflows

1st SIRIB Research Colloquium

Date

Projects

ParQ
Designed and implemented a relational database system for managing parking slots, users, and transactions

Smart UPang
Designed and developed a miniature of University of Pangasinan with a phone-controlled light switch and garage door opener with ESP32 Microcontroller.

Arduino Training Kit
Designed a Basic Training Kit using Arduino.

Lam-Ang Chronicles
Developed a Developed a 2D game using Godot Engine, inspired by a traditional Filipino epic

`

CURRICULUM VITAE

Ric Ardeil Sean C. Estrada

Email: rico.estrada.up@phinmaed.com
Mobile: +63 951 311 0254



Education

PHINMA - University of Pangasinan | B.S. Computer Engineering 2021 - Present

Experience

1st SIRIB Research Colloquium

Projects

LucidOasis
Developed a 2D Horror/Funny game using Godot.

Smart Home
Designed and developed a miniature home with an automatic garage door opener with ESP32 Microcontroller.

Arduino Training Kit
Designed a Basic Training Kit using Arduino.

Line Following Robot
Designed and developed a Robot with Logic Gates Implementation.

Power Supply
Created a standard power supply.

Skills

Hardware: Breadboarding, Arduino, Logic Circuit Design
Development Tools: Microsoft Visual Studio, Godot
Networking: Configuration
Other Skills: Problem-Solving, Collaboration, Adaptability

CURRICULUM VITAE

Richell Ann C. Gabis

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Education

PHINMA - University of Pangasinan | B.S. Computer Engineering 2021 - Present

Experience

On-the-job Training (OJT)
DOST PAGASA

1st SIRIB Research Colloquim Date

Projects

Transport Management System
Designed and developed a transport management system for tracking vehicles, routes, and schedules

Space Hunter
Developed a deck-building strategy game featuring card-based combat and progression mechanics

ParQ
Designed and implemented a relational database system for managing parking slots, users, and transactions

Smart Home
Designed and developed a miniature home with an automatic garage door opener with ESP32 Microcontroller.

Arduino Training Kit
Designed a Basic Training Kit using Arduino

CURRICULUM VITAE



Mitch T. Torio

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Education

PHINMA - University of Pangasinan | B.S. Computer Engineering 2021 - Present

Experience

1st SIRIB Research Colloquium

On-the-job Training (OJT) 2025
DOST PAGASA

Projects

Power Supply
Designed and built a power supply for circuit testing, ensuring stable voltage output for electronics applications.

2D Shooter Game
Developed a shooting game using Godot

Smart Home
Designed and developed a miniature home with an automatic garage door opener with ESP32 Microcontroller.

Arduino Training Kit
Designed a Basic Training Kit using Arduino.

Skills

Hardware: Breadboarding, Arduino, Logic Circuit Design
Programming Languages Python, Ladder Diagram (LD)
CAD: Autodesk AutoCAD
Other Skills: Problem-Solving, Adaptability

Appendix C

CODES

ESP32 Code

```
#include <Arduino.h>
#include
<Adafruit_NeoPixel.h>

//
==================
CHANGE PER NODE
==================
#define NODE_ID 1
//
==================
==================
==============

// ---------- HC12 ----------
#define HC12 Serial1
#define HC12_RX_PIN 16
#define HC12_TX_PIN 17
#define HC12_BAUD
9600

// ---------- LD2451
----------
HardwareSerial
RADAR(2);
#define RADAR_RX_PIN
26
#define RADAR_TX_PIN
27
#define RADAR_BAUD
256000

// ---------- NeoMatrix
----------
#define MATRIX_PIN 19
#define W 8
#define H 8
#define NUM_PIXELS
(W * H)
#define BRIGHTNESS
255

Adafruit_NeoPixel
matrix(NUM_PIXELS,
MATRIX_PIN,
NEO_GRB +
NEO_KHZ800);

uint16_t XY(uint8_t x,
uint8_t y) {
  if (y % 2 == 0) return y *
W + x;
  return y * W + (W - 1 -
x);
}

const uint8_t DISK[8][8]
= {
  {0,0,1,1,1,1,0,0},
  {0,1,1,1,1,1,1,0},
  {1,1,1,1,1,1,1,1},
  {1,1,1,1,1,1,1,1},
  {1,1,1,1,1,1,1,1},
  {1,1,1,1,1,1,1,1},
  {0,1,1,1,1,1,1,0},
  {0,0,1,1,1,1,0,0}
};

uint32_t white() {
  return matrix.Color(255,
255, 255);
}

void clearMatrix() {
  for (int i = 0; i <
NUM_PIXELS; i++) {
    matrix.setPixelColor(i,
0);
  }
  matrix.show();
}

void drawDisk(uint32_t c)
{
  for (int y = 0; y < H;
y++) {
    for (int x = 0; x < W;
x++) {

matrix.setPixelColor(XY(x
, y), DISK[y][x] ? c : 0);
    }
  }
  matrix.show();
}

// ---------- Blink control
----------
bool blinking = false;
bool blinkState = false;
uint32_t blinkEnd = 0;
uint32_t lastToggle = 0;
```

```
const uint32_t
BLINK_TOTAL_MS =
5000;
const uint32_t
BLINK_INTERVAL_MS
= 1000;

void startBlink() {
  uint32_t now = millis();

  if (!blinking) {
    blinking = true;
    blinkState = true;
    lastToggle = now;
    drawDisk(white());
  }

  // Extend warning
seamlessly
  blinkEnd = now +
BLINK_TOTAL_MS;
}

void stopBlink() {
  blinking = false;
  blinkState = false;
  clearMatrix();
}

void updateBlink() {
  if (!blinking) return;

  uint32_t now = millis();

  if ((int32_t)(now -
blinkEnd) >= 0) {
    stopBlink();
    return;
  }
  if (now - lastToggle >=
BLINK_INTERVAL_MS)
{
    lastToggle = now;
    blinkState = !blinkState;
    drawDisk(blinkState ?
white() : 0);
  }
}

// ---------- Command
parser ----------
bool
isNumericString(const
String& s) {
  if (s.length() == 0) return
false;
  for (int i = 0; i <
s.length(); i++) {
    if (!isDigit(s[i])) return
false;
  }
  return true;
}

void handleHC12(String
msg) {
  msg.trim();
  if (msg.length() == 0)
return;

  int sep =
msg.indexOf(':');
  if (sep == -1) return;

  String idPart =
msg.substring(0, sep);
  String cmd =
msg.substring(sep + 1);
  idPart.trim();
  cmd.trim();
  if
(!isNumericString(idPart))
return;

  int target = idPart.toInt();
  if (!(target == NODE_ID
|| target == 0)) return;

  if (cmd ==
"BLINK_YELLOW") {
    startBlink();
    HC12.println("ACK
Node " +
String(NODE_ID) + "
BLINK_YELLOW");
  }
  else if (cmd == "OFF") {
    stopBlink();
    HC12.println("ACK
Node " +
String(NODE_ID) + "
OFF");
  }
  else if (cmd == "PING")
{
    delay(NODE_ID * 60);
    HC12.println("ACK
Node " +
String(NODE_ID) + "
PONG");
  }
}

// ---------- Radar parser
----------
static const uint8_t
HDR[4]  = {0xF4, 0xF3,
0xF2, 0xF1};
static const uint8_t
TAIL[4] = {0xF8, 0xF7,
0xF6, 0xF5};
```

```
enum ParseState {
SEEK_HDR,
READ_LEN,
READ_PAYLOAD,
READ_TAIL };
ParseState state =
SEEK_HDR;

uint8_t hdrIdx = 0;
uint8_t tailIdx = 0;
uint16_t payloadLen = 0;
uint16_t payloadRead = 0;
uint8_t payload[128];

const uint8_t
MAX_INCOMING_TO_
REPORT = 3;
const uint32_t
DATA_INTERVAL_MS =
250;
uint32_t lastDataSent = 0;

struct VehicleInfo {
  uint8_t angleRaw;
  uint8_t distM;
  uint8_t dirRaw;
  uint8_t speedKmh;
  uint8_t snr;
};

VehicleInfo
incomingVehicles[MAX_I
NCOMING_TO_REPORT
];
uint8_t incomingCount =
0;

void
collectIncomingVehicles(u
int8_t targets) {
  incomingCount = 0;
  for (uint8_t i = 0; i <
targets; i++) {
    uint16_t idx = 2 + i * 5;
    if (idx + 4 >=
payloadLen) break;

    uint8_t angle =
payload[idx + 0];
    uint8_t dist  =
payload[idx + 1];
    uint8_t dir   =
payload[idx + 2];
    uint8_t speed =
payload[idx + 3];
    uint8_t snr   =
payload[idx + 4];

    if (dir != 1) continue;

    if (incomingCount <
MAX_INCOMING_TO_
REPORT) {

incomingVehicles[incomin
gCount].angleRaw =
angle;

incomingVehicles[incomin
gCount].distM = dist;

incomingVehicles[incomin
gCount].dirRaw = dir;

incomingVehicles[incomin
gCount].speedKmh =
speed;

incomingVehicles[incomin
gCount].snr = snr;
      incomingCount++;
    }
  }
}

void sendData() {
  uint32_t now = millis();
  if (now - lastDataSent <
DATA_INTERVAL_MS)
return;
  lastDataSent = now;

  String out = "DATA:" +
String(NODE_ID)
          + ",incoming=" +
String(incomingCount);

  for (uint8_t i = 0; i <
incomingCount; i++) {
    out += ";V" + String(i +
1)
        + ":a=" +
String(incomingVehicles[i]
.angleRaw)
        + ",d=" +
String(incomingVehicles[i]
.distM)
        + ",dir=" +
String(incomingVehicles[i]
.dirRaw)
        + ",s=" +
String(incomingVehicles[i]
.speedKmh)
        + ",snr=" +
String(incomingVehicles[i]
.snr);
  }

  HC12.println(out);
}

void handleFrame() {
  if (payloadLen < 2)
return;
```

```
  uint8_t targets =
payload[0];

collectIncomingVehicles(t
argets);
  sendData();
}

void
feedRadarByte(uint8_t b)
{
  switch (state) {
    case SEEK_HDR:
      if (b == HDR[hdrIdx])
{
        hdrIdx++;
        if (hdrIdx == 4) {
          hdrIdx = 0;
          state = READ_LEN;
        }
      } else {
        hdrIdx = (b ==
HDR[0]) ? 1 : 0;
      }
      break;

    case READ_LEN: {
      static uint8_t
lenBytes[2];
      static uint8_t lenIdx =
0;

      lenBytes[lenIdx++] =
b;

      if (lenIdx == 2) {
        lenIdx = 0;
        payloadLen =
(uint16_t)lenBytes[0] |
((uint16_t)lenBytes[1] <<
8);
        if (payloadLen >
sizeof(payload)) {
          state = SEEK_HDR;
        } else if (payloadLen
== 0) {
          tailIdx = 0;
          state =
READ_TAIL;
        } else {
          payloadRead = 0;
          state =
READ_PAYLOAD;
        }
      }
      break;
    }

    case
READ_PAYLOAD:

payload[payloadRead++]
= b;
      if (payloadRead >=
payloadLen) {
        tailIdx = 0;
        state = READ_TAIL;
      }
      break;

    case READ_TAIL:
      if (b == TAIL[tailIdx])
{
        tailIdx++;
        if (tailIdx == 4) {
          handleFrame();
          state = SEEK_HDR;
          tailIdx = 0;
        }
      } else {
        state = SEEK_HDR;
        tailIdx = 0;
      }
      break;
  }
}

void setup() {
  Serial.begin(115200);
  delay(200);


HC12.begin(HC12_BAUD
, SERIAL_8N1,
HC12_RX_PIN,
HC12_TX_PIN);

RADAR.begin(RADAR_
BAUD, SERIAL_8N1,
RADAR_RX_PIN,
RADAR_TX_PIN);

  matrix.begin();

matrix.setBrightness(BRI
GHTNESS);
  clearMatrix();

  HC12.println("Node " +
String(NODE_ID) + "
ONLINE");
}

void loop() {
  while
(RADAR.available()) {

feedRadarByte((uint8_t)R
ADAR.read());
  }

  if (HC12.available()) {
    String msg =
HC12.readStringUntil('\n')
;
```

```
  handleHC12(msg);                                                    }
}                                   updateBlink();
```

Raspberry Pi 5 Code

```
import serial
import time
import sys
import select
import threading
import json
import csv
import os
import base64
from collections import deque
from datetime import datetime
import paho.mqtt.client as mqtt
import numpy as np
import cv2
import psutil
from ultralytics import YOLO

# ============================
# SERIAL CONFIG
# ============================
SERIAL_PORT = "/dev/ttyAMA0"
BAUD_RATE   = 9600

# ============================
# MQTT CONFIG
# ============================
MQTT_BROKER = "100.100.146.107"
MQTT_PORT   = 1883

# ============================
# VISION CONFIG
# ============================
MODEL_PATH          = "/home/bard/lumen/bestest_ncnn_model"
RTSP_URL            = "rtsp://localhost:8554/cam"
CLASS_NAMES         = ["Accident"]
INPUT_SIZE          = 640
VISION_LOOP_DELAY   = 0.01
CONF_THRESHOLD      = 0.60
ACCIDENT_MIN_CONFIDENCE  = 0.60
ACCIDENT_MIN_BOX_AREA    = 200
ACCIDENT_REQUIRED_STREAK = 3
ACCIDENT_MISS_TOLERANCE  = 8
ACCIDENT_AVG_CONF_MIN    = 0.40
ACCIDENT_COOLDOWN        = 10.0
VISION_BLINK_INTERVAL    = 4.0
ACCIDENT_VISION_TIMEOUT  = 60.0

# ============================
# RECORDING CONFIG
# ============================
RECORDINGS_DIR     = "/home/bard/lumen/recordings"
PRE_BUFFER_SEC     = 5
POST_BUFFER_SEC    = 10
RECORD_FPS         = 10
RECORD_FRAME_SIZE  = (640, 640)
MAX_RECORDING_FILES = 50

# ============================
# LOGGING CONFIG
# ============================
```

```
CONSOLE_LOG_LEVEL = 0
LOG_DATE = datetime.now().strftime("%Y-%m-%d")
RAW_LOG_FILE = f"raw_radar_{LOG_DATE}.csv"
DECISION_LOG_FILE = f"decision_{LOG_DATE}.csv"
SUMMARY_LOG_FILE = f"summary_{LOG_DATE}.csv"
VISION_LOG_FILE = f"vision_{LOG_DATE}.csv"
EVENT_COUNTER_FILE = "event_counter.json"

# ============================================
# GLOBALS
# ============================================
running = True
start_time = time.time()
event_counter = 0
last_combined_alert_time = 0.0
confirm_updates = 1
clear_zone_delay = 1
warning_cooldown = 1
combined_alert_cooldown = 3.0

# Locks
serial_lock = threading.Lock()
print_lock = threading.Lock()
state_lock = threading.Lock()
vision_lock = threading.Lock()
csv_lock = threading.Lock()
event_lock = threading.Lock()
recording_latch_lock = threading.Lock()
combined_alert_lock = threading.Lock()
pre_buffer_lock = threading.Lock()
live_frame_lock = threading.Lock()
active_recording_lock = threading.Lock()
last_img_b64_lock = threading.Lock()

# Shared state
pre_buffer = deque(maxlen=PRE_BUFFER_SEC * RECORD_FPS)
accident_recording_event = threading.Event()
live_frame_for_recording = None
recording_latch = False
active_recording_file = None
last_accident_img_b64 = ""  # most recent encoded accident frame
ser = None

vision_state = {
    "accident_confirmed": False,
    "accident_confidence": 0.0,
    "accident_box": None,
    "last_publish_time": 0.0,
    "fps": 0,
    "inference_ms": 0.0,
    "last_logged_confirmed": False,
    "streak_confidences": [],
    "miss_count": 0,
}

node_state = {
    node: {
        "incoming_count": 0,
        "vehicles": [],
        "confirm_count": 0,
        "active_event": False,
        "last_seen": 0.0,
        "last_warn_sent": 0.0,
        "last_payload": "",
        "event_id": "",
        "event_start_time": None,
        "max_incoming_count": 0,
        "peak_speed_kmh": 0.0,
    }
    for node in (1, 2)
}

mqtt_client = mqtt.Client(mqtt.CallbackAPIVersion.VERSION2)

# ============================================
# PRINTING
# ============================================
def safe_print(*args, **kwargs):
    with print_lock:
        print(*args, **kwargs)

def log_print(level, *args, **kwargs):
    if CONSOLE_LOG_LEVEL
```

```
>= level:
        safe_print(*args,
**kwargs)

def show_prompt():
    with print_lock:

sys.stdout.write("LUMEN
> ")
        sys.stdout.flush()

#
==================
======
# CSV LOGGING
#
==================
======
def csv_timestamp():
    return
datetime.now().strftime("
%Y-%m-%d
%H:%M:%S.%f")

def init_csv_file(path,
headers):
    if not
os.path.exists(path):
        with open(path, "w",
newline="",
encoding="utf-8") as f:

csv.writer(f).writerow(hea
ders)

def init_csv_logs():

init_csv_file(RAW_LOG_
FILE, [
        "timestamp", "node",
"incoming_count",
        "v1_angle",
"v1_dist", "v1_speed",
"v1_dir", "v1_snr",
        "v2_angle",
"v2_dist", "v2_speed",
"v2_dir", "v2_snr",
        "v3_angle",
"v3_dist", "v3_speed",
"v3_dir", "v3_snr",
        "raw_line",
    ])

init_csv_file(DECISION_
LOG_FILE, ["timestamp",
"event_id", "node",
"action", "details"])

init_csv_file(SUMMARY
_LOG_FILE,  ["event_id",
"start_time", "end_time",
"detector_node",

"warned_node",
"duration_sec",
"max_incoming_count",

"peak_speed_kmh"])

init_csv_file(VISION_LO
G_FILE,   ["timestamp",
"event_id", "status",
"confidence",

"box", "details"])

def log_raw_radar(node,
incoming_count, vehicles,
raw_line):
    row = [csv_timestamp(),
node, incoming_count]
    for i in range(3):
        v = vehicles[i] if i <
len(vehicles) else {}

row.extend([v.get("angle_d
eg",""),
v.get("distance_m",""),

v.get("speed_kmh",""),
v.get("dir_raw",""),
v.get("snr","")])
    row.append(raw_line)
    with csv_lock:
        with
open(RAW_LOG_FILE,
"a", newline="",
encoding="utf-8") as f:

csv.writer(f).writerow(row
)

def log_decision(event_id,
node, action, details=""):
    with csv_lock:
        with
open(DECISION_LOG_FI
LE, "a", newline="",
encoding="utf-8") as f:

csv.writer(f).writerow([csv
_timestamp(), event_id,
node, action, details])

def
log_summary(event_id,
start_time_val, end_time,
detector_node,
                warned_node,
max_incoming_count,
peak_speed_kmh):
    with csv_lock:
        with
open(SUMMARY_LOG_
FILE, "a", newline="",
encoding="utf-8") as f:

csv.writer(f).writerow([
                event_id,

datetime.fromtimestamp(st
art_time_val).strftime("%
Y-%m-%d
%H:%M:%S.%f"),

datetime.fromtimestamp(e
nd_time).strftime("%Y-%
m-%d %H:%M:%S.%f"),
                detector_node,
warned_node,
                round(end_time
- start_time_val, 3),

max_incoming_count,
```

```
round(peak_speed_kmh,
2),
        ])

def log_vision(event_id,
status, confidence=0.0,
box=None, details=""):
    with csv_lock:
        with
open(VISION_LOG_FILE
, "a", newline="",
encoding="utf-8") as f:

csv.writer(f).writerow([
            csv_timestamp(),
event_id, status,

round(float(confidence), 4)
if confidence != "" else "",
            json.dumps(box)
if box else "", details,
        ])

#
==================
======
# EVENT COUNTER
#
==================
======
def load_event_counter():
    try:
        if
os.path.exists(EVENT_CO
UNTER_FILE):
            with
open(EVENT_COUNTER
_FILE) as f:
                return
json.load(f).get("counter",
0)
    except Exception:
        pass
    return 0

def
save_event_counter(value)
:
    try:
        with
open(EVENT_COUNTER
_FILE, "w") as f:

json.dump({"counter":
value}, f)
    except Exception as e:

safe_print(f"[COUNTER]
Failed to save: {e}")

def next_event_id():
    global event_counter
    with event_lock:
        event_counter += 1

save_event_counter(event
_counter)
        return
f"EVT{event_counter:04d
}"

#
==================
======
# STATE HELPERS
#
==================
======
def
get_active_event_id_safe()
:
    with state_lock:
        active = [

(node_state[n]["last_seen"
],
node_state[n]["event_id"])
            for n in (1, 2)
            if
node_state[n]["active_eve
nt"] and
node_state[n]["event_id"]
        ]
    return sorted(active,
reverse=True)[0][1] if
active else ""

def
get_node_snapshot(node):
    with state_lock:
        return
dict(node_state[node])

#
==================
======
# MQTT
#
==================
======
def on_disconnect(client,
userdata, rc,
properties=None):
    safe_print("[MQTT]
Disconnected,
reconnecting...")
    while running:
        try:
            client.reconnect()

safe_print("[MQTT]
Reconnected")
            break
        except Exception:
            time.sleep(3)

mqtt_client.on_disconnect
= on_disconnect

def publish_node(node):
    snap  =
get_node_snapshot(node)
    topic =
f"traffic/node{node}"
    if
snap["incoming_count"]
== 0:
        payload = "CLEAR
0.00 0.00"
    else:
        v       =
snap["vehicles"][0]
        status  =
"WARNING" if
```

```
snap["active_event"] else
"CLEAR"
        payload = f"{status}
{v['distance_m']:.2f}
{v['speed_kmh']:.2f}"

    with state_lock:
        if payload ==
node_state[node]["last_pa
yload"]:
            return

node_state[node]["last_pa
yload"] = payload


mqtt_client.publish(topic,
payload, qos=0)
    log_print(2, f"[MQTT]
[{topic}]: {payload}")

def publish_rpi_stats():
    try:
        with
open("/sys/class/thermal/th
ermal_zone0/temp") as f:
            temp =
f.read().strip()

mqtt_client.publish("syste
m/rpi", json.dumps({
            "cpu":
psutil.cpu_percent(),
            "ram":
psutil.virtual_memory().pe
rcent,
            "temp": temp,
        }), qos=0)
    except Exception as e:
        safe_print(f"[STATS]
Error: {e}")

def
maybe_publish_combined
_alert():
    global
last_combined_alert_time
    with
combined_alert_lock:
        with state_lock:
            radar_active =
node_state[1]["active_eve
nt"] or
node_state[2]["active_eve
nt"]
        with vision_lock:
            confirmed  =
vision_state["accident_con
firmed"]
            confidence =
vision_state["accident_con
fidence"]
        now = time.time()
        if radar_active and
confirmed and now -
last_combined_alert_time
>=
combined_alert_cooldown
:
            with
last_img_b64_lock:
                img_b64 =
last_accident_img_b64

mqtt_client.publish("lume
n/alert", json.dumps({
                "event":
"HIGH_PRIORITY_ACCI
DENT",
                "confidence":
round(confidence, 2),
                "timestamp":
time.strftime("%H:%M:%
S"),
                "image":
img_b64,
            }), qos=0)

log_decision(get_active_e
vent_id_safe(), 0,
"HIGH_PRIORITY_ACCI
DENT",

f"confidence={round(confi
dence, 2)}")

last_combined_alert_time
= now

#
==================
======
# SERIAL HELPERS
#
==================
======
def send_raw(msg):
    global ser
    with serial_lock:
        if ser is not None:
            ser.write((msg +
"\n").encode("utf-8"))
    safe_print(f"\n[TX]
{msg}")
    show_prompt()

def send_cmd(node, cmd):

send_raw(f"{node}:{cmd}
")

def send_cmd_both(cmd):
    global ser
    with serial_lock:
        if ser is not None:

ser.write(f"1:{cmd}\n2:{c
md}\n".encode("utf-8"))
    safe_print(f"\n[TX]
1:{cmd}  2:{cmd}")
    show_prompt()

def
warn_opposite(source_nod
e):
    now        = time.time()
    target_node = 2 if
source_node == 1 else 1
    with state_lock:
        if now -
node_state[source_node]["
last_warn_sent"] <
warning_cooldown:
            return
```

```
node_state[source_node]["
last_warn_sent"] = now
        event_id =
node_state[source_node]["
event_id"]

    safe_print(f"\n[WARN]
Node {source_node} →
BLINK_YELLOW on
Node {target_node}")
    send_cmd(target_node,
"BLINK_YELLOW")
    if event_id:

log_decision(event_id,
source_node,
"BLINK_YELLOW",
f"sent to node
{target_node}")

#
==================
======
# DATA PARSING
#
==================
======
def
parse_vehicle_part(part):
    _, data = part.split(":",
1)
    m = {k.strip(): v.strip()
for item in data.split(",")
        if "=" in item for k, v
in [item.split("=", 1)]}
    a = int(m.get("a", 0))
    return {
        "angle_raw":  a,
        "angle_deg":  a - 128,
        "distance_m":
int(m.get("d",   0)),
        "dir_raw":
int(m.get("dir", -1)),
        "speed_kmh":
int(m.get("s",   0)),
        "snr":
int(m.get("snr", 0)),
    }

def
process_node_event_logic
(node):
    now           =
time.time()
    event_confirmed =
False
    event_id_for_log = ""
    confirmed_count  = 0

    with state_lock:
        s       =
node_state[node]
        incoming =
s["incoming_count"] > 0

        if incoming:
            s["last_seen"]
= now

s["max_incoming_count"]
=
max(s["max_incoming_co
unt"],
s["incoming_count"])
            if s["vehicles"]:

s["peak_speed_kmh"] =
max(s["peak_speed_kmh"]
,

max(v["speed_kmh"] for v
in s["vehicles"]))

        if not
s["active_event"]:
            if not incoming:

s["confirm_count"] = 0
            elif
s["confirm_count"] <
confirm_updates:

s["confirm_count"] += 1

        if not
s["active_event"] and
s["confirm_count"] >=
confirm_updates:
            s["active_event"]
= True
            s["confirm_count"]
= 0
            s["event_id"]
= next_event_id()

s["event_start_time"]   =
now

s["max_incoming_count"]
= s["incoming_count"]

s["peak_speed_kmh"]     =
max((v["speed_kmh"] for
v in s["vehicles"]),
default=0.0)
            event_confirmed
= True
            event_id_for_log
= s["event_id"]
            confirmed_count
= s["incoming_count"]

        active_after  =
s["active_event"]
        incoming_after =
s["incoming_count"] > 0

    if event_confirmed:

safe_print(f"\n[EVENT]
Node {node} ACTIVE |
ID: {event_id_for_log}")

log_decision(event_id_for
_log, node,
"EVENT_CONFIRMED",

f"incoming_count={confir
med_count}")
        warn_opposite(node)

    if active_after and
incoming_after:
```

```
        warn_opposite(node)

    publish_node(node)

def parse_data(line):
    try:
        parts  = line.split(";")
        header =
parts[0].replace("DATA:",
"", 1).split(",")
        node   =
int(header[0].strip())
        if node not in (1, 2):
            raise
ValueError(f"Invalid node:
{node}")

        incoming_count =
int(header[1].split("=")[1].
strip())
        vehicles       =
[parse_vehicle_part(p) for
p in parts[1:] if ":" in p]

        with state_lock:

node_state[node]["incomin
g_count"] =
incoming_count

node_state[node]["vehicles
"]        = vehicles

        log_raw_radar(node,
incoming_count, vehicles,
line)

process_node_event_logic
(node)

        if
CONSOLE_LOG_LEVEL
>= 2:
            snap =
get_node_snapshot(node)

safe_print(f"\n[NODE
{node}]
incoming={incoming_cou
nt} "

f"confirm={snap['confirm
_count']}/{confirm_update
s} "

f"active={snap['active_eve
nt']}")
            for i, v in
enumerate(vehicles, 1):
                safe_print(f"
V{i} | {v['distance_m']}m
| {v['speed_kmh']}km/h | "

f"{'Approaching' if
v['dir_raw']==1 else
'Unknown'} |
SNR:{v['snr']}")
            show_prompt()

    except Exception as e:

safe_print(f"\n[PARSE
ERROR] {line} — {e}")
        show_prompt()

def handle_line(line):
    line = line.strip()
    if not line:
        return
    if
line.startswith("DATA:"):
        log_print(2, f"\n[RX]
{line}")
        parse_data(line)
    elif
line.startswith("ACK "):
        log_print(2, f"\n[RX]
{line}")
        show_prompt()
    elif "ONLINE" in line:
        log_print(1, f"\n[RX]
{line}")
        show_prompt()
    else:
        log_print(1,
f"\n[INFO] Unhandled:
{line}")
        show_prompt()

def print_config():

safe_print("\n=========
= CONFIG
==========")

safe_print(f"confirm_upda
tes         =
{confirm_updates}")

safe_print(f"clear_zone_de
lay         =
{clear_zone_delay}")

safe_print(f"warning_cool
down        =
{warning_cooldown}")

safe_print(f"console_log_l
evel        =
{CONSOLE_LOG_LEVE
L}")

safe_print(f"accident_requ
ired_streak =
{ACCIDENT_REQUIRE
D_STREAK}")

safe_print(f"accident_miss
_tolerance  =
{ACCIDENT_MISS_TOL
ERANCE}")

safe_print(f"accident_visio
n_timeout   =
{ACCIDENT_VISION_TI
MEOUT}")

safe_print(f"recordings_dir
=
{RECORDINGS_DIR}")

safe_print(f"pre_buffer_se
c           =
{PRE_BUFFER_SEC}")
```

```
safe_print(f"post_buffer_sec        = {POST_BUFFER_SEC}")

safe_print("==========================\n")
    show_prompt()

# ============================
# RECORDING
# ============================
def cleanup_old_recordings():
    if not os.path.exists(RECORDINGS_DIR):
        return
    with active_recording_lock:
        active = active_recording_file
    files = sorted(

(os.path.getmtime(os.path.join(RECORDINGS_DIR, f)),

os.path.join(RECORDINGS_DIR, f))
        for f in os.listdir(RECORDINGS_DIR)
        if f.lower().endswith(".mp4")
    )
    for _, path in files[:-MAX_RECORDING_FILES]:
        if active and os.path.abspath(path) == os.path.abspath(active):
            continue
        try:
            os.remove(path)
            log_print(1, f"[RECORD] Deleted: {os.path.basename(path)}")
        except Exception as e:

safe_print(f"[RECORD] Delete failed: {e}")

# ============================
# THREADS
# ============================
def open_serial():
    global ser
    while running:
        try:

safe_print(f"[INFO] Opening {SERIAL_PORT}...")
            ser = serial.Serial(SERIAL_PORT, BAUD_RATE, timeout=0.1)
            time.sleep(2)
            safe_print("[INFO] Serial opened OK")
            return True
        except Exception as e:

safe_print(f"[INFO] Serial failed: {e} — retrying in 2s")
            time.sleep(2)
    return False

def base_thread():
    global ser
    if not open_serial():
        return
    safe_print("[INFO] LUMEN Base Ready")
    safe_print("[INFO] Commands: confirm <n> | clear <s> | cooldown <s> | "
               "loglevel 0|1|2 | config | simulate <1|2> | exit\n")
    print_config()
    show_prompt()

    last_stats = time.time()
    while running:
        try:
            if time.time() - last_stats > 10:

publish_rpi_stats()
                last_stats = time.time()

            with serial_lock:
                has_data = ser.in_waiting if ser else 0
            if has_data:
                with serial_lock:
                    line = ser.readline().decode("utf-8", errors="ignore")
                handle_line(line)

maybe_publish_combined_alert()

            time.sleep(0.01)

        except serial.SerialException as e:

safe_print(f"\n[BASE] Serial lost: {e} — reconnecting...")
            with serial_lock:
                try:
                    if ser:
                        ser.close()
```

```
            except
Exception:
                pass
            ser = None
        if not
open_serial():
            break
    except Exception as
e:

safe_print(f"\n[BASE
ERROR] {e}")
        time.sleep(1)

def cli_thread():
    global confirm_updates,
clear_zone_delay,
warning_cooldown,
CONSOLE_LOG_LEVEL
, running

    while running:
        try:
            if not
select.select([sys.stdin], [],
[], 0.1)[0]:
                continue
            cmd =
sys.stdin.readline().strip()

            if
cmd.startswith("confirm"):
                try:

confirm_updates =
int(cmd.split()[1])

safe_print(f"\n[CONFIG]
confirm_updates =
{confirm_updates}")

log_decision("", 0,
"CONFIG_CHANGE",
f"confirm_updates={confir
m_updates}")
                except
Exception:

safe_print("\nUsage:
confirm <number>")

            elif
cmd.startswith("clear"):
                try:

clear_zone_delay =
float(cmd.split()[1])

safe_print(f"\n[CONFIG]
clear_zone_delay =
{clear_zone_delay}")

log_decision("", 0,
"CONFIG_CHANGE",
f"clear_zone_delay={clear
_zone_delay}")
                except
Exception:

safe_print("\nUsage: clear
<seconds>")

            elif
cmd.startswith("cooldown
"):
                try:

warning_cooldown =
float(cmd.split()[1])

safe_print(f"\n[CONFIG]
warning_cooldown =
{warning_cooldown}")

log_decision("", 0,
"CONFIG_CHANGE",
f"warning_cooldown={wa
rning_cooldown}")
                except
Exception:

safe_print("\nUsage:
cooldown <seconds>")

            elif
cmd.startswith("loglevel"):
                try:
                    level =
int(cmd.split()[1])
                    if level not in
(0, 1, 2):
                        raise
ValueError

CONSOLE_LOG_LEVEL
= level

safe_print(f"\n[CONFIG]
console_log_level =
{CONSOLE_LOG_LEVE
L}")

log_decision("", 0,
"CONFIG_CHANGE",
f"console_log_level={CO
NSOLE_LOG_LEVEL}")
                except
Exception:

safe_print("\nUsage:
loglevel 0|1|2")

            elif cmd ==
"config":
                print_config()

            elif
cmd.startswith("simulate")
:
                try:

warn_opposite(int(cmd.spl
it()[1]))
                except
Exception:

safe_print("\nUsage:
simulate 1 or simulate 2")

            elif cmd in ("exit",
"quit"):

safe_print("\n[INFO]
Shutting down
```

```
LUMEN...")
            running = False
            break

        elif cmd:
            send_raw(cmd)

        show_prompt()

    except Exception as
e:
        safe_print(f"\n[CLI
ERROR] {e}")
        time.sleep(1)

def heartbeat_thread():
  while running:
    try:

mqtt_client.publish("syste
m/heartbeat",
json.dumps({
            "status":
"alive",
            "timestamp":
time.strftime("%H:%M:%
S"),
            "uptime":
round(time.time() -
start_time, 0),
        }), qos=0)
    except Exception as
e:

safe_print(f"[HEARTBEA
T] Error: {e}")
    time.sleep(15)

def recording_thread():
  global recording_latch,
active_recording_file

os.makedirs(RECORDIN
GS_DIR, exist_ok=True)
  with
recording_latch_lock:
    recording_latch =
False
accident_recording_event.
clear()

  while running:

accident_recording_event.
wait()

accident_recording_event.
clear()

cleanup_old_recordings()

    timestamp =
datetime.now().strftime("
%Y-%m-%d_%H-%M-%S
")
    filename  =
os.path.join(RECORDING
S_DIR,
f"accident_{timestamp}.m
p4")
    writer    =
cv2.VideoWriter(filename,
cv2.VideoWriter_fourcc(*
"avc1"),

RECORD_FPS,
RECORD_FRAME_SIZE
)

    if not
writer.isOpened():

safe_print("[RECORD]
avc1 failed, trying
XVID...")
        filename =
filename.replace(".mp4",
".avi")
        writer   =
cv2.VideoWriter(filename,
cv2.VideoWriter_fourcc(*
"XVID"),

RECORD_FPS,
RECORD_FRAME_SIZE
)
    if not
writer.isOpened():

safe_print("[RECORD]
ERROR: Cannot open
VideoWriter")
        continue

    with
active_recording_lock:

active_recording_file =
filename

safe_print(f"[RECORD]
Saving: {filename}")

    with pre_buffer_lock:
      pre_frames =
list(pre_buffer)
    for frame in
pre_frames:

writer.write(cv2.cvtColor(f
rame,
cv2.COLOR_RGB2BGR))

    post_end =
time.time() +
POST_BUFFER_SEC
    frames_written = 0
    last_hash = None
    while time.time() <
post_end and running:
        with
live_frame_lock:
            frame =
live_frame_for_recording.
copy() if
live_frame_for_recording
is not None else None
        if frame is not
None:
            h =
hash(frame.tobytes())
            if h != last_hash:
```

```
writer.write(cv2.cvtColor(f
rame,
cv2.COLOR_RGB2BGR))

frames_written += 1
                last_hash = h
          time.sleep(1.0 /
RECORD_FPS)

       writer.release()
       with
active_recording_lock:

active_recording_file =
None

       total =
len(pre_frames) +
frames_written

safe_print(f"[RECORD]
Saved {total} frames
({total/RECORD_FPS:.1f
}s) → {filename}")

log_decision(get_active_e
vent_id_safe(), 0,
"RECORDING_SAVED",

f"file={os.path.basename(f
ilename)}
pre={len(pre_frames)}
post={frames_written}")

def
clear_checker_thread():
   while running:
      try:
         now          =
time.time()
         clear_actions = []

         with state_lock:
            for node in (1,
2):
               s    =
node_state[node]
               idle = now -
s["last_seen"]
                  if not
s["active_event"] or idle
<= clear_zone_delay:
                     continue

clear_actions.append({
                     "node":
node, "event_id":
s["event_id"],

"start_time":
s["event_start_time"],

"warned_node": 2 if node
== 1 else 1,

"max_incoming_count":
s["max_incoming_count"],

"peak_speed_kmh":
s["peak_speed_kmh"],
"idle": idle,
                  })
                  s.update({

"active_event": False,
"incoming_count": 0,
"vehicles": [],

"confirm_count": 0,
"event_id": "",
"event_start_time": None,

"max_incoming_count": 0,
"peak_speed_kmh": 0.0,
                  })

            for item in
clear_actions:
               node, event_id =
item["node"],
item["event_id"]

safe_print(f"\n[RESET]
Node {node} cleared (idle
{item['idle']:.1f}s)")

log_decision(event_id,
node,
"EVENT_CLEARED",
f"idle_for={item['idle']:.2f
}s")
               if event_id and
item["start_time"]:

log_summary(event_id,
item["start_time"], now,
node,

item["warned_node"],
item["max_incoming_cou
nt"],

item["peak_speed_kmh"])

publish_node(node)
               show_prompt()

         time.sleep(0.10)
      except Exception as
e:

safe_print(f"\n[CLEAR
CHECKER ERROR]
{e}")
         time.sleep(1)

def vision_thread():
   global
live_frame_for_recording,
recording_latch,
last_accident_img_b64

   if not
os.path.exists(MODEL_P
ATH):

safe_print(f"[VISION]
FATAL: Model not found:
{MODEL_PATH}")
      return

   safe_print("[VISION]
Loading model...")
   model =
```

```
YOLO(MODEL_PATH)
    safe_print("[VISION]
Model loaded OK")

    safe_print(f"[VISION]
Connecting to RTSP:
{RTSP_URL}")
    cap =
cv2.VideoCapture(RTSP_
URL)

cap.set(cv2.CAP_PROP_B
UFFERSIZE, 1)

    for attempt in range(1,
11):
        if cap.isOpened():
            break

safe_print(f"[VISION]
RTSP not ready, retrying
({attempt}/10)...")
        time.sleep(2)
        cap =
cv2.VideoCapture(RTSP_
URL)

    if not cap.isOpened():
        safe_print("[VISION]
Failed to open RTSP
stream.")
        return

    safe_print("[VISION]
RTSP connected OK\n")

    fps_counter         = 0
    fps_timer           =
time.time()
    accident_streak     = 0
    miss_count          = 0
    streak_confs        = []
    consecutive_failures = 0
    MAX_FAILURES
= 30

    try:
        while running:
            # Drain buffer for
low latency
            drain_until =
time.time() + 0.05
            while time.time() <
drain_until:
                cap.grab()

            ret, frame =
cap.read()
            if not ret:

consecutive_failures += 1
                log_print(1,
f"[VISION] Frame fail
{consecutive_failures}/{M
AX_FAILURES}")
                if
consecutive_failures >=
MAX_FAILURES:

safe_print("[VISION]
Reconnecting RTSP...")
                    cap.release()
                    time.sleep(3)
                    cap =
cv2.VideoCapture(RTSP_
URL)

cap.set(cv2.CAP_PROP_B
UFFERSIZE, 1)

consecutive_failures = 0
                time.sleep(0.1)
                continue


consecutive_failures = 0
            frame_rgb =
cv2.cvtColor(frame,
cv2.COLOR_BGR2RGB)
            if
frame_rgb.shape[:2] !=
(INPUT_SIZE,
INPUT_SIZE):
                frame_rgb =
cv2.resize(frame_rgb,
(INPUT_SIZE,
INPUT_SIZE))

            frame_copy =
frame_rgb.copy()
            with
pre_buffer_lock:

pre_buffer.append(frame_c
opy)
            with
live_frame_lock:

live_frame_for_recording
= frame_copy

            # Inference
            t0      = time.time()
            results =
model(frame_rgb,
imgsz=INPUT_SIZE,
verbose=False)
            elapsed =
time.time() - t0

            detections = [
                {"conf":
float(box.conf[0]), "box":
list(map(int,
box.xyxy[0]))}
                for r in results
for box in r.boxes
                if
float(box.conf[0]) >=
CONF_THRESHOLD
            ]

            fps_counter += 1
            if time.time() -
fps_timer >= 1.0:
                with
vision_lock:

vision_state["fps"]         =
fps_counter

vision_state["inference_ms
"] = elapsed * 1000.0
                log_print(2,
```

```
f"[VISION]
FPS={fps_counter}
Inference={elapsed*1000:.
1f}ms")
            fps_counter = 0
            fps_timer   =
time.time()

        best          =
max(detections,
key=lambda d: d["conf"])
if detections else None
        now           =
time.time()
        frame_qualifies =
False
        conf = box =
box_area = 0

        if best:
            conf    =
best["conf"]
            box     =
best["box"]
            box_area =
max(0, box[2]-box[0]) *
max(0, box[3]-box[1])
            frame_qualifies
= (conf >=
ACCIDENT_MIN_CONF
IDENCE and

box_area >=
ACCIDENT_MIN_BOX_
AREA)

        if frame_qualifies:
            miss_count      =
0
            accident_streak
+= 1
            streak_confs    =
streak_confs[-9:] + [conf]

            if
accident_streak >=
ACCIDENT_REQUIRED
_STREAK:
                avg_conf =
sum(streak_confs) /
len(streak_confs)

                if avg_conf
>=
ACCIDENT_AVG_CONF
_MIN:
                    with
vision_lock:

vision_state.update({

"accident_confirmed":
True,

"accident_confidence":
round(avg_conf, 4),

"accident_box":       box,

"streak_confidences":
list(streak_confs),

"miss_count":         0,
                        })

already_logged =
vision_state["last_logged_
confirmed"]
                    last_pub
=
vision_state["last_publish_
time"]

                    if not
already_logged:
                        event_id
=
get_active_event_id_safe()

log_vision(event_id,
"ACCIDENT_CONFIRM
ED", avg_conf, box,

f"streak={accident_streak}
avg={avg_conf:.2f} "
f"area={box_area}")

safe_print(f"\n[ACCIDEN
T]
Confidence={avg_conf:.2f
} Box={box}")
                    with
vision_lock:

vision_state["last_logged_
confirmed"] = True

                if now -
last_pub >
ACCIDENT_COOLDOW
N:
                    # Encode
current frame as base64
JPEG for dashboard
                    img_b64
= ""
                    try:

snap_bgr =
cv2.cvtColor(frame_copy,
cv2.COLOR_RGB2BGR)

ret_enc, buf =
cv2.imencode(

".jpg", snap_bgr,

[cv2.IMWRITE_JPEG_Q
UALITY, 70]
                        )
                        if
ret_enc:

img_b64 =
base64.b64encode(buf).de
code("utf-8")
                            with
last_img_b64_lock:

last_accident_img_b64 =
img_b64
                    except
Exception as e:
```

```
safe_print(f"[VISION] Image encode error: {e}")

                payload = json.dumps({

"event":    "ACCIDENT",

"confidence": round(avg_conf, 2),
                    "box": box,

"timestamp": time.strftime("%H:%M:%S"),

"image":    img_b64,
                })

mqtt_client.publish("vision/accident", payload)

safe_print(f"[MQTT] vision/accident published "

f"(conf={avg_conf:.2f}, img={'yes' if img_b64 else 'no'})")

                with recording_latch_lock:
                    if not recording_latch:

accident_recording_event.set()

recording_latch = True


log_vision(get_active_event_id_safe(), "ACCIDENT_PUBLISHED",

avg_conf, box,
f"streak={accident_streak}")
                with vision_lock:

vision_state["last_publish_time"] = now
            else:
                log_print(1, f"[VISION] Suppressed: avg_conf={avg_conf:.2f}"
                          f"<{ACCIDENT_AVG_CONF_MIN}")

        else:
            miss_count += 1
            with vision_lock:

vision_state["miss_count"] = miss_count

            if miss_count > ACCIDENT_MISS_TOLERANCE:

accident_streak = 0
                streak_confs = []
                miss_count = 0

                with vision_lock:

was_confirmed = vision_state["accident_confirmed"]
                    was_logged = vision_state["last_logged_confirmed"]
                    old_conf = vision_state["accident_confidence"]
                    old_box = vision_state["accident_box"]

vision_state.update({

"accident_confirmed": False,
"accident_confidence": 0.0,

"accident_box": None,
"streak_confidences": [],

"miss_count": 0,
"last_logged_confirmed": False,
                })

                if was_logged or was_confirmed:

log_vision(get_active_event_id_safe(), "ACCIDENT_CLEARED",

old_conf, old_box,

f"lost after {ACCIDENT_MISS_TOLERANCE+1} misses")

                with recording_latch_lock:

recording_latch = False


time.sleep(VISION_LOOP_DELAY)

    except Exception as e:
        safe_print(f"[VISION ERROR] {e}")
    finally:
        cap.release()
```

```
        safe_print("[VISION] Stream closed")

def vision_blink_thread():
    safe_print("[VISION BLINK] Started")
    while running:
        try:
            with vision_lock:
                confirmed = vision_state["accident_confirmed"]
                last_pub  = vision_state["last_publish_time"]

            # Auto-expire if camera died and no new detections
            if confirmed and (time.time() - last_pub) > ACCIDENT_VISION_TIMEOUT:
                with vision_lock:
                    vision_state["accident_confirmed"]    = False
                    vision_state["last_logged_confirmed"] = False
                with recording_latch_lock:
                    global recording_latch
                    recording_latch = False
                log_decision(get_active_event_id_safe(), 0, "VISION_BLINK_TIMEOUT",
                             f"auto-expired after {ACCIDENT_VISION_TIMEOUT}s")
                safe_print(f"[VISION BLINK] Auto-expired after {ACCIDENT_VISION_TIMEOUT}s")
                continue

            if confirmed:
                send_cmd_both("BLINK_YELLOW")
                log_decision(get_active_event_id_safe(), 0, "VISION_BLINK_YELLOW",
                             "sent to nodes 1 and 2")
                elapsed = 0.0
                while elapsed < VISION_BLINK_INTERVAL and running:
                    time.sleep(0.25)
                    elapsed += 0.25
                    with vision_lock:
                        if not vision_state["accident_confirmed"]:
                            break
                else:
                    time.sleep(0.25)

        except Exception as e:
            safe_print(f"[VISION BLINK ERROR] {e}")
            time.sleep(1)

# ==========================
# MAIN
# ==========================
def main():
    global running, event_counter

    init_csv_logs()
    event_counter = load_event_counter()
    safe_print(f"[INFO] Event counter restored: {event_counter}")

    try:
        mqtt_client.connect(MQTT_BROKER, MQTT_PORT)
        mqtt_client.loop_start()
        safe_print(f"[MQTT] Connected to {MQTT_BROKER}")
    except Exception as e:
        safe_print(f"[MQTT] Connection failed: {e}")

    threads = [
        threading.Thread(target=base_thread,          daemon=True, name="base"),
        threading.Thread(target=cli_thread,           daemon=True, name="cli"),
        threading.Thread(target=clear_checker_thread, daemon=True, name="clear"),
        threading.Thread(target=vision_thread,        daemon=True, name="vision"),
        threading.Thread(target=re
```

```
cording_thread,
daemon=True,
name="recording"),

threading.Thread(target=h
eartbeat_thread,
daemon=True,
name="heartbeat"),

threading.Thread(target=vi
sion_blink_thread,
daemon=True,
name="vision_blink"),
    ]
    for t in threads:
        t.start()
    safe_print("[INFO] All
threads started.")

    try:
        while running:
            time.sleep(1)
    except
KeyboardInterrupt:
        safe_print("\n[INFO]
Stopped by user.")
        running = False
        time.sleep(1)
    finally:
        try:
            if ser:
                with serial_lock:
                    ser.close()
            except Exception:
                pass

mqtt_client.loop_stop()

mqtt_client.disconnect()
        safe_print("[INFO]
LUMEN stopped.")

if __name__ ==
"__main__":
    main()
```

Dashboard Front-end

```
<Window
xmlns="https://github.com
/avaloniaui"

xmlns:x="http://schemas.
microsoft.com/winfx/2006
/xaml"

xmlns:vm="using:Dashboa
rd.ViewModels"

xmlns:vlc="clr-namespace
:LibVLCSharp.Avalonia;a
ssembly=LibVLCSharp.A
valonia"

x:Class="Dashboard.View
s.MainWindow"

x:DataType="vm:MainWi
ndowViewModel"

Title="LUMENDASH -
Live Monitor"
        MinWidth="900"
MinHeight="600"

WindowState="Maximize
d"

Background="#070B12">

            <Grid>
                    <!-- Main
content -->
                    <Grid
RowDefinitions="Auto,
Auto, Auto, Auto, *"
Margin="16">


<!-- Status Bar -->

<Border Grid.Row="0"
Background="#0C1220"
CornerRadius="10"

Padding="16,10"
Margin="0,0,0,10"

BorderBrush="#1A2A4A"
BorderThickness="1">

            <Grid
ColumnDefinitions="Auto
, *, Auto, Auto, Auto">


<StackPanel
Orientation="Horizontal"
Spacing="20">


<StackPanel
Orientation="Horizontal"
Spacing="8">


            <Ellipse
Width="9" Height="9"
Fill="{Binding
StatusColor}"/>
```

```
            <TextBlock
Text="{Binding
ConnectionStatus}"
Foreground="White"

FontWeight="SemiBold"
VerticalAlignment="Cente
r" FontSize="12"/>


</StackPanel>


<TextBlock
Text="{Binding
MqttStatus}"
Foreground="#00CFFF"

FontSize="11"
VerticalAlignment="Cente
r" Opacity="0.85"/>


</StackPanel>


                    <!--
Traffic detections counter
-->

                    <Button
Grid.Column="2"
Command="{Binding
ShowTrafficLogCommand
}"

Background="Transparent
" BorderThickness="0"

Padding="0"
Margin="0,0,8,0"
VerticalAlignment="Cente
r"

Cursor="Hand">

<Border
Background="#0A1A0A"
CornerRadius="6"
Padding="10,4">


            <StackPanel
Orientation="Horizontal"
Spacing="6">


<TextBlock
Text="TRAFFIC"
Foreground="#44BB44"
FontSize="10"

FontWeight="Bold"
VerticalAlignment="Cente
r" LetterSpacing="1"/>


<TextBlock
Text="{Binding
TotalAlertCount}"
Foreground="#44FF44"

FontSize="13"
FontWeight="Black"
VerticalAlignment="Cente
r"/>


            </StackPanel>


</Border>

                    </Button>


                    <!--
Accident counter -->

                    <Button
Grid.Column="3"
Command="{Binding
ShowAccidentLogComma
nd}"

Background="Transparent
" BorderThickness="0"

Padding="0"
Margin="0,0,12,0"
VerticalAlignment="Cente
r"

Cursor="Hand">


<Border
Background="#1A0A0A"
CornerRadius="6"
Padding="10,4">


            <StackPanel
Orientation="Horizontal"
Spacing="6">


<TextBlock
Text="ACCIDENTS"
Foreground="#FF6644"
FontSize="10"

FontWeight="Bold"
VerticalAlignment="Cente
r" LetterSpacing="1"/>


<TextBlock
Text="{Binding
AccidentCount}"
Foreground="#FF4422"

FontSize="13"
FontWeight="Black"
VerticalAlignment="Cente
r"/>
```

```
            </StackPanel>

</Border>
                    </Button>


<TextBlock
Grid.Column="4"
Text="LUMEN
MONITORING"

Foreground="#4A8AAA"
VerticalAlignment="Cente
r" FontSize="11"/>

            </Grid>

</Border>


<!-- Device Stats Bar -->

<Border Grid.Row="1"
Background="#0C1220"
CornerRadius="10"
Padding="14,8"

Margin="0,0,0,12"
BorderBrush="#1A2A4A"
BorderThickness="1">

            <StackPanel
Orientation="Horizontal"
Spacing="20">


<TextBlock
Text="Raspberry Pi Stats"
Foreground="#00CFFF"
FontSize="10"

FontWeight="Bold"
LetterSpacing="2"
VerticalAlignment="Cente
r"/>


<StackPanel
Orientation="Horizontal"
Spacing="5">


<TextBlock Text="CPU"
Foreground="#4A8AAA"
FontSize="10"

FontWeight="SemiBold"
VerticalAlignment="Cente
r"/>


<TextBlock
Text="{Binding RpiCpu}"
Foreground="#E0F4FF"

FontSize="11"
FontWeight="Bold"
VerticalAlignment="Cente
r"/>


</StackPanel>


<StackPanel
Orientation="Horizontal"
Spacing="5">


<TextBlock Text="RAM"
Foreground="#4A8AAA"
FontSize="10"

FontWeight="SemiBold"
VerticalAlignment="Cente
r"/>


<TextBlock
Text="{Binding RpiRam}"
Foreground="#E0F4FF"

FontSize="11"
FontWeight="Bold"
VerticalAlignment="Cente
r"/>


</StackPanel>


<StackPanel
Orientation="Horizontal"
Spacing="5">


<TextBlock Text="TEMP"
Foreground="#4A8AAA"
FontSize="10"

FontWeight="SemiBold"
VerticalAlignment="Cente
r"/>


<TextBlock
Text="{Binding
RpiTemp}"
Foreground="#E0F4FF"

FontSize="11"
FontWeight="Bold"
VerticalAlignment="Cente
r"/>


</StackPanel>

            </StackPanel>

</Border>


<!-- Accident Alert Banner
(passive - no buttons) -->

<Border Grid.Row="2"
IsVisible="{Binding
ShowAccidentBanner}"
```

```
Background="{Binding
AccidentBannerBg}"
CornerRadius="8"

BorderBrush="{Binding
AccidentBannerColor}"
BorderThickness="2"

Padding="14,10"
Margin="0,0,0,10">

        <StackPanel
Orientation="Vertical"
Spacing="3">


<TextBlock
Text="{Binding
AccidentBannerMessage}"

Foreground="{Binding
AccidentBannerColor}"

FontSize="13"
FontWeight="Bold"

VerticalAlignment="Cente
r"/>


<TextBlock
Text="{Binding
AccidentConfidenceText}"

Foreground="#FFD580"
FontSize="11"

VerticalAlignment="Cente
r"/>

        </StackPanel>

</Border>


<!-- MQTT Error Banner
-->

<Border Grid.Row="3"
IsVisible="{Binding
ShowMqttError}"

Background="#1A0A0A"
CornerRadius="8"

BorderBrush="#FF4444"
BorderThickness="1"

Padding="14,8"
Margin="0,0,0,10">

        <Grid
ColumnDefinitions="*,
Auto">


<StackPanel
Orientation="Horizontal"
Spacing="12">


<TextBlock Text="⚠"
Foreground="#FF4444"

FontSize="14"
VerticalAlignment="Cente
r"/>


<TextBlock
Text="{Binding
MqttErrorMessage}"

Foreground="#FFAAAA"
FontSize="11"

VerticalAlignment="Cente
r"/>


</StackPanel>

                <Button
Grid.Column="1"
Content="↻ RETRY
MQTT"
Command="{Binding
RetryMqttCommand}"

Background="#2A1A1A"

Foreground="#FF6666"

FontSize="11"
FontWeight="Bold"

Padding="14,6"
CornerRadius="6"/>

        </Grid>

</Border>


<!-- Main Layout -->

<Grid Grid.Row="4"
ColumnDefinitions="3*,
2*">


        <!-- LEFT: Video
+ Log -->

        <Grid
Grid.Column="0"
RowDefinitions="*, 12,
Auto" Margin="0,0,12,0">


                <!-- Video
-->

                <Border
Grid.Row="0"
Background="#000"
CornerRadius="14"

BorderBrush="#1A2A4A"
BorderThickness="1">


<Grid>
```

```
            <vlc:VideoView
x:Name="VideoView"

IsVisible="{Binding
IsVideoVisible}"

HorizontalAlignment="Str
etch"

VerticalAlignment="Stretc
h"/>


            <Border
IsVisible="{Binding
ShowStreamError}"

Background="#CC070B12
" CornerRadius="14"

HorizontalAlignment="Str
etch"

VerticalAlignment="Stretc
h">



<StackPanel
HorizontalAlignment="Ce
nter"

VerticalAlignment="Cente
r"

Spacing="16">



<TextBlock Text="⚠
STREAM
UNAVAILABLE"

Foreground="#FF4444"

FontSize="16"
FontWeight="Bold"

HorizontalAlignment="Ce
nter"/>



<TextBlock
Text="{Binding
StreamErrorMessage}"

Foreground="#A0BFCC"

FontSize="11"

TextAlignment="Center"

HorizontalAlignment="Ce
nter"/>



<Button Content="↻
RETRY STREAM"

Command="{Binding
RetryStreamCommand}"

HorizontalAlignment="Ce
nter"

Background="#1A2A4A"

Foreground="#00CFFF"

FontSize="11"
FontWeight="Bold"

Padding="20,8"
CornerRadius="8"/>



</StackPanel>


            </Border>
</Grid>
            </Border>


            <!-- Alert
Log (recent 50) -->

            <Border
Grid.Row="2"
Background="#0C1220"
CornerRadius="14"

BorderBrush="#1A2A4A"
BorderThickness="1"
MaxHeight="250">


<Grid
RowDefinitions="38, *">


            <Border
Grid.Row="0"
Background="#0A0F1A"
CornerRadius="14,14,0,0"

Padding="14,0"
BorderBrush="#1A2A4A"
BorderThickness="0,0,0,1
">


                <Grid
ColumnDefinitions="*,
Auto, Auto, Auto">


<TextBlock Text="ALERT
LOG"
Foreground="#00CFFF"

FontSize="10"
FontWeight="Bold"
LetterSpacing="3"
```

```
VerticalAlignment="Center"/>

<Border Grid.Column="1"
Background="#0F1A2F"
CornerRadius="5"

Padding="8,3"
Margin="0,0,8,0"
VerticalAlignment="Center">

        <StackPanel
Orientation="Horizontal"
Spacing="5">

            <Ellipse
Width="6" Height="6"
Fill="{Binding
Node1Color}"

VerticalAlignment="Center"/>

<TextBlock Text="N1"
Foreground="#4A8AAA"
FontSize="9"

FontWeight="SemiBold"
VerticalAlignment="Center"/>

<TextBlock
Text="{Binding
Node1TrafficCount}"

Foreground="{Binding
Node1Color}"

FontSize="11"
FontWeight="Bold"

VerticalAlignment="Center"/>

        </StackPanel>

</Border>

<Border Grid.Column="2"
Background="#0F1A2F"
CornerRadius="5"

Padding="8,3"
Margin="0,0,8,0"
VerticalAlignment="Center">

        <StackPanel
Orientation="Horizontal"
Spacing="5">

            <Ellipse
Width="6" Height="6"
Fill="{Binding
Node2Color}"

VerticalAlignment="Center"/>

<TextBlock Text="N2"
Foreground="#4A8AAA"
FontSize="9"

FontWeight="SemiBold"
VerticalAlignment="Center"/>

<TextBlock
Text="{Binding
Node2TrafficCount}"

Foreground="{Binding
Node2Color}"

FontSize="11"
FontWeight="Bold"

VerticalAlignment="Center"/>

        </StackPanel>

</Border>

<TextBlock
Grid.Column="3"
Text="LAST 50
EVENTS"

Foreground="#4A8AAA"
FontSize="9"

VerticalAlignment="Center"/>

    </Grid>
```

```
        </Border>

        <ScrollViewer Grid.Row="1" VerticalScrollBarVisibility="Auto">

<ItemsControl ItemsSource="{Binding AlertLog}">

<ItemsControl.ItemTemplate>

        <DataTemplate>

                <Border Padding="12,9" BorderBrush="#111E33" BorderThickness="0,0,0,1">

<Grid RowDefinitions="Auto, Auto">

        <Grid Grid.Row="0" ColumnDefinitions="65, 75, *, 70">

<TextBlock Grid.Column="0" Text="{Binding Time}"

Foreground="#4A7A9A" FontSize="10"

VerticalAlignment="Center"

FontFamily="Courier New"/>

                        <Border Grid.Column="1" Background="#112030"

CornerRadius="4" Padding="6,3"

Margin="0,0,8,0" VerticalAlignment="Center">

<TextBlock Text="{Binding Node}"

Foreground="#60AACC" FontSize="9"

FontWeight="Bold"

VerticalAlignment="Center"/>

                        </Border>

<TextBlock Grid.Column="2" Text="{Binding Message}"

Foreground="{Binding Color}" FontSize="11"

FontWeight="Bold"

VerticalAlignment="Center"/>

<TextBlock Grid.Column="3" Text="{Binding Distance}"

Foreground="#80B4CC" FontSize="10"

FontWeight="SemiBold"

HorizontalAlignment="Right"

VerticalAlignment="Center"/>

        </Grid>

        <Grid
```

```
Grid.Row="1"
ColumnDefinitions="*,
Auto"

Margin="0,5,0,0">




<TextBlock
Grid.Column="0"
Text="{Binding
RawLine}"

Foreground="#3A6A8A"
FontSize="9"

FontFamily="Courier
New"

TextWrapping="Wrap"/>



                <Border
Grid.Column="1"
Background="#0A1E30"

CornerRadius="4"
Padding="7,2"

Margin="8,0,0,0"
VerticalAlignment="Cente
r">




<TextBlock
Text="{Binding Speed}"

Foreground="#00CFFF"
FontSize="10"

FontWeight="Bold"
VerticalAlignment="Cente
r"/>

                </Border>

            </Grid>

        </Border>

    </DataTemplate>

</ItemsControl.ItemTempl
ate>

</ItemsControl>

        </ScrollViewer>

</Grid>

                </Border>

        </Grid>

        <!-- RIGHT:
Radar Panels -->

        <Grid
Grid.Column="1"
RowDefinitions="*, 10,
*">

            <!--
NODE 1 -->

            <Border
Grid.Row="0"
CornerRadius="14"
ClipToBounds="True"

Background="#0C1220"
BorderBrush="{Binding
Node1Color}"
BorderThickness="1">

<Grid
RowDefinitions="44, *">

        <Grid
Grid.Row="0"
ColumnDefinitions="Auto
, *, Auto, Auto, Auto"

Margin="14,0,14,0">

            <Ellipse
Grid.Column="0"
Width="8" Height="8"

Fill="{Binding
Node1Color}"

VerticalAlignment="Cente
r" Margin="0,0,8,0"/>


<TextBlock
Grid.Column="1"
```

```
Text="NODE 1"

Foreground="#4A8AAA"
FontSize="10"

FontWeight="Bold"
LetterSpacing="2"

VerticalAlignment="Cente
r"/>



<TextBlock
Grid.Column="2"
Text="{Binding
Node1Speed}"

Foreground="#00CFFF"
FontSize="11"

FontWeight="Bold"
VerticalAlignment="Cente
r"

Margin="0,0,10,0"/>



<TextBlock
Grid.Column="3"
Text="{Binding
Node1Distance}"

Foreground="#A0D4EE"
FontSize="12"

FontWeight="SemiBold"
VerticalAlignment="Cente
r"

Margin="0,0,10,0"/>



<TextBlock
Grid.Column="4"
Text="{Binding
Node1Status}"

Foreground="{Binding
Node1Color}"

FontSize="11"
FontWeight="Bold"

VerticalAlignment="Cente
r"/>

        </Grid>

        <Viewbox
Grid.Row="1"
Stretch="Fill">

            <Canvas
Width="300"
Height="200"
Background="#090E18">

<Rectangle
Canvas.Left="0"
Canvas.Top="0"

Width="300" Height="90"
Fill="#0A0F1A"/>



<Rectangle
Canvas.Left="60"
Canvas.Top="84"

Width="180" Height="8"

Fill="#0A2A4A"
Opacity="0.6"/>



<Path Fill="#111E2E"
Data="M 80,90 L 220,90
L 300,200 L 0,200 Z"/>



<Path Stroke="#1A3A5A"
StrokeThickness="1.5"
Fill="Transparent"

Data="M 80,90 L 0,200"/>



<Path Stroke="#1A3A5A"
StrokeThickness="1.5"
Fill="Transparent"

Data="M 220,90 L
300,200"/>



<Path Stroke="#2A5A7A"
StrokeThickness="1"
Fill="Transparent"

Data="M 85,90 L 5,200"/>



<Path Stroke="#2A5A7A"
StrokeThickness="1"
Fill="Transparent"

Data="M 215,90 L
295,200"/>



<Rectangle
Canvas.Left="80"
Canvas.Top="88"

Width="140" Height="3"

Fill="#0A2A4A"
Opacity="0.8"/>
```

```
<Path Stroke="#111E2E"
StrokeThickness="1"
Fill="Transparent"

Data="M 100,110 L
200,110 L 240,140 L
60,140 Z"/>



<Path Stroke="#111E2E"
StrokeThickness="1"
Fill="Transparent"

Data="M 110,120 L
190,120"/>



<Ellipse
Canvas.Left="120"
Canvas.Top="182"

Width="60" Height="20"

Fill="#00CFFF"
Opacity="0.06"/>



<Ellipse
Canvas.Left="138"
Canvas.Top="188"

Width="24" Height="10"

Fill="#00CFFF"
Opacity="0.15"/>



<Ellipse
Canvas.Left="133"
Canvas.Top="128"
Width="34" Height="18"

Fill="{Binding
Node1Color}"

Opacity="{Binding
Node1VehicleOpacity}"/>



<Rectangle
Canvas.Left="141"
Canvas.Top="132"

Width="18" Height="10"
RadiusX="3"
RadiusY="3"

Fill="{Binding
Node1Color}"

Opacity="{Binding
Node1VehicleOpacity}"/>



<Ellipse
Canvas.Left="131"
Canvas.Top="126"

Width="38" Height="22"

Stroke="{Binding
Node1Color}"
StrokeThickness="1"

Fill="Transparent"

Opacity="{Binding
Node1VehicleOpacity}"/>


        </Canvas>


    </Viewbox>
</Grid>

        </Border>

        <!--
NODE 2 -->

        <Border
Grid.Row="2"
CornerRadius="14"
ClipToBounds="True"

Background="#0C1220"
BorderBrush="{Binding
Node2Color}"
BorderThickness="1">


<Grid
RowDefinitions="44, *">


    <Grid
Grid.Row="0"
ColumnDefinitions="Auto
, *, Auto, Auto, Auto"

Margin="14,0,14,0">


        <Ellipse
Grid.Column="0"
Width="8" Height="8"

Fill="{Binding
Node2Color}"

VerticalAlignment="Cente
r" Margin="0,0,8,0"/>


<TextBlock
Grid.Column="1"
Text="NODE 2"

Foreground="#4A8AAA"
```

```
FontSize="10"

FontWeight="Bold"
LetterSpacing="2"

VerticalAlignment="Cente
r"/>



<TextBlock
Grid.Column="2"
Text="{Binding
Node2Speed}"

Foreground="#00CFFF"
FontSize="11"

FontWeight="Bold"
VerticalAlignment="Cente
r"

Margin="0,0,10,0"/>



<TextBlock
Grid.Column="3"
Text="{Binding
Node2Distance}"

Foreground="#A0D4EE"
FontSize="12"

FontWeight="SemiBold"
VerticalAlignment="Cente
r"

Margin="0,0,10,0"/>



<TextBlock
Grid.Column="4"
Text="{Binding
Node2Status}"

Foreground="{Binding
Node2Color}"

FontSize="11"
FontWeight="Bold"

VerticalAlignment="Cente
r"/>


        </Grid>


        <Viewbox
Grid.Row="1"
Stretch="Fill">


            <Canvas
Width="300"
Height="200"
Background="#090E18">



<Rectangle
Canvas.Left="0"
Canvas.Top="0"

Width="300" Height="90"
Fill="#0A0F1A"/>



<Rectangle
Canvas.Left="60"
Canvas.Top="84"

Width="180" Height="8"

Fill="#0A2A4A"
Opacity="0.6"/>



<Path Fill="#111E2E"

Data="M 80,90 L 220,90
L 300,200 L 0,200 Z"/>

<Path Stroke="#1A3A5A"
StrokeThickness="1.5"
Fill="Transparent"

Data="M 80,90 L 0,200"/>


<Path Stroke="#1A3A5A"
StrokeThickness="1.5"
Fill="Transparent"

Data="M 220,90 L
300,200"/>


<Path Stroke="#2A5A7A"
StrokeThickness="1"
Fill="Transparent"

Data="M 85,90 L 5,200"/>


<Path Stroke="#2A5A7A"
StrokeThickness="1"
Fill="Transparent"

Data="M 215,90 L
295,200"/>


<Rectangle
Canvas.Left="80"
Canvas.Top="88"

Width="140" Height="3"

Fill="#0A2A4A"
Opacity="0.8"/>
```

```
<Path Stroke="#111E2E"
StrokeThickness="1"
Fill="Transparent"

Data="M 100,110 L
200,110 L 240,140 L
60,140 Z"/>



<Path Stroke="#111E2E"
StrokeThickness="1"
Fill="Transparent"

Data="M 110,120 L
190,120"/>



<Ellipse
Canvas.Left="120"
Canvas.Top="182"

Width="60" Height="20"

Fill="#00CFFF"
Opacity="0.06"/>



<Ellipse
Canvas.Left="138"
Canvas.Top="188"

Width="24" Height="10"

Fill="#00CFFF"
Opacity="0.15"/>



<Ellipse
Canvas.Left="133"
Canvas.Top="128"

Width="34" Height="18"

Fill="{Binding
Node2Color}"

Opacity="{Binding
Node2VehicleOpacity}"/>



<Rectangle
Canvas.Left="141"
Canvas.Top="132"

Width="18" Height="10"
RadiusX="3"
RadiusY="3"

Fill="{Binding
Node2Color}"

Opacity="{Binding
Node2VehicleOpacity}"/>



<Ellipse
Canvas.Left="131"
Canvas.Top="126"

Width="38" Height="22"

Stroke="{Binding
Node2Color}"
StrokeThickness="1"

Fill="Transparent"

Opacity="{Binding
Node2VehicleOpacity}"/>


                </Canvas>


            </Viewbox>


</Grid>

                </Border>
            </Grid>

</Grid>
                </Grid>

                <!-- Full
Log Overlay -->
                <Border
IsVisible="{Binding
ShowLogOverlay}"

Background="#E5070B12
"

HorizontalAlignment="Str
etch"

VerticalAlignment="Stretc
h">

<Border
Background="#0C1220"
CornerRadius="16"

BorderBrush="{Binding
LogOverlayAccentColor}"
BorderThickness="1"
                Width="620"
MaxHeight="700"

HorizontalAlignment="Ce
nter"
VerticalAlignment="Cente
r">

            <Grid
RowDefinitions="50, *">

                <Border
Grid.Row="0"
Background="#0A0F1A"

CornerRadius="16,16,0,0"

BorderBrush="#1A2A4A"
BorderThickness="0,0,0,1
```

```
"

Padding="20,0">


<Grid
ColumnDefinitions="*,
Auto">


        <TextBlock
Text="{Binding
LogOverlayTitle}"

Foreground="{Binding
LogOverlayAccentColor}"

FontSize="11"
FontWeight="Bold"

LetterSpacing="2"

VerticalAlignment="Cente
r"/>


        <Button
Grid.Column="1"
Content="✕  CLOSE"

Command="{Binding
CloseLogOverlayComman
d}"

Background="#1A2A3A"

Foreground="#A0BFCC"

FontSize="11"
FontWeight="Bold"

Padding="14,6"
CornerRadius="6"

VerticalAlignment="Cente
r"/>
</Grid>

                </Border>


<ScrollViewer
Grid.Row="1"
VerticalScrollBarVisibility
="Auto" Padding="0,4">


<ItemsControl
ItemsSource="{Binding
ActiveOverlayLog}">


<ItemsControl.ItemTempla
te>


<DataTemplate>


<Border Padding="20,10"
BorderBrush="#111E33"

BorderThickness="0,0,0,1
">


        <Grid
RowDefinitions="Auto,
Auto">


                <Grid
Grid.Row="0"
ColumnDefinitions="75,
90, *, 80">
<TextBlock
Grid.Column="0"
Text="{Binding Time}"

Foreground="#5A9ABF"
FontSize="11"

VerticalAlignment="Cente
r"

FontFamily="Courier
New"/>


<Border Grid.Column="1"
Background="#112030"

CornerRadius="4"
Padding="8,3"

Margin="0,0,10,0"
VerticalAlignment="Cente
r">


        <TextBlock
Text="{Binding Node}"

Foreground="#70BBDD"
FontSize="10"

FontWeight="Bold"

VerticalAlignment="Cente
r"/>


</Border>
```

```
<TextBlock
Grid.Column="2"
Text="{Binding
Message}"

Foreground="{Binding
Color}" FontSize="12"

FontWeight="Bold"

VerticalAlignment="Cente
r"/>



<TextBlock
Grid.Column="3"
Text="{Binding
Distance}"

Foreground="#90C4DD"
FontSize="11"

FontWeight="SemiBold"

HorizontalAlignment="Ri
ght"

VerticalAlignment="Cente
r"/>


            </Grid>


            <Grid
Grid.Row="1"
ColumnDefinitions="*,
Auto"

Margin="0,6,0,0">
```

```
<TextBlock
Grid.Column="0"
Text="{Binding
RawLine}"

Foreground="#4A7A9A"
FontSize="9"

FontFamily="Courier
New"

TextWrapping="Wrap"/>



<Border Grid.Column="1"
Background="#0A1E30"

CornerRadius="4"
Padding="8,3"

Margin="10,0,0,0"
VerticalAlignment="Cente
r">


        <TextBlock
Text="{Binding Speed}"

Foreground="#00CFFF"
FontSize="11"

FontWeight="Bold"

VerticalAlignment="Cente
r"/>


</Border>
```

```
            </Grid>


        </Border>


</DataTemplate>


</ItemsControl.ItemTempl
ate>


</ItemsControl>


</ScrollViewer>


        </Grid>


</Border>
            </Border>


            <!--
Accident Image Modal -->
            <Border
IsVisible="{Binding
ShowAccidentModal}"

Background="#E5070B12
"

HorizontalAlignment="Str
etch"

VerticalAlignment="Stretc
h">

<Border
Background="#0C1220"
CornerRadius="16"
```

```
BorderBrush="{Binding
AccidentModalColor}"
BorderThickness="2"
                Width="660"
MaxHeight="780"

HorizontalAlignment="Ce
nter"
VerticalAlignment="Cente
r">

          <Grid
RowDefinitions="50, *,
Auto">


                    <!--
Header -->

                    <Border
Grid.Row="0"
Background="#0A0F1A"

CornerRadius="16,16,0,0"

BorderBrush="#1A2A4A"
BorderThickness="0,0,0,1
"

Padding="20,0">


<Grid
ColumnDefinitions="*,
Auto">


          <TextBlock
Text="{Binding
AccidentModalMessage}"

Foreground="{Binding
AccidentModalColor}"

FontSize="11"
FontWeight="Bold"

LetterSpacing="2"
VerticalAlignment="Cente
r"/>


                <Button
Grid.Column="1"
Content="✕  CLOSE"

Command="{Binding
CloseAccidentModalCom
mand}"

Background="#1A2A3A"

Foreground="#A0BFCC"

FontSize="11"
FontWeight="Bold"

Padding="14,6"
CornerRadius="6"

VerticalAlignment="Cente
r"/>


</Grid>

                    </Border>


                    <!-- Image
area -->

                    <Border
Grid.Row="1"
Margin="20,16,20,0"

CornerRadius="10"
ClipToBounds="True"

Background="#060A10"

BorderBrush="#1A2A4A"
BorderThickness="1"

MinHeight="200">

<Panel>


          <Image
Source="{Binding
AccidentModalImage}"

Stretch="Uniform"

MaxHeight="500"

IsVisible="{Binding
AccidentModalImage,
Converter={x:Static
ObjectConverters.IsNotNu
ll}}"/>


          <StackPanel
HorizontalAlignment="Ce
nter"

VerticalAlignment="Cente
r"

Spacing="10"

IsVisible="{Binding
AccidentModalImage,
Converter={x:Static
ObjectConverters.IsNull}}
">


<TextBlock Text="📷"
FontSize="32"
HorizontalAlignment="Ce
nter"/>


<TextBlock Text="No
image received"

Foreground="#3A6A8A"
```

```
FontSize="13"

HorizontalAlignment="Ce
nter"/>



<TextBlock Text="The
RPi did not include an
image in this alert."

Foreground="#2A4A6A"
FontSize="10"

HorizontalAlignment="Ce
nter"/>


        </StackPanel>


</Panel>

                </Border>


                <!--
Footer with confidence,
timestamp, confirm and
dismiss -->

                <Border
Grid.Row="2"
Padding="20,14"

BorderBrush="#1A2A4A"
BorderThickness="0,1,0,0
"

Margin="0,14,0,0">


<Grid
ColumnDefinitions="*,
Auto">

            <StackPanel
Orientation="Vertical"
Spacing="4">



<TextBlock
Text="{Binding
AccidentModalConfidence
}"

Foreground="{Binding
AccidentModalColor}"

FontSize="13"
FontWeight="Bold"/>



<TextBlock
Text="{Binding
AccidentModalTimestamp
}"

Foreground="#80B4CC"
FontSize="11"/>


        </StackPanel>


        <StackPanel
Grid.Column="1"
Orientation="Horizontal"

Spacing="8"
VerticalAlignment="Cente
r">


                <Button
Content="✔ CONFIRM"

Command="{Binding
ConfirmAccidentComman
d}"

Background="#0A2A0A"

Foreground="#44FF44"

FontSize="11"
FontWeight="Bold"

Padding="14,6"
CornerRadius="6"/>


                <Button
Content="✕ DISMISS"

Command="{Binding
DismissAccidentComman
d}"

Background="#2A1500"

Foreground="{Binding
AccidentModalColor}"

FontSize="11"
FontWeight="Bold"

Padding="14,6"
CornerRadius="6"/>


        </StackPanel>


</Grid>

                </Border>


        </Grid>

</Border>
                </Border>


        </Grid>
</Window>
```

Dashboard Back-end

```
using CommunityToolkit.Mvvm.ComponentModel;

using CommunityToolkit.Mvvm.Input;

using LibVLCSharp.Shared;

using MQTTnet;

using System;

using System.Collections.ObjectModel;

using System.Threading;

using System.Threading.Tasks;

using Avalonia.Threading;

using System.Text.Json;

using System.IO;

using Avalonia.Media.Imaging;



namespace Dashboard.ViewModels;



public partial class AlertEntry : ObservableObject

{

    [ObservableProperty]
private string _time = "";

    [ObservableProperty]
private string _node = "";

    [ObservableProperty]
private string _message = "";

    [ObservableProperty]
private string _distance = "";

    [ObservableProperty]
private string _color = "#FF4444";

    [ObservableProperty]
private string _speed = "";

    [ObservableProperty]
private string _direction = "";

    [ObservableProperty]
private string _snr = "";

    [ObservableProperty]
private string _rawLine = "";

}



public partial class MainWindowViewModel : ObservableObject, IDisposable

{

    [ObservableProperty]
private string _node1Status = "WAITING...";

    [ObservableProperty]
private string _node1Distance = "N/A";

    [ObservableProperty]
private string _node1Color = "#808080";



    [ObservableProperty]
private string _node2Status = "WAITING...";

    [ObservableProperty]
private string _node2Distance = "N/A";

    [ObservableProperty]
private string _node2Color = "#808080";



    [ObservableProperty]
private string _connectionStatus = "DISCONNECTED";

    [ObservableProperty]
private string _statusColor
```

```
= "#FF4444";

    [ObservableProperty]
private string _mqttStatus
= "MQTT: OFF";



    [ObservableProperty]
private string
_node1Speed = "0 km/h";

    [ObservableProperty]
private string
_node2Speed = "0 km/h";



    [ObservableProperty]
private int
_node1TrafficCount = 0;

    [ObservableProperty]
private int
_node2TrafficCount = 0;

    [ObservableProperty]
private int
_totalAlertCount = 0;

    [ObservableProperty]
private int _accidentCount
= 0;



    [ObservableProperty]
private string _rpiCpu =
"0%";

    [ObservableProperty]
private string _rpiRam =
"0%";

    [ObservableProperty]
private string _rpiTemp =
"0°C";
```

```
    [ObservableProperty]
private double
_node1VehicleOpacity =
0.0;

    [ObservableProperty]
private double
_node2VehicleOpacity =
0.0;



    [ObservableProperty]
private bool
_showStreamError = false;

    [ObservableProperty]
private string
_streamErrorMessage =
"";



    [ObservableProperty]
private bool
_showMqttError = false;

    [ObservableProperty]
private string
_mqttErrorMessage = "";



    // ---- Accident banner
----

    [ObservableProperty]
private bool
_showAccidentBanner =
false;

    [ObservableProperty]
private string
_accidentBannerMessage
= "";

    [ObservableProperty]
private string
_accidentBannerColor =
```

```
"#FF6600";

    [ObservableProperty]
private string
_accidentBannerBg =
"#1A0D00";

    [ObservableProperty]
private string
_accidentConfidenceText
= "";



    // ---- Accident modal
----

    [ObservableProperty]
private bool
_showAccidentModal =
false;

    [ObservableProperty]
private string
_accidentModalMessage =
"";

    [ObservableProperty]
private string
_accidentModalConfidenc
e = "";

    [ObservableProperty]
private string
_accidentModalTimestamp
= "";

    [ObservableProperty]
private string
_accidentModalColor =
"#FF8800";

    [ObservableProperty]
private Bitmap?
_accidentModalImage =
null;
```

```
    // ---- Full log overlay
----

    [ObservableProperty]
private bool
_showLogOverlay = false;

    [ObservableProperty]
private string
_logOverlayTitle = "";

    [ObservableProperty]
private string
_logOverlayAccentColor
= "#00CFFF";

    [ObservableProperty]
private
ObservableCollection<Ale
rtEntry>
_activeOverlayLog =
new();



    // Pending accident held
until user confirms or
dismisses

    private AlertEntry?
_pendingAccidentEntry =
null;



    // Recent 50 (alert log
strip)

    public
ObservableCollection<Ale
rtEntry> AlertLog { get; }
= new();



    // Full unbounded logs

    public
ObservableCollection<Ale
rtEntry> FullTrafficLog {
get; } = new();

    public
ObservableCollection<Ale
rtEntry> FullAccidentLog
{ get; } = new();



    private string
_lastNode1Status = "";

    private string
_lastNode2Status = "";



    private LibVLC?
_libVLC;

    private Media?
_currentMedia;

    private IMqttClient?
_mqttClient;

    private
MqttClientOptions?
_mqttOptions;

    private
CancellationTokenSource
_mqttCts = new();



    private
System.Timers.Timer
_node1Timer =
new(5000);

    private
System.Timers.Timer
_node2Timer =
new(5000);

    private
System.Timers.Timer
_accidentDismissTimer =
new(10000);



    [ObservableProperty]
private MediaPlayer?
_mediaPlayer;

    [ObservableProperty]
private bool
_isVideoVisible = true;



    public
MainWindowViewModel()

    {

        InitializeVlc();

        _ =
SetupMqttWithRetry(_mqt
tCts.Token);



_node1Timer.AutoReset =
false;

        _node1Timer.Elapsed
+= (s, e) =>

Dispatcher.UIThread.Post(
() => ResetNode(1));



_node2Timer.AutoReset =
false;

        _node2Timer.Elapsed
+= (s, e) =>
```

```
Dispatcher.UIThread.Post(
() => ResetNode(2));



_accidentDismissTimer.Au
toReset = false;

_accidentDismissTimer.El
apsed += (s, e) =>

Dispatcher.UIThread.Post(
() =>
        {

ShowAccidentBanner =
false;

AccidentBannerMessage =
"";

AccidentConfidenceText =
"";
        });



    _node1Timer.Start();

    _node2Timer.Start();

  }



  private void
ResetNode(int node)

  {

    if (node == 1)
    {

      Node1Status =
"CLEAR";

      Node1Distance =
"N/A";

      Node1Speed = "0
km/h";

      Node1Color =
"#228B22";


Node1VehicleOpacity =
0.0;

      _lastNode1Status =
"CLEAR";

    }

    else

    {

      Node2Status =
"CLEAR";

      Node2Distance =
"N/A";

      Node2Speed = "0
km/h";

      Node2Color =
"#228B22";


Node2VehicleOpacity =
0.0;

      _lastNode2Status =
"CLEAR";

    }

  }

  private void
InitializeVlc()

  {

    Core.Initialize();

    _libVLC = new
LibVLC("--network-cachi
ng=500", "--rtsp-tcp");

    MediaPlayer = new
MediaPlayer(_libVLC);

    MediaPlayer.Scale =
0;



    MediaPlayer.Playing
+= (s, e) =>


Dispatcher.UIThread.Post(
() =>
        {


ConnectionStatus =
"VIDEO LIVE";

          StatusColor =
"#44FF44";


ShowStreamError = false;


StreamErrorMessage = "";

        });



MediaPlayer.Encountered
```

```
Error += (s, e) =>

Dispatcher.UIThread.Post(
() =>
        {

ConnectionStatus =
"STREAM ERROR";
            StatusColor =
"#FF4444";

ShowStreamError = true;

StreamErrorMessage =
"Stream connection
failed.\nCheck that
MediaMTX is running on
the RPi.";
        });


    MediaPlayer.Stopped
+= (s, e) =>

Dispatcher.UIThread.Post(
() =>
        {

ConnectionStatus =
"STREAM STOPPED";
            StatusColor =
"#FFAA00";

ShowStreamError = true;

StreamErrorMessage =
"Stream stopped.";
        });
    }


    public void StartVideo()
    {
        if (_libVLC == null ||
MediaPlayer == null)
return;
        if
(MediaPlayer.IsPlaying)
MediaPlayer.Stop();


        ShowStreamError =
false;
        StreamErrorMessage
= "";


_currentMedia?.Dispose();
        _currentMedia = new
Media(_libVLC,
"rtsp://100.100.146.107:85
54/cam",
FromType.FromLocation);

_currentMedia.AddOption
(":rtsp-tcp");

_currentMedia.AddOption
(":network-caching=500");
        MediaPlayer.Media =
_currentMedia;
        MediaPlayer.Play();
    }


    [RelayCommand]
    public void
RetryStream()
    {
        ShowStreamError =
false;
        StreamErrorMessage
= "";
        ConnectionStatus =
"CONNECTING...";
        StatusColor =
"#FFAA00";
        StartVideo();
    }


    [RelayCommand]
    public async Task
RetryMqtt()
    {
        ShowMqttError =
false;
        MqttErrorMessage =
"";
        MqttStatus =
"MQTT:
RECONNECTING...";
```

```
        _mqttCts.Cancel();

        _mqttCts.Dispose();

        _mqttCts = new
CancellationTokenSource(
);



        if (_mqttClient is not
null)

        {

            try { await
_mqttClient.DisconnectAs
ync(); } catch { }


_mqttClient.Dispose();

            _mqttClient = null;

        }



        await
SetupMqttWithRetry(_mqt
tCts.Token);

    }



    // CONFIRM —
commit pending accident
to logs and increment
counter

    [RelayCommand]

    public void
ConfirmAccident()

    {

        if
(_pendingAccidentEntry is
not null)

        {

            AccidentCount++;


AddToRecentLog(_pendin
gAccidentEntry.Node,
_pendingAccidentEntry.M
essage,


_pendingAccidentEntry.Di
stance,
_pendingAccidentEntry.Sp
eed,


_pendingAccidentEntry.Co
lor,
_pendingAccidentEntry.Ra
wLine);


AddToFullLog(FullAccide
ntLog,
_pendingAccidentEntry.N
ode,


_pendingAccidentEntry.M
essage,
_pendingAccidentEntry.Di
stance,


_pendingAccidentEntry.Sp
eed,
_pendingAccidentEntry.Co
lor,


_pendingAccidentEntry.Ra
wLine);

_pendingAccidentEntry =
null;

        }



_accidentDismissTimer.St
op();

        ShowAccidentBanner
= false;


AccidentBannerMessage =
"";


AccidentConfidenceText =
"";



        ShowAccidentModal
= false;


AccidentModalImage?.Dis
pose();

        AccidentModalImage
= null;


AccidentModalMessage =
"";


AccidentModalConfidence
= "";


AccidentModalTimestamp
= "";

    }
```

```
    // DISMISS — discard pending accident silently
    [RelayCommand]
    public void DismissAccident()
    {
        _pendingAccidentEntry = null;

        _accidentDismissTimer.Stop();
        ShowAccidentBanner = false;
        AccidentBannerMessage = "";
        AccidentConfidenceText = "";

        ShowAccidentModal = false;
        AccidentModalImage?.Dispose();
        AccidentModalImage = null;
        AccidentModalMessage = "";
        AccidentModalConfidence = "";
        AccidentModalTimestamp = "";
    }

    [RelayCommand]
    public void CloseAccidentModal()
    {
        // Closing without choosing = dismiss
        _pendingAccidentEntry = null;

        ShowAccidentModal = false;
        AccidentModalImage?.Dispose();
        AccidentModalImage = null;
        AccidentModalMessage = "";
        AccidentModalConfidence = "";
        AccidentModalTimestamp = "";
    }

    [RelayCommand]
    public void ShowTrafficLog()
    {
        LogOverlayTitle = "TRAFFIC DETECTION LOG";
        LogOverlayAccentColor = "#44FF44";
        ActiveOverlayLog = FullTrafficLog;
        ShowLogOverlay = true;
        IsVideoVisible = false;
    }

    [RelayCommand]
    public void ShowAccidentLog()
    {
        LogOverlayTitle = "ACCIDENT DETECTION LOG";
        LogOverlayAccentColor = "#FF4422";
        ActiveOverlayLog = FullAccidentLog;
        ShowLogOverlay =
```

```
true;
        IsVideoVisible =
false;
    }


    [RelayCommand]
    public void
CloseLogOverlay()
    {
        ShowLogOverlay =
false;
        IsVideoVisible = true;
    }


    private async Task
SetupMqttWithRetry(Canc
ellationToken ct)
    {
        var mqttFactory =
new MqttClientFactory();
        _mqttClient =
mqttFactory.CreateMqttCli
ent();


        _mqttOptions = new
MqttClientOptionsBuilder(
)

.WithTcpServer("100.100.
146.107", 1883)
.WithKeepAlivePeriod(Ti
meSpan.FromSeconds(10)
)

.WithTimeout(TimeSpan.F
romSeconds(5))
            .Build();



_mqttClient.ApplicationM
essageReceivedAsync +=
e =>
        {
            var topic =
e.ApplicationMessage.Top
ic;
            var payload =
e.ApplicationMessage.Con
vertPayloadToString();

Dispatcher.UIThread.Post(
() =>
HandleMqttMessage(topic,
payload));
            return
Task.CompletedTask;
        };



_mqttClient.Disconnected
Async += async e =>
        {
            if
(ct.IsCancellationRequeste
d) return;

Dispatcher.UIThread.Post(
() =>
            {
                MqttStatus =
"MQTT:
DISCONNECTED";
                ShowMqttError
= true;

MqttErrorMessage =
"MQTT broker
disconnected.\nCheck
Tailscale and Mosquitto on
the RPi.";
            });
            await
Task.Delay(3000,
ct).ContinueWith(_ => {
});
            if
(!ct.IsCancellationRequest
ed)
                await
TryConnectMqtt(_mqttOpt
ions!, ct);
        };


        await
TryConnectMqtt(_mqttOpt
ions!, ct);
    }


    private async Task
TryConnectMqtt(MqttClie
```

```
ntOptions options,
CancellationToken ct)

    {

        try

        {

            await
_mqttClient!.ConnectAsyn
c(options, ct);

            await
_mqttClient!.SubscribeAsy
nc(

                new
MqttClientSubscribeOptio
nsBuilder()


.WithTopicFilter("traffic/#
")


.WithTopicFilter("system/
#")


.WithTopicFilter("vision/#
")


.WithTopicFilter("lumen/#
")

                    .Build(), ct);


Dispatcher.UIThread.Post(
() =>

            {

                MqttStatus =
"MQTT: CONNECTED";
ConnectionStatus =
"CONNECTED";

                StatusColor =
"#44FF44";

                ShowMqttError
= false;


MqttErrorMessage = "";

            });

        }

        catch
(OperationCanceledExcept
ion) { }

        catch

        {


Dispatcher.UIThread.Post(
() =>

            {

                MqttStatus =
"MQTT: CONN FAIL";


ConnectionStatus =
"DISCONNECTED";

                StatusColor =
"#FF4444";

                ShowMqttError
= true;


MqttErrorMessage =
"Cannot reach MQTT
broker at
100.100.146.107:1883.\nC
heck Tailscale VPN and
Mosquitto service.";

            });

        }

    }


    private void
HandleMqttMessage(strin
g topic, string payload)

    {

        if (topic ==
"traffic/node1" || topic ==
"traffic/node2")

        {

            var parts =
payload.Trim().Split(' ');

            var status =
parts[0].ToUpper();

            var distance =
parts.Length > 1 ?
$"{parts[1]}m" : "N/A";

            var speed =
parts.Length > 2 ?
$"{parts[2]} km/h" :
"N/A";

            var isDanger =
status.Contains("WARNIN
G");

            var color =
isDanger ? "#FF4444" :
"#228B22";


            if (topic ==
"traffic/node1")
```

```
        {

_node1Timer.Stop();

_node1Timer.Start();
            Node1Status =
status;
            Node1Distance
= distance;
            Node1Speed =
speed;
            Node1Color =
color;

Node1VehicleOpacity =
isDanger ? 1.0 : 0.0;

AddToRecentLog("NODE
1", status, distance, speed,
color, payload);
            if (isDanger &&
_lastNode1Status !=
"WARNING")
            {

Node1TrafficCount++;

TotalAlertCount++;

AddToFullLog(FullTraffic
Log, "NODE 1", status,
distance, speed, color,
payload);
            }
_lastNode1Status = status;
        }
        else
        {

_node2Timer.Stop();

_node2Timer.Start();
            Node2Status =
status;
            Node2Distance
= distance;
            Node2Speed =
speed;
            Node2Color =
color;

Node2VehicleOpacity =
isDanger ? 1.0 : 0.0;

AddToRecentLog("NODE
2", status, distance, speed,
color, payload);
            if (isDanger &&
_lastNode2Status !=
"WARNING")
            {

Node2TrafficCount++;

TotalAlertCount++;

AddToFullLog(FullTraffic
Log, "NODE 2", status,
distance, speed, color,
payload);
            }

_lastNode2Status = status;
        }
    }
    else if (topic ==
"system/rpi")
    {
        try
        {
            var json =
JsonDocument.Parse(paylo
ad).RootElement;
            RpiCpu =
$"{json.GetProperty("cpu"
).GetDouble():0.0}%";
            RpiRam =
$"{json.GetProperty("ram"
).GetDouble():0.0}%";
            var tempStr =
json.GetProperty("temp").
GetString();
            if
(double.TryParse(tempStr,
out double tempVal))
                RpiTemp =
$"{tempVal /
1000.0:0.0}°C";
        }
```

```
        catch { }

    }

    else if (topic ==
"vision/accident")

    {

        try

        {

            var json =
JsonDocument.Parse(paylo
ad).RootElement;

            var conf =
json.GetProperty("confide
nce").GetDouble();

            var confPct =
(int)(conf * 100);

            var ts =
json.TryGetProperty("time
stamp", out var t) ?
t.GetString() ?? "" : "";



            // Store pending
— not logged until
confirmed


_pendingAccidentEntry =
new AlertEntry

            {

                Time =
DateTime.Now.ToString("
HH:mm:ss"),

                Node =
"VISION",

                Message =
"ACCIDENT",

                Distance = "",

                Speed =
$"{confPct}%",

                Color =
"#FF8800",

                RawLine =
$"conf={confPct}%
ts={ts}",

            };



AccidentBannerColor =
"#FF8800";


AccidentBannerBg =
"#1A0D00";


AccidentBannerMessage =
"⚠  ACCIDENT
DETECTED";


AccidentConfidenceText =
$"Confidence: {confPct}%
•  {ts}";


ShowAccidentBanner =
true;


_accidentDismissTimer.St
op();


_accidentDismissTimer.St
art();


AccidentModalImage?.Dis
pose();


AccidentModalImage =
null;

            if
(json.TryGetProperty("ima
ge", out var imgProp))

            {

                var base64 =
imgProp.GetString();

                if
(!string.IsNullOrEmpty(ba
se64))

                {

                    var bytes =
Convert.FromBase64Strin
g(base64);

                    using var
ms = new
MemoryStream(bytes);


AccidentModalImage =
new Bitmap(ms);

                }

            }



AccidentModalMessage =
"⚠  ACCIDENT
DETECTED";


AccidentModalConfidence
= $"Confidence:
```

```
{confPct}%";

AccidentModalTimestamp = ts;

AccidentModalColor = "#FF8800";

ShowAccidentModal = true;
        }
        catch { }
    }
    else if (topic == "lumen/alert")
    {
        try
        {
            var json = JsonDocument.Parse(payload).RootElement;
            var conf = json.GetProperty("confidence").GetDouble();
            var confPct = (int)(conf * 100);
            var ts = json.TryGetProperty("timestamp", out var t) ? t.GetString() ?? "" : "";

            // Store pending — not logged until confirmed
            _pendingAccidentEntry = new AlertEntry
            {
                Time = DateTime.Now.ToString("HH:mm:ss"),
                Node = "RADAR+CAM",
                Message = "HIGH PRIORITY",
                Distance = "",
                Speed = $"{confPct}%",
                Color = "#FF2222",
                RawLine = $"conf={confPct}% ts={ts}",
            };

            AccidentBannerColor = "#FF2222";

            AccidentBannerBg = "#1A0000";

            AccidentBannerMessage = "🚨 HIGH PRIORITY ACCIDENT";

            AccidentConfidenceText = $"Confidence: {confPct}% • {ts}";

            ShowAccidentBanner = true;

            _accidentDismissTimer.Stop();

            _accidentDismissTimer.Start();



            AccidentModalImage?.Dispose();

            AccidentModalImage = null;
            if (json.TryGetProperty("image", out var imgProp))
            {
                var base64 = imgProp.GetString();
                if (!string.IsNullOrEmpty(base64))
                {
                    var bytes = Convert.FromBase64String(base64);
                    using var ms = new MemoryStream(bytes);
```

```
AccidentModalImage =
new Bitmap(ms);
                }

            }



AccidentModalMessage =
"🚨 HIGH PRIORITY
ACCIDENT";


AccidentModalConfidence
= $"Confidence:
{confPct}%";


AccidentModalTimestamp
= ts;


AccidentModalColor =
"#FF2222";


ShowAccidentModal =
true;
        }
        catch { }
    }
  }


  private void
AddToRecentLog(string
node, string status, string
distance,
                    string
speed, string color, string
rawLine = "")
  {
    var entry = new
AlertEntry
    {
      Time =
DateTime.Now.ToString("
HH:mm:ss"),
      Node = node,
      Message = status,
      Distance =
distance,
      Speed = speed,
      Color = color,
      RawLine =
rawLine,
    };


    if (AlertLog.Count
>= 50)

AlertLog.RemoveAt(Alert
Log.Count - 1);


    AlertLog.Insert(0,
entry);
  }


  private void
AddToFullLog(Observable
Collection<AlertEntry>
log,
                    string
node, string status, string
distance,
                    string
speed, string color, string
rawLine = "")
  {
    var entry = new
AlertEntry
    {
      Time =
DateTime.Now.ToString("
HH:mm:ss"),
      Node = node,
      Message = status,
      Distance =
distance,
      Speed = speed,
      Color = color,
      RawLine =
rawLine,
    };
    log.Insert(0, entry);
  }


  public void Dispose()
  {
    _mqttCts.Cancel();
    _mqttCts.Dispose();
```

```
        _node1Timer.Stop();
        _node1Timer.Dispose();
        _node2Timer.Stop();
        _node2Timer.Dispose();
        _accidentDismissTimer.Stop();
        _accidentDismissTimer.Dispose();

        if (_mqttClient is not null)
        {
            try { _mqttClient.DisconnectAsync().GetAwaiter().GetResult(); } catch { }
            _mqttClient.Dispose();
        }

        if (MediaPlayer is not null)
        {
            if (MediaPlayer.IsPlaying)
                MediaPlayer.Stop();
            MediaPlayer.Dispose();
        }

        AccidentModalImage?.Dispose();
        _currentMedia?.Dispose();
        _libVLC?.Dispose();
    }
}
```